KRATOS

Kratos: an agentic automation system for model calibration, estimation and verification

Application to human monoclonal-antibody mean pharmacokinetic profiles (two-part ID split) and a synthetic verification study

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> **Note**
>
> **Data provenance.** Runs labelled SYNTHETIC use records simulated from known parameters and serve only to verify the platform. The human monoclonal-antibody (mAb) runs use published *mean* concentration–time profiles. Every number in this document is read from saved run artifacts at render time; nothing is typed by hand or re-estimated during rendering.

Abstract

Background. Fitting target-mediated drug disposition (TMDD) models involves many repetitive, error-prone steps: dataset preparation, unit handling, choosing starting values, numerical optimisation, diagnostic review and reporting.

Objective. To build and evaluate Kratos, a constrained and auditable agentic workflow that automates these steps from raw data to a reproducible report, while keeping every scientific decision explicit.

Methods. A human mAb dataset of 143 published mean profiles (36 molecules) was split by ID, grouped by molecule, into Part A (72 profiles) and Part B (71 profiles); each part was analysed with a two-compartment full TMDD model (SC absorption depot) by bounded multistart pooled calibration followed by nlmixr2 FOCEi population estimation, heuristic diagnostics and a visual predictive check; each part's model then predicted the other part's unseen molecules; platform accuracy was verified on a synthetic dataset with known parameters and by Python–R cross-engine agreement; the adaptive planner was compared with a fixed multistart baseline on replicated synthetic data.

Results. On synthetic data all 6 estimated typical values were recovered within 9% of truth; Python and R predictions agreed to a maximum relative difference of 1.2e-06; 54 of 54 automated tests passed; Part A FOCEi converged (OFV 4005.5; covariance step succeeded) with typical CL 0.188 L/day, Vc 3.08 L, ka 0.209 1/day and F 0.71; Part B FOCEi converged (OFV 4890.7; covariance step failed) with typical CL 0.268 L/day, Vc 3.50 L, ka 0.304 1/day and F 0.84; the Part A model predicted Part B with median fold error 1.29 (76% within 2-fold); the Part B model predicted Part A with median fold error 1.35 (84% within 2-fold); the adaptive and fixed planners reached equivalent optima in 10/10 replicates.

Conclusions. The workflow ran end to end without analyst intervention, reproduced known parameters, kept numerical failures visible, and produced a complete audit trail and report. Its accuracy and reproducibility claims are supported on synthetic data; a reduction in analyst effort relative to a manual workflow and any advantage of adaptive over fixed planning were not demonstrated. For the pooled cross-molecule mAb data, linear PK and absorption parameters were well determined, whereas target-mediated parameters were not identifiable.

1 Hypothesis

Central hypothesis. A constrained, auditable agentic workflow — in which a deterministic controller plans and executes bounded numerical actions, validates every proposal and result, and records all evidence — can reduce the analyst effort needed to calibrate and estimate TMDD models **while preserving** (i) estimation accuracy, (ii) numerical reliability and (iii) reproducibility.

This was treated as a testable hypothesis, not an established claim. It was operationalised as four sub-hypotheses:

	Sub-hypothesis	Evidence used in this report
H1	**Accuracy** — on data with known generating parameters, the automated workflow recovers identifiable typical values within a prespecified tolerance (±25%).	Synthetic parameter recovery; bootstrap intervals
H2	**Numerical reliability** — the two independent solvers agree, invalid inputs and failed numerics are rejected or flagged rather than hidden, and convergence/covariance problems remain visible.	Cross-engine agreement; automated test suite; review flags on real data
H3	**Reproducibility** — identical data, configuration and seeds reproduce the analysis, and all results can be regenerated from saved artifacts without refitting.	Seed-reproducibility and regeneration tests; run manifests
H4	**Effort / efficiency** — the adaptive (agentic) planner reaches acceptable fits with fewer attempts or less compute than a fixed multistart baseline under the same budget, and the complete analysis requires no manual intervention.	Planner comparison on replicated synthetic data; intervention counts

2 Specific aims

1. **Aim 1 — Reproducible TMDD foundation.** Implement explicit TMDD equations, dose-event handling (IV bolus, infusion, repeat and SC dosing), unit normalisation with declared molecular weights, assay mappings and data validation, with matching Python and R simulation engines.
2. **Aim 2 — Adaptive bounded calibration.** Implement multistart numerical calibration in transformed parameter space with an adaptive planner, diagnostic feedback and transparent stopping rules.
3. **Aim 3 — Population estimation.** Use calibrated starting values for nlmixr2 FOCEi population estimation with uncertainty assessment and documented convergence review.
4. **Aim 4 — Automated reporting.** Automatically produce raw-data graphics, model diagnostics, parameter tables, standalone results scripts and this Quarto report.
5. **Application aim.** Apply the platform to the human mAb mean-profile dataset split into two parts by ID, compare the parts, and test out-of-part prediction.

3 Methods

3.1 Data and two-part split

The source dataset (`mAb_dataset.csv`) contains 143 IDs; each ID is a published **mean** concentration–time profile of one antibody at one dose and route (single doses; columns ID, TIME, AMT, DV, ADM). IDs were assigned wholly to one of two parts using a seeded (seed 20260929), balanced assignment of *molecules*, so all dose levels and routes of an antibody fall in the same part and the parts share no molecule. Part A: 72 profiles, 19 molecules, 13 SC; Part B: 71 profiles, 17 molecules, 13 SC. One spelling variant (Benralizuman → Benralizumab) was aliased for grouping only; IDs were not edited.

Declared data interpretation: TIME in h, AMT in ug, DV in ug/mL, molecular weight 150000 g/mol for every antibody, assay measures **free drug**, ADM 1 = IV bolus and 2 = SC, dose/observation convention *post_dose*. These assumptions are not documented in the source and are recorded in the configuration.

Synthetic verification data. 36 simulated subjects in six design groups (IV bolus 0.1–1 mg/kg, 1-h infusion 3 and 10 mg/kg, and 1 mg/kg every 2 weeks ×3 with pre-dose troughs) were generated from known typical parameters with log-normal between-subject variability (ω^2 CL 0.09, Vc 0.04, R0 0.09) and log-normal residual error (SD 0.1). Samples below 0.05 nmol/L were not recorded (111 samples).

Validation. The data agent maps columns, derives dose events, converts to hours, nmol and nmol/L, and records every conversion. It rejects unsupported or inconsistent records with row- and field-specific errors (e.g. unsupported event codes, steady-state shortcuts, inconsistent infusions, censored records, non-positive concentrations). It flags suspicious records for review but never deletes them.

3.2 TMDD model

Two-compartment full TMDD model with central binding and peripheral free-drug distribution (approved template `tmdd2c_full` v1.1.0, 1:1 binding):

$$
\begin{aligned}
\frac{dA_d}{dt} &= -k_a A_d \quad (\text{SC depot; dose} \times F) \\
\frac{dA_c}{dt} &= k_a A_d + \text{Input}(t) - CL\,C - Q\,(C - C_p) - V_c\,k_{on}\,C\,R + V_c\,k_{off}\,RC \\
\frac{dA_p}{dt} &= Q\,(C - C_p) \\
\frac{dR}{dt} &= k_{syn} - k_{deg} R - k_{on}\,C\,R + k_{off}\,RC \\
\frac{dRC}{dt} &= k_{on}\,C\,R - (k_{off} + k_{int})\,RC
\end{aligned}
$$

where $C = A_c/V_c$, $C_p = A_p/V_p$ and $k_{syn} = k_{deg} R_0$ (drug-naive baseline $R = R_0$). The observed concentration was mapped to free drug C.

For the mAb analyses, **kon = 1 L/(nmol*h), koff = 0.1 1/h, kdeg = 0.1 1/h** were fixed at generic values, because binding and target turnover cannot be estimated from drug-only mean data; CL, Vc, Q, Vp, kint, R0, ka, F were estimated. Random effects (log-normal) were placed on CL, Vc, R0.

Statistical model. $\theta_i = \theta_{pop}\exp(\eta_i)$, with $\eta \sim N(0, \Omega)$ diagonal; residual error log-normal, $\log y = \log f + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$. For the mAb data each "individual" is a mean profile, so Ω describes **between-profile (largely between-molecule) variability**.

3.3 Calibration (Aim 2)

Pooled calibration (one parameter set for all profiles) supplied starting values. Starts were the scientist's initial values plus Latin-hypercube points in log (logit for F) space, within declared bounds. Each attempt minimised the sum of squared log residuals, which is consistent with the log-normal residual model, using a bounded trust-region least-squares optimiser. The rules planner used failed and successful attempts to propose further starts: after failures it moved starts halfway toward the centre of the bounds; otherwise it refined locally around the best fit, pulled near-bound parameters back from the bound, and added exploration points. The loop stopped at the attempt or time budget, on a plateau of the best objective, or once the best objective had been reproduced from several independent starts. Every attempt, including failures, was recorded with its start values, fit, evaluations, elapsed time and flags.

3.4 Population estimation (Aim 3)

Up to three distinct, ranked calibration optima were translated into an nlmixr2 model (log/logit-scale thetas with the declared bounds; fixed parameters via `fix()`) and fitted with **FOCEi** (`est = "focei"`, covariance method `r,s`, LSODA with rtol 1e-8, atol 1e-10). Retries were bounded, and the first fit with a finite objective function value was accepted. Saved outputs include the model code, fit object, estimates, standard errors, Wald confidence intervals, Ω, residual SD, individual parameters, predictions and residuals.

3.5 Diagnostics and review

The following were computed from the saved fit: - optimiser convergence and covariance-step status; - delta-method relative standard errors on the parameter scale; - proximity of estimates to their bounds, parameter correlations and η-shrinkage; - plausibility ranges; - trends in conditional weighted residuals (CWRES); - a visual predictive check (VPC): 200 replicates from `vpcSim` preserving each profile's dose history, between-profile variability and residual error, stratified by route, dose-normalised, with bins at observed-time quantiles.

Thresholds (RSE > 50%, shrinkage > 30%, |correlation| > 0.95, estimate within 2% of a bound, more than 20% of VPC percentile points outside the simulated intervals) are **heuristic review flags**, not acceptance criteria. On the synthetic run, a subject-level bootstrap refitted FOCEi to resampled subjects, with each subject's dose history preserved.

3.6 Verification, out-of-part prediction and hypothesis evaluation

- **Cross-engine agreement.** Python (SciPy stiff solver) and R (rxode2) predictions were compared on identical events and parameters against a prespecified criterion: |difference| ≤ 1e-6 nmol/L + 1e-4 × |value|.
- **Automated tests.** These covered hand-checkable dosing cases, mass balance, rejection of invalid inputs, planner and LLM guardrails, cancellation, resume, secrets and reproducibility.
- **Out-of-part prediction.** The typical values, Ω and residual SD from one part were used to simulate the other part's design with 200 replicates, giving population predictions (fold error) and 90% prediction-interval coverage.

- **Hypothesis H4.** The adaptive rules planner was compared with a non-adaptive Latin-hypercube multistart on five replicated synthetic datasets. Both arms used the same model, bounds, evaluation budgets and stopping rules. There were two settings: default, and a *stress* setting with a poor user start inside a numerically failing region and a smaller first round.

3.7 Software and reproducibility

The implementation used Python 3.12 (NumPy, SciPy, pandas, Matplotlib) and R 4.5.1 (rxode2 5.1.6, nlmixr2 4.0.1, nlmixr2est 4.1.0), with reports rendered by Quarto 1.10. Each run is stored in its own directory holding: - the original inputs, normalised data and subject-ID map; - a configuration snapshot, seeds, and hashes of the code and inputs; - the audit log, calibration history, fit object, tables, figures (with the data behind each) and a checksummed manifest.

4 Results

4.1 Platform verification (H1–H3)

Cross-engine agreement. All 1362 comparisons in 4 cases passed the prespecified criterion (maximum relative difference 1.2e-06).

Python vs R prediction agreement

case	comparisons	passed	max relative difference	max absolute difference (nmol/L)
typical parameters, pre-dose	325	325	3.176e-07	4.966e-05
typical parameters, post-dose	325	325	3.918e-09	1.466e-07
high-capacity target, pre-dose	325	325	1.184e-06	2.829e-05
synthetic demo dataset, 36 individual parameter sets	387	387	2.940e-09	4.303e-07

Automated tests. 54 passed, 0 failed, 0 skipped (19 min), including seed reproducibility, regeneration of all figures without refitting, calibration-only operation without R, cancellation preserving records, and exclusion of secrets from exports.

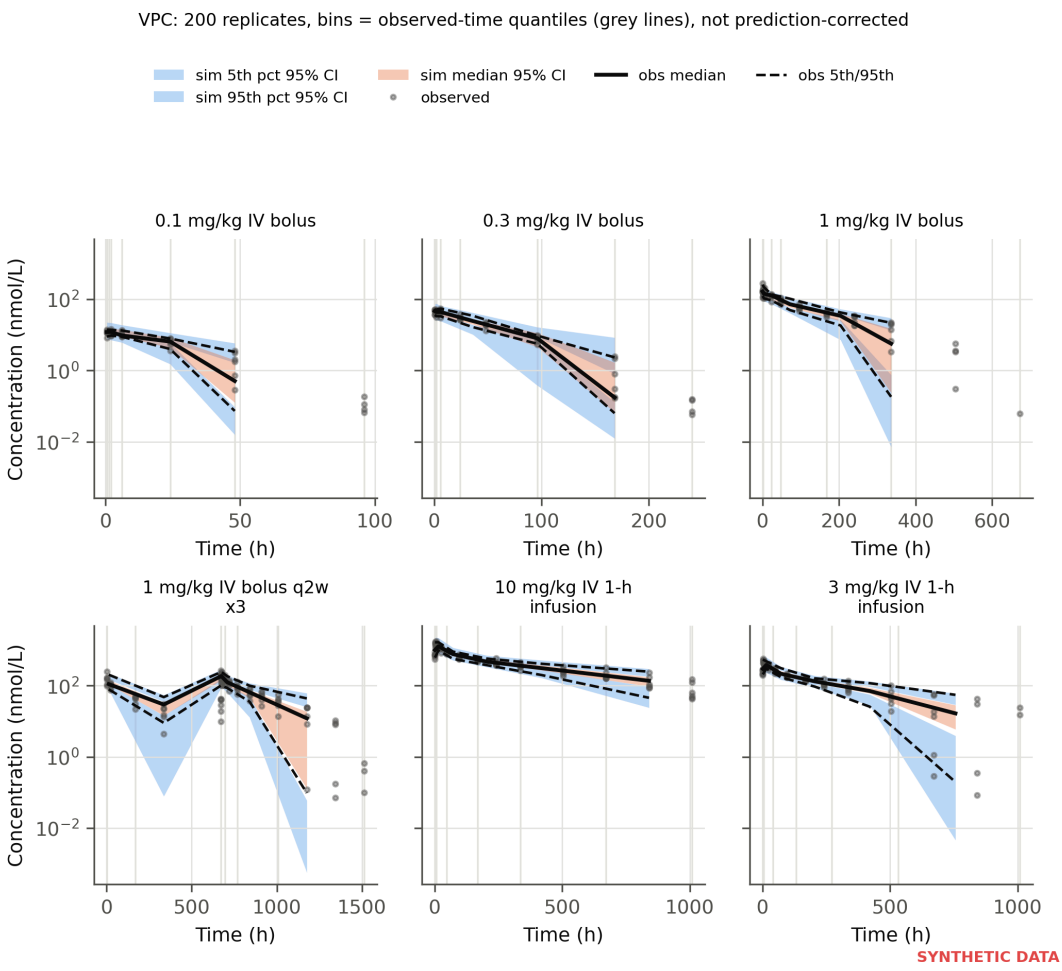
Synthetic parameter recovery (H1). FOCEi converged (OFV 2054.96; relative convergence (4); covariance successful) and **all 6 typical values were within 8.6%** of truth, inside the prespecified ±25%. The true value lay inside the Wald 95% CI for 5/6 parameters. The VPC showed 4% of observed percentile points outside the simulated intervals, and no heuristic review flags were raised.

Recovery of known parameters from the SYNTHETIC dataset

parameter	unit	true	estimate	ci95_lower	ci95_upper	relative error (%)	true value in 95% CI	bootstrap 95% interval
CL	L/h	0.01	0.01008	0.009186	0.01106	0.7935	True	0.009506–0.0105
Vc	L	3	3.045	2.864	3.237	1.489	True	2.899–3.235
Q	L/h	0.03	0.03259	0.03071	0.03459	8.639	False	0.03122–0.03369
Vp	L	2.5	2.611	2.455	2.778	4.46	True	2.569–2.685
kint	1/h	0.05	0.05254	0.0491	0.05622	5.075	True	0.05093–0.05592
R0	nmol/L	2	1.881	1.704	2.075	-5.969	True	1.738–2.047

Between-subject variances were recovered less precisely (estimated vs true ω^2: CL 0.034 vs 0.09; Vc 0.033 vs 0.04; R0 0.073 vs 0.09); the CL variance was underestimated, a common finding with 6 subjects per design group.

In the bootstrap, 5/8 replicates converged (the remainder ended in ‘false convergence’ and were excluded, not repaired). The percentile intervals were consistent with the Wald intervals.



SYNTHETIC run: visual predictive check by design group

4.2 mAb dataset: validation and calibration

Both parts passed validation with review flags only. The flagged items were: dose rows that also carried a time-0 peak concentration (split into dose + observation records), small time inversions from digitisation (sorted, original unchanged), one IV profile with a late rise (Part A), and the aggregate-data warning. No record was deleted.

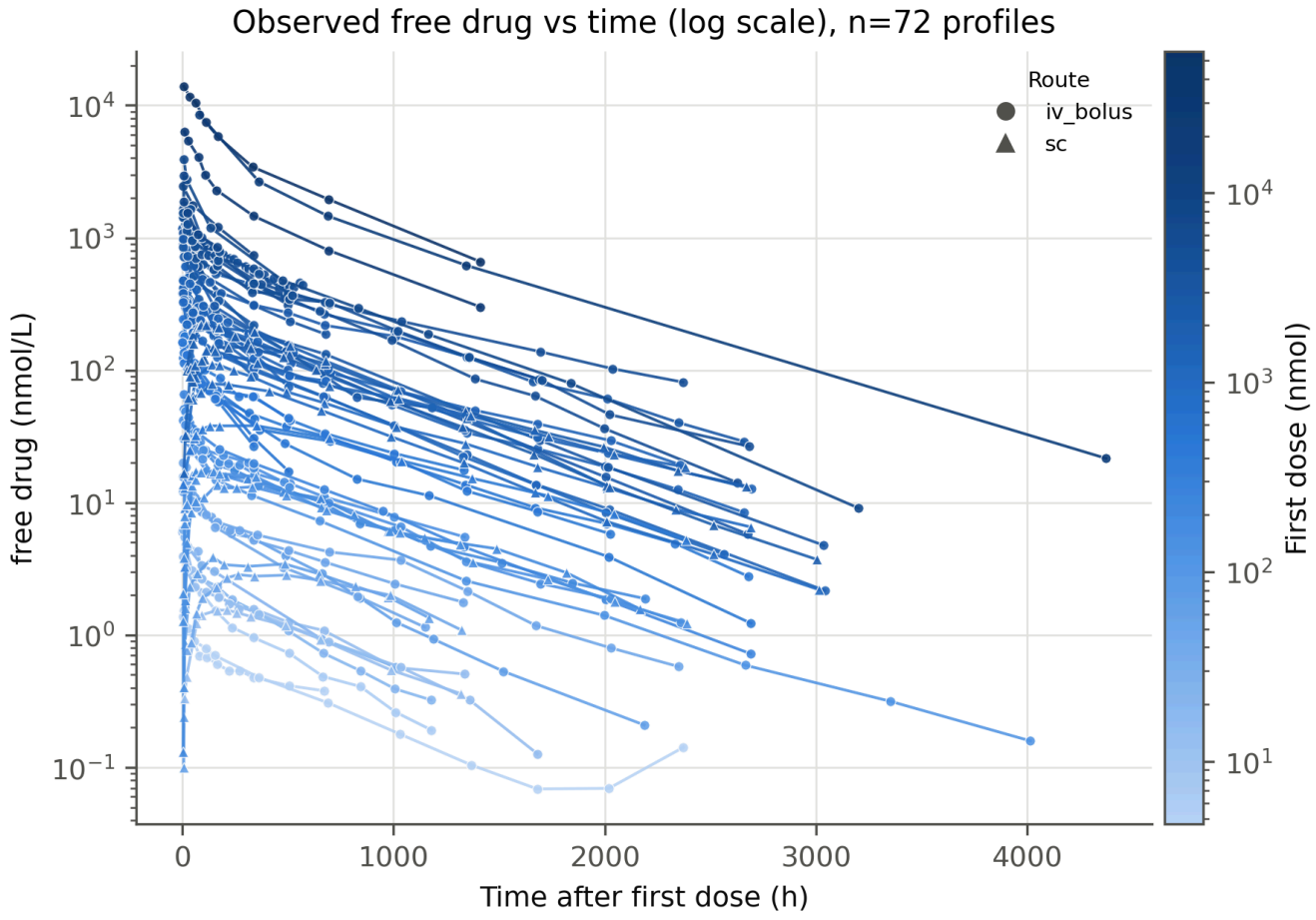
mAb data and calibration summary

part	profiles	observations	IV / SC profiles	dose range (nmol)	review flags	calibration attempts	stop reason
Part A	72	744	59 / 13	4.67–5.6e+04	aggregate_data, coincident_dose_observation, combined_dose_observation_split, suspicious_rise, unsorted_time	8	numerical_success

part	profiles	observations	IV / SC profiles	dose range (nmol)	review flags	calibration attempts	stop reason
Part B	71	679	58 / 13	14–7.2e+04	aggregate_data, coincident_dose_observation, combined_dose_observation_split, unsorted_time	8	plateau

Part A: 8/8 calibration attempts succeeded and the best pooled objective was reproduced from independent starts (numerical_success). The pooled optimum put most elimination on the linear pathway (R0 = 0.0184 nmol/L, kint = 0.00058 1/h), as expected when molecules with different targets share one parameter set.

Part B: 7/8 calibration attempts succeeded and the best pooled objective was reproduced from independent starts (plateau). The pooled optimum put most elimination on the linear pathway (R0 = 0.283 nmol/L, kint = 0.0001 1/h), as expected when molecules with different targets share one parameter set.



Part A: observed mean profiles (colour = dose; marker = route)

4.3 Population estimates (Part A and Part B)

FOCEi fit summary for the two parts

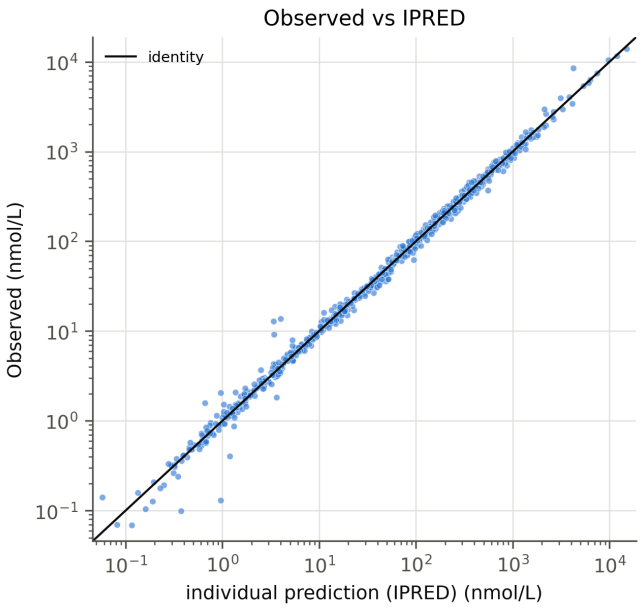
part	OFV	convergence	covariance step	runtime (min)	CWRES mean / SD	VPC points outside PI	critical / warning flags
Part A	4005	relative convergence (4)	success	63.09	0.02 / 0.98	21%	0 / 4
Part B	4891	relative convergence (4)	FAILED	33.63	-0.01 / 1.02	43%	1 / 4

Typical values are summarised below (kon = 1 L/(nmol·h), koff = 0.1 1/h and kdeg = 0.1 1/h were fixed).

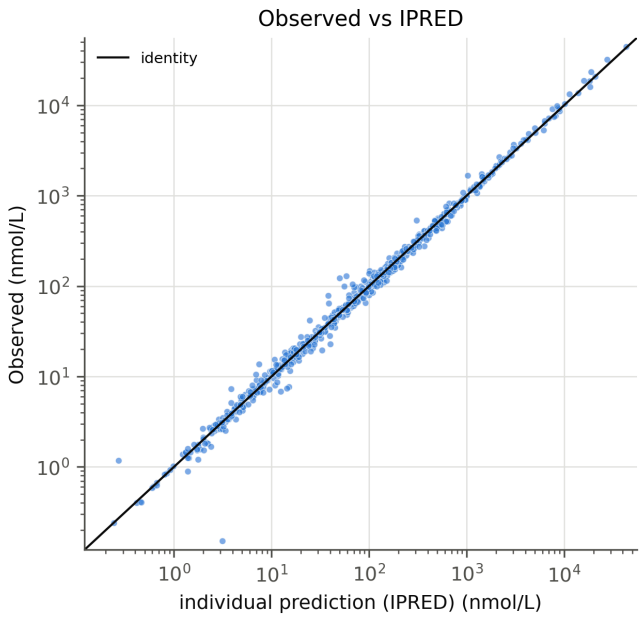
Population (typical) parameter estimates; BSV = between-profile variability

parameter	unit	Part A estimate (95% CI)	Part A RSE %	Part A BSV CV% / shrinkage %	Part B estimate (95% CI)	Part B RSE %	Part B BSV CV% / shrinkage %
CL	L/h	0.00783 (0.00727–0.00843)	3.782	26 / 9	0.0112 (SE unavailable)	–	68 / 2
Vc	L	3.08 (2.88–3.28)	3.296	28 / 9	3.5 (SE unavailable)	–	69 / 2
Q	L/h	0.0214 (0.0161–0.0285)	14.48	–	0.0184 (SE unavailable)	–	–
Vp	L	2.63 (2.39–2.91)	4.995	–	2.26 (SE unavailable)	–	–
kint	1/h	0.000396 (6.4e-05–0.00245)	93.02	–	0.0001 (SE unavailable)	–	–
R0	nmol/L	0.0058 (0.00159–0.0211)	65.92	162 / 86	0.0382 (SE unavailable)	–	119 / 67
ka	1/h	0.00869 (0.00696–0.0108)	11.31	–	0.0127 (SE unavailable)	–	–
F	fraction	0.705 (0.618–0.78)	5.892	–	0.835 (SE unavailable)	–	–
lnorm.sd	log-scale SD	0.197	–	–	0.221	–	–

Linear disposition and absorption parameters were similar between the two independent parts: CL 0.188 vs 0.268 L/day, Vc 3.08 vs 3.50 L, Vp 2.63 vs 2.26 L, ka 0.209 vs 0.304 1/day and F 0.71 vs 0.84. Target-mediated parameters were not: in Part A kint = 0.0004 1/h (RSE 93%) and R0 = 0.0058 nmol/L (RSE 66%, η-shrinkage 86%). In Part B kint converged to its lower bound (0.0001 1/h), which made the covariance step fail (critical flag), and R0 η-shrinkage was 67%. Between-profile variability of CL and Vc was 26% and 28% (CV) in Part A, but 68% and 69% in Part B.

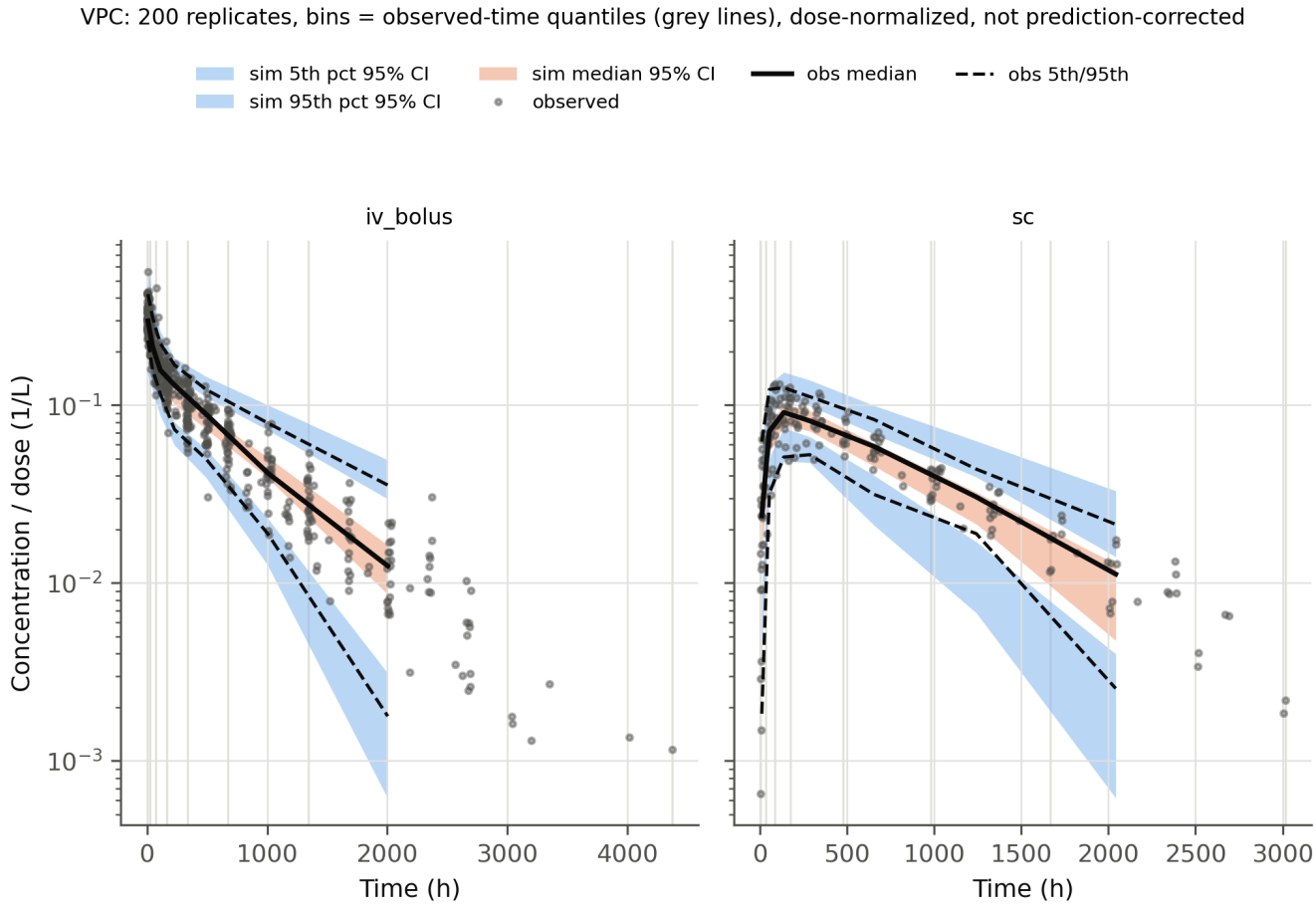


Part A: observed vs individual predictions

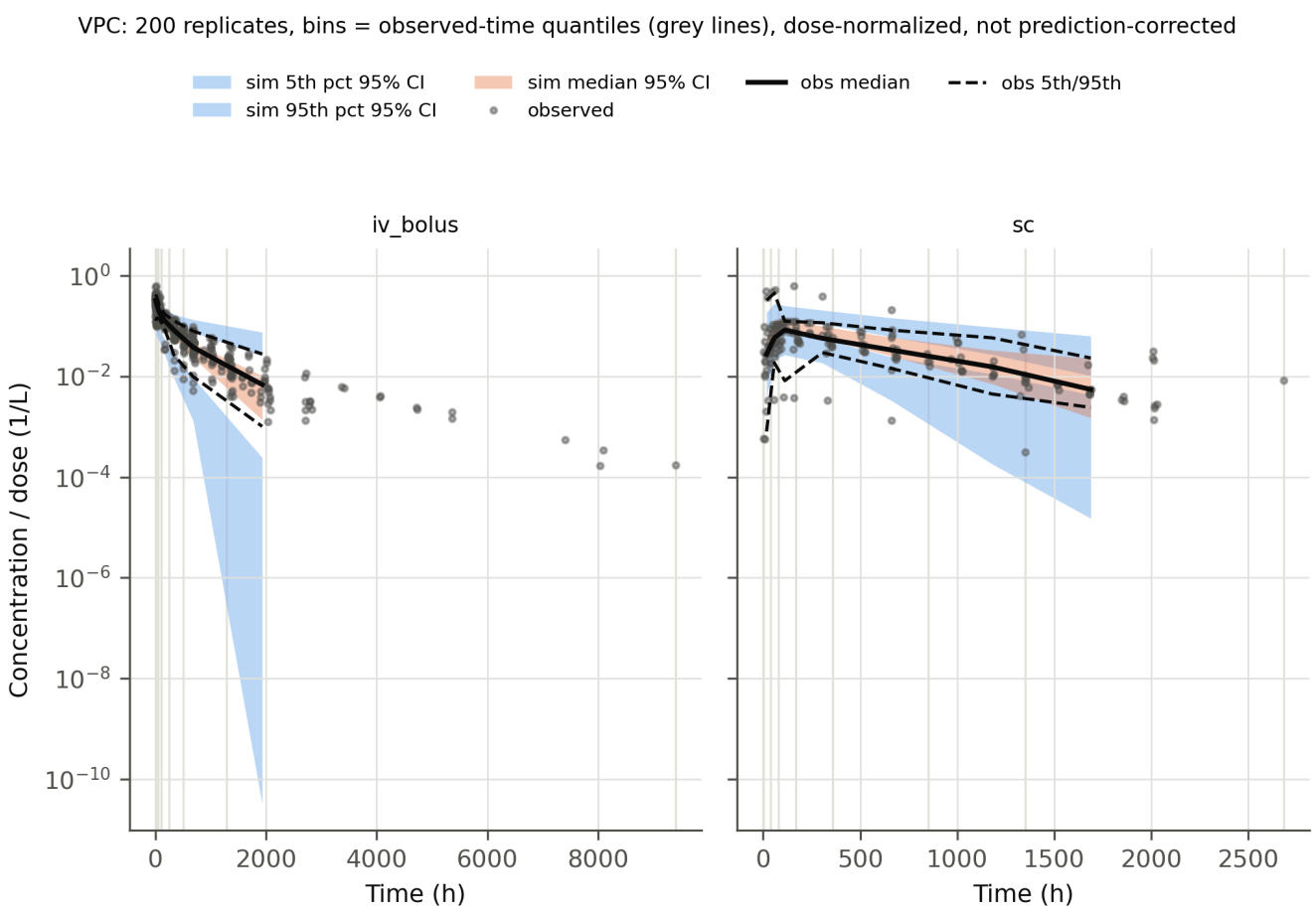


Part B: observed vs individual predictions

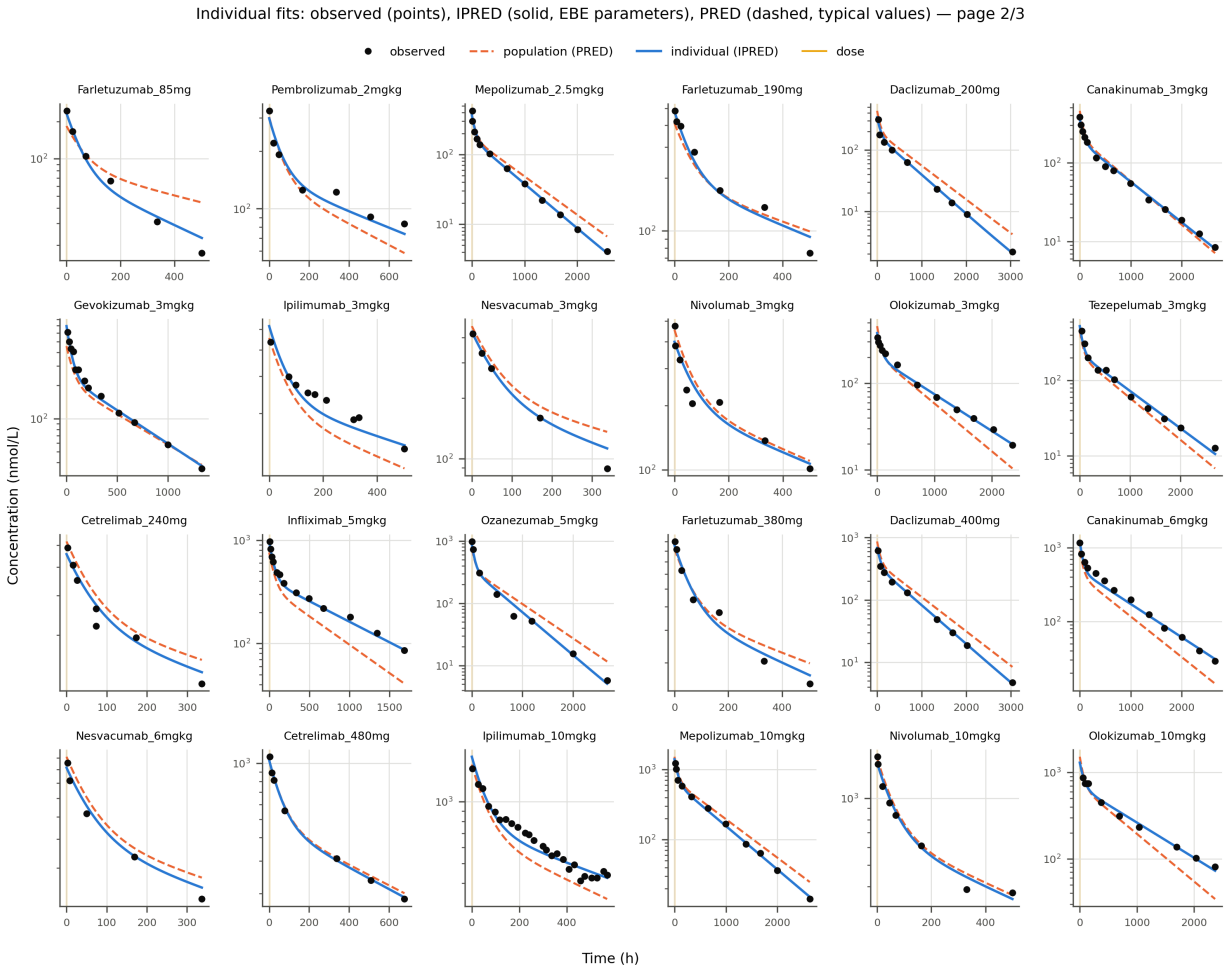
Individual predictions tracked the observed profiles closely in both parts. Population predictions and the VPC were less satisfactory: 21% (Part A) and 43% (Part B) of observed percentile points fell outside the simulated 95% intervals. The lower simulated percentiles flared at late times, because a subset of simulated low-dose profiles is cleared rapidly by the target pathway.



Part A: dose-normalised VPC stratified by route (200 replicates)



Part B: dose-normalised VPC stratified by route (200 replicates)



Part A: individual fits, page 2 (points observed; solid IPRED; dashed PRED)

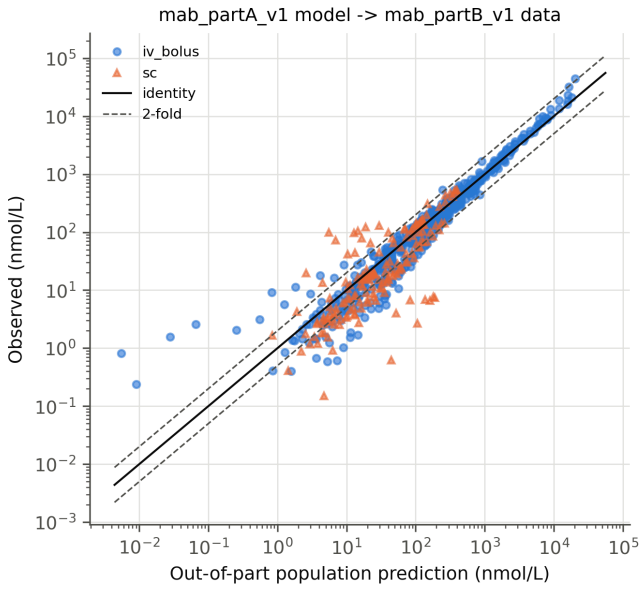
4.4 Out-of-part prediction

Each part's population model was applied to the other part's molecules, none of which it had seen.

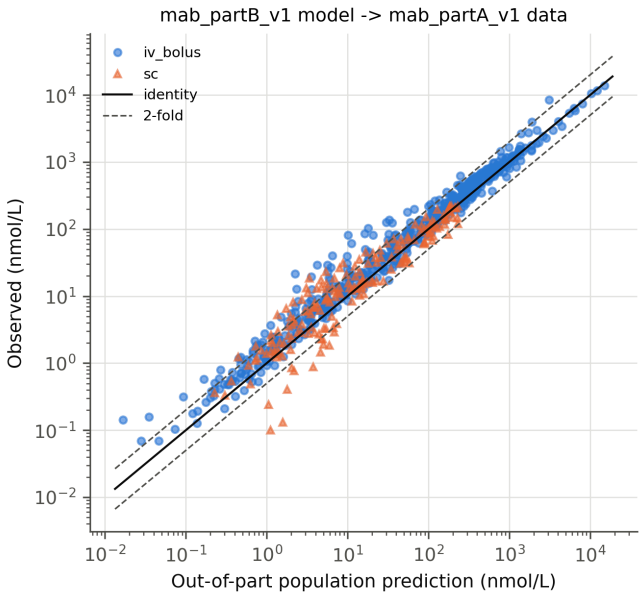
Out-of-part population prediction

model → data	observations	median fold error	bias (median log error)	within 2-fold	within 3-fold	90% PI coverage	IV coverage	SC coverage
Part A → Part B	679	1.294	-0.09838	76%	89%	75%	84%	46%
Part B → Part A	744	1.352	0.244	84%	93%	97%	98%	96%

Population predictions were typically within about 1.3-fold. Interval coverage from the Part A model was adequate for IV profiles (84%) but poor for SC profiles (46%), because Part A's between-profile variances are smaller than Part B's and absorption varies between molecules. The Part B model, with larger variances, gave wider intervals and high coverage.



out-of-part prediction: mab partA v1 on mab partB v1



out-of-part prediction: mab partB v1 on mab partA v1

4.5 Hypothesis evaluation of the adaptive planner (H4)

Adaptive planner vs fixed multistart on replicated SYNTHETIC data

setting	planner	replicates	successful-fit rate	attempts (mean)	failed attempts (mean)	attempts to best (mean)	median	parameter error	
default	adaptive rules	5	100%	4	0	3.2	0.2491	133 ± 32	0
default	fixed multistart	5	100%	4	0	3.2	0.2491	130 ± 32	0
stress	adaptive rules	5	100%	4	0.4	2.6	0.2457	191 ± 37	0
stress	fixed multistart	5	100%	4	0.4	2.2	0.2458	208 ± 54	0

Default setting: best objectives tied (within 0.1%) in 5/5 replicates; attempt counts tied in 5/5; paired runtime difference (rules − fixed) 3 ± 28 s. LLM-assisted arm: not_run: no Anthropic credentials in environment.

Stress setting: best objectives tied (within 0.1%) in 5/5 replicates; attempt counts tied in 5/5; paired runtime difference (rules − fixed) -18 ± 24 s. LLM-assisted arm: not requested.

In both settings, both planners found the same optimum within the first two rounds. The poor user start in the stress setting failed and was recorded as expected, but the Latin-hypercube starts shared by both arms recovered. The small runtime differences are within their variability and were measured on a heavily loaded machine.

5 Discussion

5.1 Evaluation of the hypothesis

sub-hypothesis	verdict
H1 accuracy	**Supported** on synthetic data: every identifiable typical value was within 9% of truth (tolerance 25%), with consistent bootstrap intervals. Variance components were recovered less precisely. This is a single simulated design, not a replicated simulation–estimation study.
H2 numerical reliability	**Supported.** The two engines agreed far inside the prespecified tolerance; invalid inputs were rejected with actionable messages; failed calibration starts, false-convergence bootstrap replicates, a failed covariance step (Part B) and boundary estimates were all surfaced as records and flags rather than hidden or repaired.
H3 reproducibility	**Supported.** Identical seeds reproduced calibration histories exactly; figures, tables and this report were regenerated from saved artifacts without refitting; every run carries hashes, seeds, environment versions and a checksummed manifest.
H4 effort / efficiency	**Not demonstrated.** The complete analysis (validation → report) ran with zero analyst interventions, but effort was not compared with a manual analyst workflow. On the synthetic problems tested, the adaptive planner was no better than a fixed multistart of the same budget, because the first round of starts already reached the optimum. A benefit of adaptive or LLM-assisted planning could only appear on harder, multimodal problems, and the LLM arm could not be run.

Overall, the evidence supports the **'preserving accuracy, reliability and reproducibility'** part of the hypothesis. The **'reducing analyst effort'** part is supported only qualitatively (full automation with an audit trail) and remains to be tested against a measured manual baseline.

5.2 Interpretation of the mAb analyses

The two independent parts gave similar linear disposition estimates: CL 0.19–0.27 L/day, central volume about 3.1–3.5 L and peripheral volume about 2.3–2.6 L. SC absorption was ka ≈ 0.21–0.30 1/day with bioavailability ≈ 0.71–0.84. These values lie in the ranges commonly reported for IgG monoclonal antibodies (clearance of the order of 0.2–0.5 L/day, central volume of about 3 L, absorption rate 0.1–0.5 1/day and SC bioavailability of roughly 50–80%; see, e.g., Dirks & Meibohm, *Clin Pharmacokinet* 2010). This agreement suggests the platform recovers physiologically sensible 'generic antibody' behaviour from pooled literature means.

The target-mediated part of the model was not identifiable in this setting, for three reasons:

- Each molecule binds a different target.
- Only drug concentrations were available.
- kon, koff and kdeg were fixed at generic values.

The fits therefore used R0 (with a very large between-profile variance) and kint to absorb molecule-specific non-linearity. As a result, kint was imprecise (Part A) or at its bound (Part B), and R0 η-shrinkage was high. These parameters should not be interpreted biologically. The larger between-profile variances in Part B, and the SC

under-coverage when Part A predicted Part B, point the same way: a single set of pooled typical values is a coarse description of a heterogeneous set of antibodies.

Because each ID is a published mean curve, the random effects capture differences between molecules and dose groups, not between patients. Standard errors ignore the number of patients behind each mean, so they are likely too small for population inference about patients.

5.3 Limitations

- Aggregate data: every ID in the mAb dataset is a published MEAN profile of a different molecule/dose/route. The nlmixr2 fits are pooled cross-molecule 'generic mAb' TMDD descriptions; random effects are between-profile (largely between-molecule) variability, and standard errors do not reflect patient numbers. They are not individual-level population PK analyses of any antibody (brief section 2).
- Declared assumptions for the mAb data that the source does not document: AMT in ug, DV in ug/mL, TIME in h; one molecular weight of 150 kDa for all antibodies; assay = free drug; IV doses treated as bolus (no infusion durations recorded); drug-naive baseline. These must be confirmed per molecule.
- Target parameters: kon (1 L/nmol/h), koff (0.1 1/h) and kdeg (0.1 1/h) are FIXED generic values; targets differ between molecules and no target data exist, so R0 and kint are only weakly identified. Part A: kint 4.0e-4 1/h with 95% CI spanning ~40-fold, R0 eta shrinkage 86%. Part B: kint converged to its lower bound (1e-4 1/h), the covariance step failed (critical flag) and R0 eta shrinkage was 66%. Target-mediated parameters should not be interpreted biologically.
- Recommended next model revision (not run; requires a scientist decision recorded in model.structure_decisions): fix kint (or reparameterize the target-mediated pathway, e.g. QSS/Michaelis-Menten) and/or estimate target parameters per molecule with molecule-level random effects, and add molecule-specific molecular weights.
- Four source rows carried both a dose and a time-0 DV equal to the profile peak; they were split into dose + observation records and interpreted as immediately post-dose (declared policy). Seven profiles contained small time inversions (digitization); the normalized copy is sorted by time and the profiles are flagged. Nothing was deleted.
- The VPC is dose-normalized and stratified by route but not prediction-corrected; with heterogeneous molecules pooled, 43% (Part B) of observed percentile points fall outside the simulated intervals, i.e. the pooled model does not reproduce the spread of the mean profiles well.
- Cross-part prediction tests transfer to unseen molecules (not to new patients). SC predictions transfer less well (Part A -> Part B SC 90% PI coverage 46%).
- Censored (BLQ) data are rejected, not modelled (no tested M3 likelihood). The synthetic design therefore omits samples observed below 0.05 nmol/L, which is a mild informative-omission bias documented in the truth file.
- Population-estimation validation is limited to one synthetic dataset fitted with FOCEi (recovery within 9% for all typical values; Q 95% CI excluded the truth) plus a small subject-level bootstrap; a replicated simulation-estimation study of bias and variability was not run because each FOCEi fit takes 10-60 min on this machine.
- Hypothesis evaluation: across 5 synthetic replicates (default setting) and 5 stress replicates, the adaptive rules planner and the fixed multistart baseline reached identical optima with identical attempt counts; no efficiency or accuracy advantage of the agentic planner is claimed. The LLM-assisted arm was not run (no credentials; anthropic package not installed). Analyst interventions were zero in both arms by construction (fully automated runs).
- Optional LLM planner guardrails were tested with mocked responses only; no live LLM call was made.

- Deviations from the brief: calibration uses the Python/SciPy engine only (no R calibration adapter); the PDF report is produced by printing the rendered HTML with headless Microsoft Edge (so it matches the HTML exactly) rather than through LaTeX; renv.lock was not written (renv not installed) — exact R versions are recorded in R/r-packages-lock.json; RcppParallel 6.2.1 was installed into the project library .Rlib to repair a stringfish/R 4.5.1 binary mismatch; the pre-dose convention is realized in rxode2 by shifting coincident doses by +1e-7 h (verified against the exact Python engine to <3.2e-7 relative).
- The Streamlit interface was smoke-tested (both pages render) but not exercised interactively end-to-end in a browser in this session.

5.4 Recommended next steps

1. Confirm the declared data assumptions (units, per-molecule molecular weights, assay format, infusion durations) and record them.
2. Revise the model (a scientist decision, versioned): fix or reparameterise the target pathway (e.g. quasi-steady-state or Michaelis–Menten elimination), or introduce molecule-level parameters or covariates (target type, dose unit mg/kg vs flat) instead of one pooled set.
3. Run a replicated simulation–estimation study (≥ 50 datasets) to quantify bias and precision of FOCEi under this design, and a larger bootstrap for the mAb fits.
4. Evaluate the planner on harder, multimodal problems and, with credentials, the LLM-assisted arm. Measure analyst time against a manual baseline to test the effort claim directly.
5. Add a tested censored-data (M3) likelihood so that below-quantification observations can be used.

6 Conclusions

Kratos implements the full workflow as an auditable pipeline: - data validation and unit normalisation; - a versioned TMDD model with matching Python and R engines; - adaptive bounded calibration; - nlmixr2 FOCEi estimation; - heuristic diagnostics and VPC; - a report built from saved artifacts only.

It recovered known parameters on synthetic data, and it made numerical problems in the real analysis visible rather than hiding them. Applied to the human mAb mean profiles split by molecule into two parts, it produced consistent, physiologically plausible linear PK and SC absorption estimates across the independent parts, and reasonable out-of-part predictions. It also showed that target-mediated parameters cannot be identified from pooled cross-molecule drug-only means. The hypothesis is supported for accuracy, reliability and reproducibility; the claimed efficiency gain of agentic planning was not demonstrated and requires further study.

7 Appendix: verification evidence

7.1 Cross-engine agreement (Python SciPy vs R rxode2)

Prespecified criterion: $|py - r| \leq a_{endpoint} + 10^{-4}|r|$ with $a = 10^{-6}$ nmol/L for every endpoint (declared before comparison). Cases span IV bolus, 1-h infusion, repeat dosing with pre-dose troughs, SC depot, several parameter sets and all five observation mappings.

case	status	n	n_pass	all_pass	max_rel_diff_nonnegligible	max_abs_diff
typical parameters, pre-dose	succeeded	325	325	True	3.176e-07	4.966e-05
typical parameters, post-dose	succeeded	325	325	True	3.918e-09	1.466e-07
high-capacity target, pre-dose	succeeded	325	325	True	1.184e-06	2.829e-05
synthetic demo dataset, 36 individual parameter sets	succeeded	387	387	True	2.940e-09	4.303e-07

7.2 Automated tests

`pytest` run on 2026-09-29T15:12:01: **54 passed, 0 failed, 0 skipped** (1148 s).

test	outcome
test_cross_engine::test_all_cases_agree	passed
test_cross_engine::test_all_endpoints_covered	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[missing_time-missing_time]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[negative_amt-invalid_dose_amount]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[unsupported_evid-unsupported_evid]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[steady_state-unsupported_ss]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[censored-unsupported_censoring]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[inconsistent_infusion-inconsistent_infusion]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[unknown_route-unsupported_route]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[nonpositive_dv-non_positive_dv]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[unsorted_time-unsorted_time]	passed

test	outcome
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[no_dose-no_dose_history]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[dv_on_dose-dv_on_dose]	passed
test_data_validation::test_invalid_inputs_rejected_with_actionable_codes[non_numeric-non_numeric]	passed
test_data_validation::test_censoring_is_unsupported_not_imputed	passed
test_data_validation::test_error_names_rows_and_fields	passed
test_data_validation::test_unsorted_can_be_flagged_and_sorted	passed
test_data_validation::test_combined_dose_observation_split_is_explicit	passed
test_data_validation::test_synthetic_normalization_units_and_provenance	passed
test_data_validation::test_mass_units_require_molecular_weight	passed
test_data_validation::test_schema_rejects_unknown_fields	passed
test_model_math::test_unit_factors	passed
test_model_math::test_linear_limit_bolus	passed
test_model_math::test_bolus_jump_pre_and_post	passed
test_model_math::test_infusion_stop	passed
test_model_math::test_repeat_doses_superposition_and_trough_conventions	passed
test_model_math::test_baseline_target_equilibrium	passed
test_model_math::test_sc_depot_bioavailability	passed
test_model_math::test_total_drug_mass_balance[iv]	passed
test_model_math::test_total_drug_mass_balance[sc]	passed
test_model_math::test_nonfinite_parameters_rejected	passed
test_model_math::test_endpoint_mappings	passed
test_model_spec_and_planner::test_spec_dimensions_and_derived_ksyn	passed
test_model_spec_and_planner::test_depot_params_without_sc_rejected	passed
test_model_spec_and_planner::test_sc_route_requires_template_1_1	passed
test_model_spec_and_planner::test_refuse_estimating_all_binding_parameters	passed
test_model_spec_and_planner::test_action_contract	passed
test_model_spec_and_planner::test_rules_planner_is_bounded_and_deterministic	passed
test_model_spec_and_planner::test_stopping_rules	passed
test_model_spec_and_planner::test_fixed_planner_is_non_adaptive	passed
test_model_spec_and_planner::test_llm_malformed_and_out_of_bounds_rejected_with_fallback	passed
test_model_spec_and_planner::test_llm_without_credentials_falls_back	passed
test_r_side_and_ui::test_r_unit_tests	passed

test	outcome
test_r_side_and_ui::test_r_bridge_allowlist	passed
test_r_side_and_ui::test_streamlit_app_renders	passed
test_workflow::test_missing_r_gives_calibration_only	passed
test_workflow::test_failed_step_visible_in_report	passed
test_workflow::test_audit_and_manifest	passed
test_workflow::test_secrets_not_exported	passed
test_workflow::test_regenerate_without_estimation	passed
test_workflow::test_seed_reproducibility	passed
test_workflow::test_cancellation_preserves_records	passed
test_workflow::test_resume_requires_compatibility	passed
test_workflow::test_existing_run_directory_not_overwritten	passed

8 Appendix: run record `synthetic_demo_v1` — SYNTHETIC

SYNTHETIC demonstration: data simulated from known TMDD parameters (not real subjects)

Analysis mode: population estimation (nlmixr2 FOCEi) + pooled calibration · **outcome:** completed

8.1 Stage status

stage	state	detail
validate	succeeded	
raw_plots	succeeded	
model	succeeded	
calibrate	succeeded	numerical_success: best objective reproduced by 4 independent starts (rel tol 0.001)
estimate	succeeded	
diagnose	succeeded	no heuristic flags raised — scientist review still required
report	succeeded	

8.2 Results narrative

Numerical findings (sourced from artifacts)

- 36 subjects with 387 observations and 48 dose records were analysed.
- Data validation: passed with 1 review flags.
- Pooled calibration ran 6 attempts (2 failed, 0 timed out); stop reason: numerical_success: best objective reproduced by 4 independent starts (rel tol 0.001). Best objective 187 (attempt 2).
- nlmixr2 FOCEi converged to OFV 2054.9588 using calibration start rank 1 (1 estimation attempt(s)); covariance step succeeded.
- Diagnostics: 0 critical and 0 warning heuristic flags; no heuristic flags raised — scientist review still required.
- VPC (200 replicates): 4% of observed percentile points outside simulated 95% intervals.
- Typical values: CL = 0.01008 L/h; Vc = 3.045 L; Q = 0.03259 L/h; Vp = 2.611 L; kint = 0.05254 1/h; R0 = 1.881 nmol/L

Interpretation

- All data are SYNTHETIC; parameter recovery can be compared with the known generating values.
- The pooled calibration provides starting values only; its objective is not comparable with the nlmixr2 OFV.
- Heuristic checks are review aids; Kratos does not declare the model scientifically valid.

8.3 Data provenance and validation

item	value
source data	C:/Users/devamdesai/Downloads/Kratos/data/synthetic/synthetic_tmdd.csv
data sha256	f7f04979c042eb9d…
configuration	synthetic_demo.yaml
config sha256	46c05ad66b3ca648…
Kratos code sha256	5e5f89f17e09a5c2…
seeds	{"global": 20260929, "calibration": 7, "estimation": 11, "vpc": 13}

item	value
n_subjects	36
n_records	435
n_observations	387
n_dose_records	48
n_doses_expanded_from_addl	12
routes	{"iv_bolus": 36, "iv_infusion": 12}
subjects_by_route	{"iv_bolus": 24, "iv_infusion": 12}
doses_per_subject	{"min": 1, "max": 3}
first_dose_nmol	{"min": 46.66666666666667, "max": 4666.666666666667, "n_levels": 5}
obs_per_subject	{"min": 5, "median": 11.0, "max": 17}
time_range_h	0, 1512
dv_range_nmol_per_L	0.05802, 1778
endpoints	{"free_drug": 387}
aggregate_data	False
synthetic	True

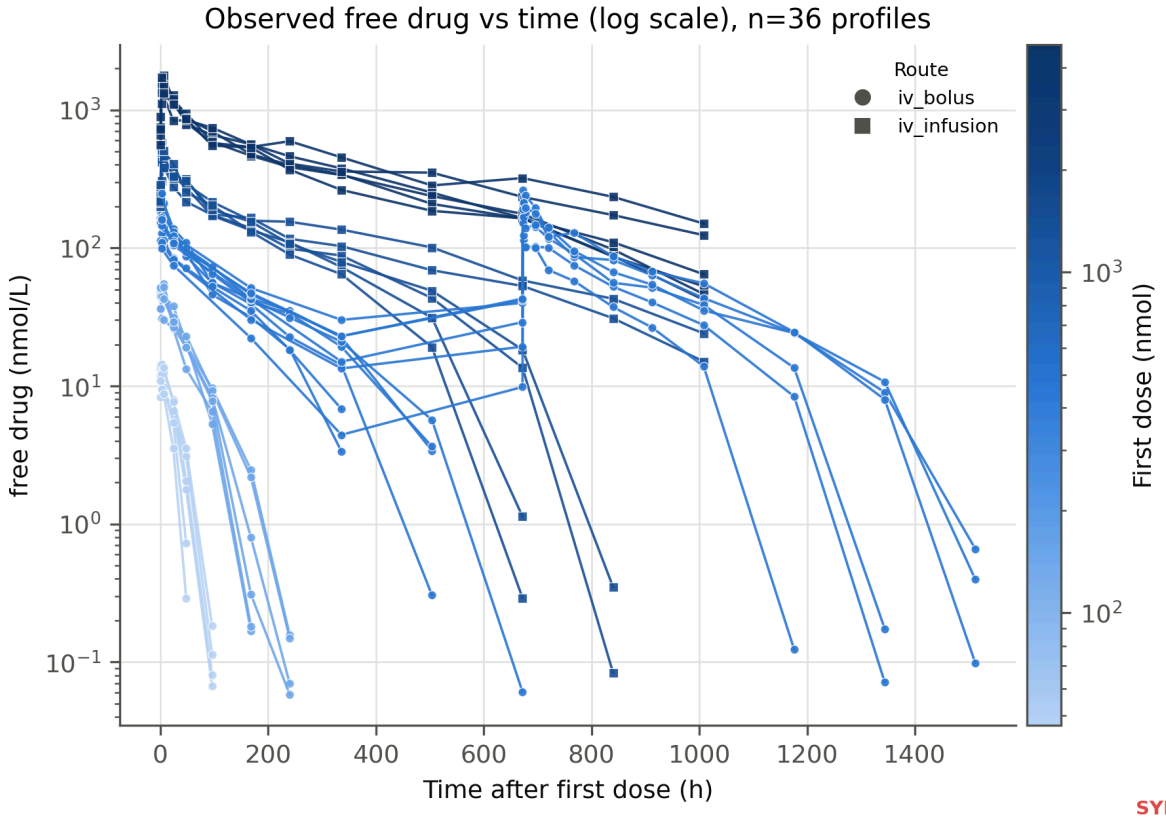
Unit conversions applied

quantity	from	to	factor	basis
TIME	h	h	1	definition
AMT	mg	nmol	6.667	drug molecular weight 150000.0 g/mol
DV[default:free_drug]	ug/mL	nmol/L	6.667	drug molecular weight 150000.0 g/mol
RATE	mg/h	nmol/h	6.667	amount factor divided by time factor

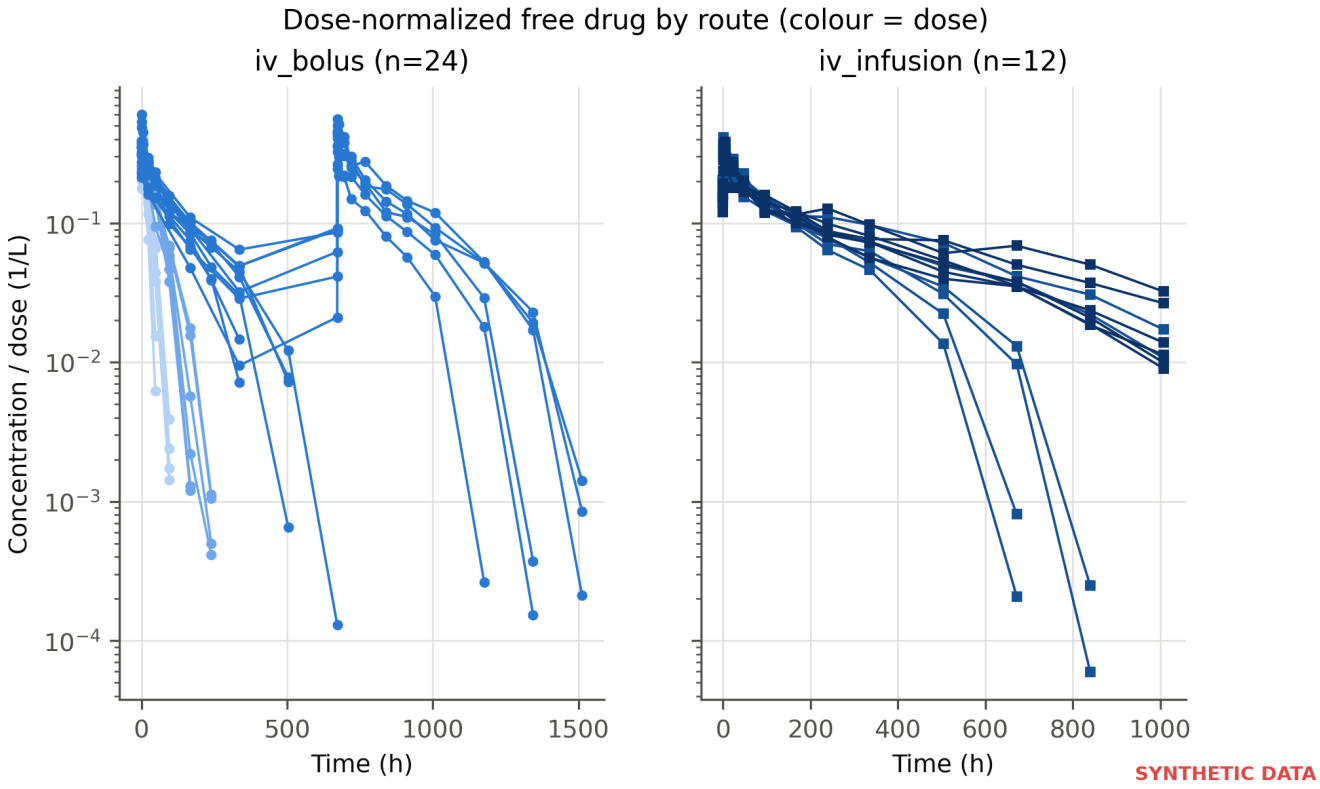
Validation: passed — flags raised for scientist review (nothing deleted):

level	code	message	where
flag	coincident_dose_observation	12 observations share a time with a dose; convention 'pre_dose' applied identically in Python and R	

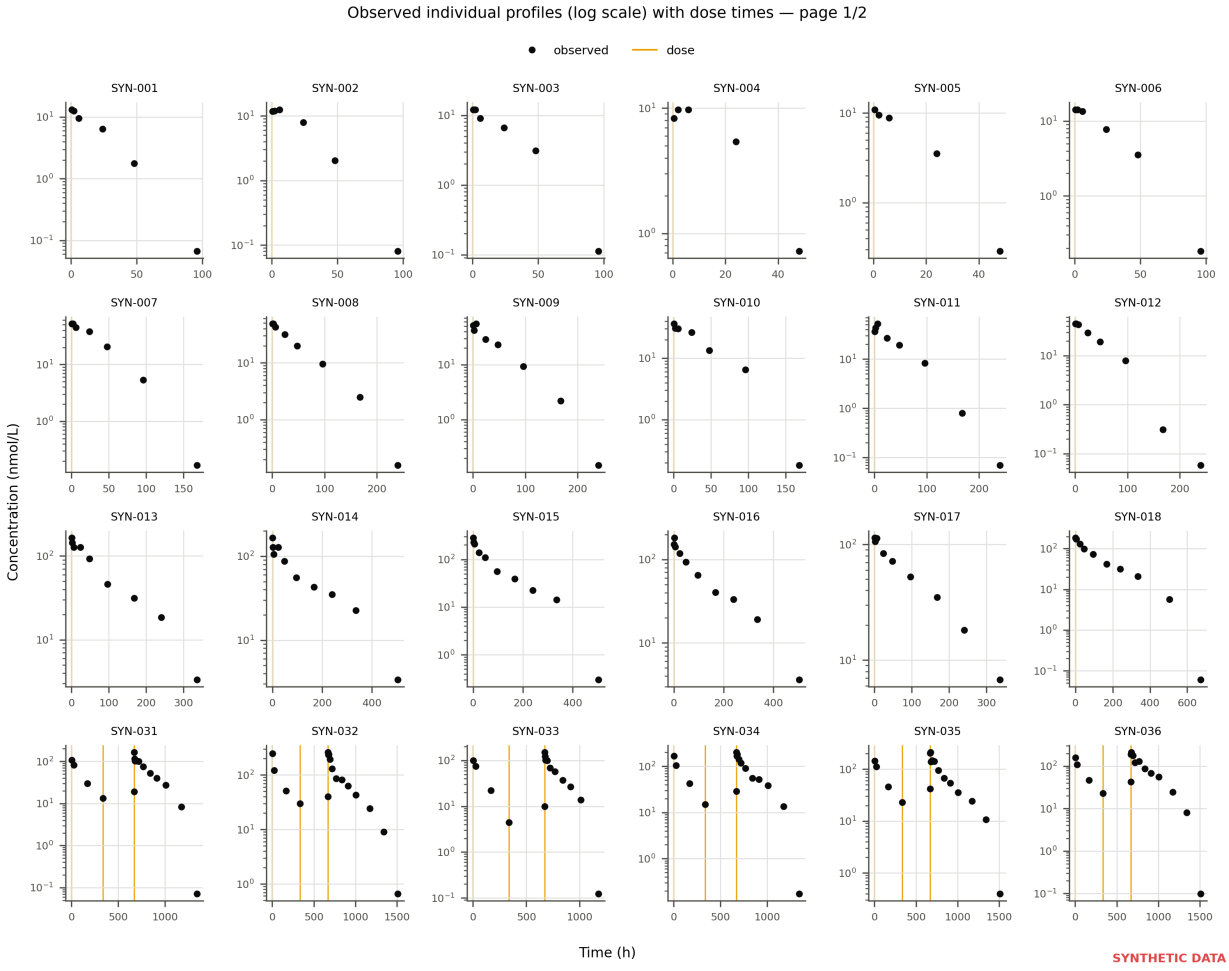
8.3.1 Raw data graphics



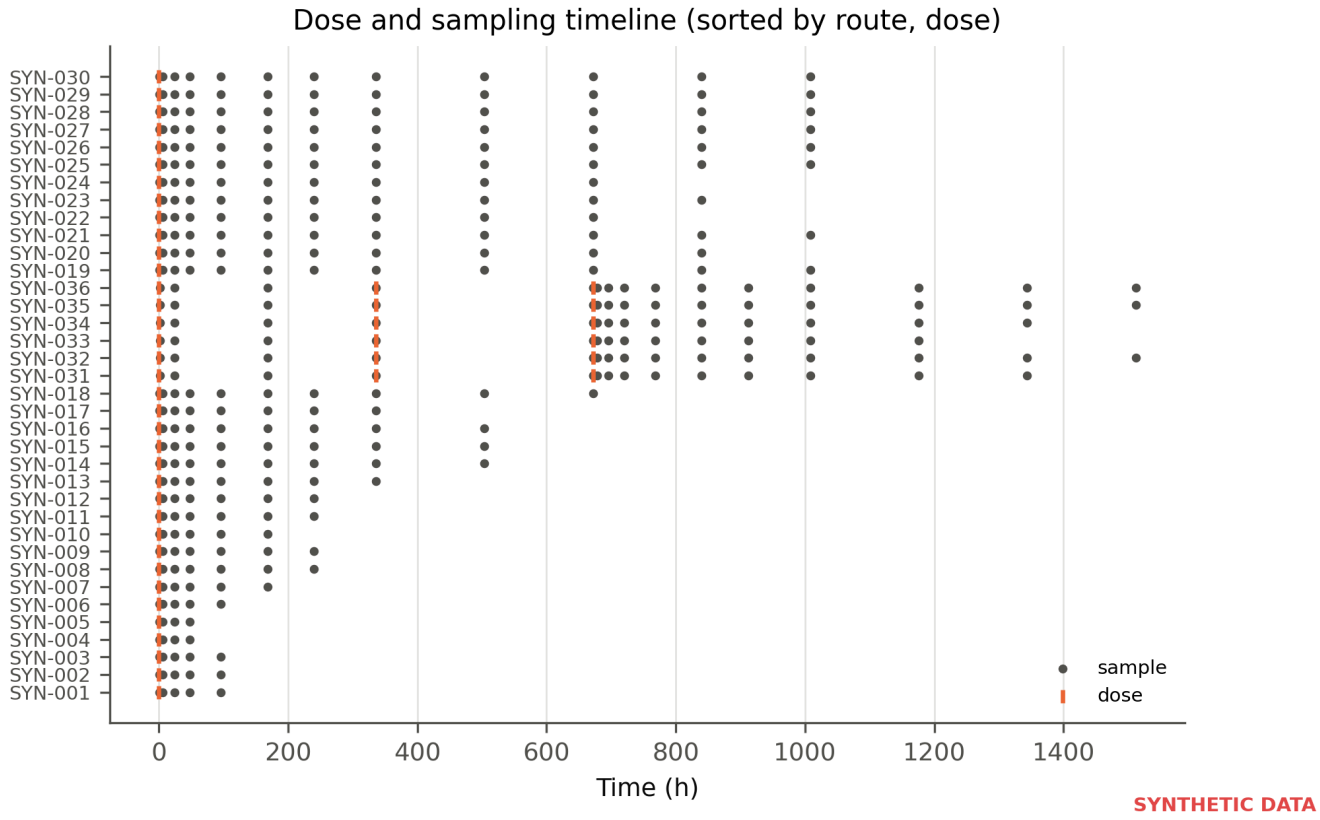
raw free drug log



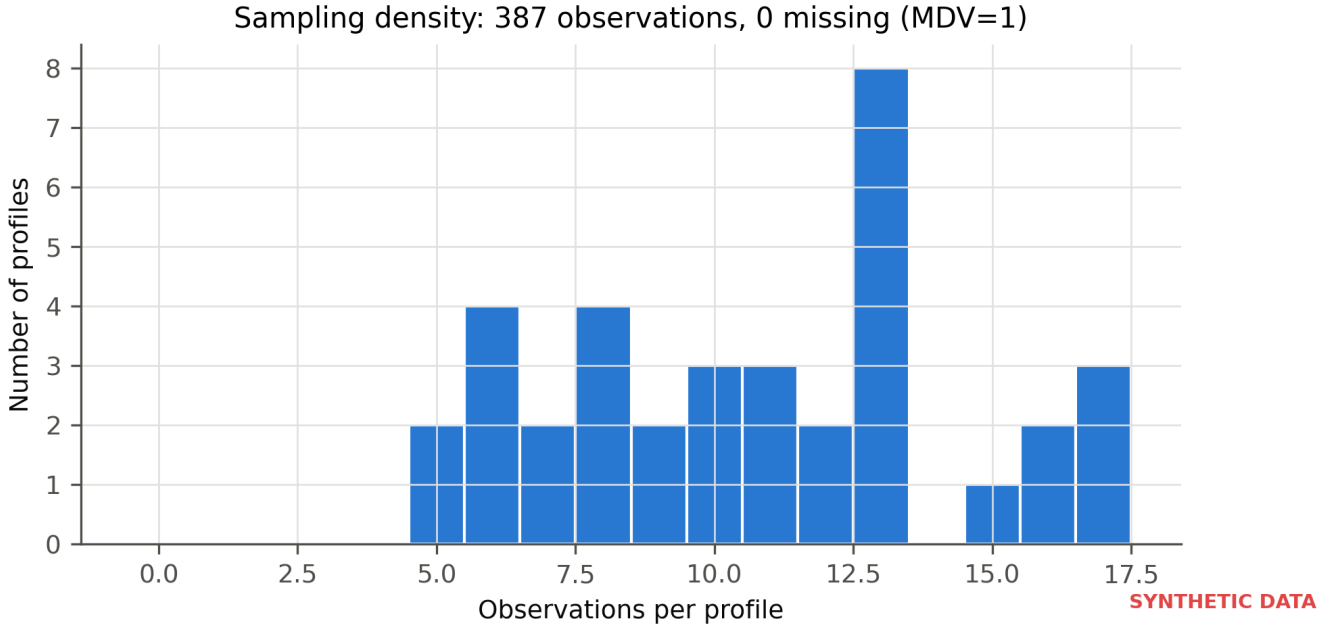
raw free drug dose normalized



raw individual p01



raw timeline



raw missingness

8.4 Model configuration

parameter	unit	status	scale	initial	lower	upper	IIV ω² init	justification
CL	L/h	estimated	log	0.02	1.000e-04	1	0.1	
Vc	L	estimated	log	4	0.5	20	0.1	

parameter	unit	status	scale	initial	lower	upper	IIV ω² init	justification
Q	L/h	estimated	log	0.05	1.000e-04	1	–	
Vp	L	estimated	log	3	0.5	30	–	
kon	L/(nmol*h)	FIXED	log	1	0.01	100	–	fixed binding constant (synthetic truth); not identifiable from drug-only data
koff	1/h	FIXED	log	0.1	0.001	10	–	fixed binding constant (synthetic truth)
kint	1/h	estimated	log	0.1	0.001	5	–	
kdeg	1/h	FIXED	log	0.1	0.001	5	–	fixed target turnover (synthetic truth)
R0	nmol/L	estimated	log	1	0.001	100	0.1	

Residual model: `{'model': 'lognormal', 'sd_init': 0.2, 'units': 'log(nmol/L)'}` · endpoints: `{'default': 'free_drug'}` · SC depot active: False · coincident dose/observation convention: **pre_dose** · template tmdd2c_full v1.1.0 (sha256 18d11fe5facf…).

8.5 Calibration (pooled; starting values only)

Objective **lognormal_sse**: SSE = sum_i (log y_i - log f_i)^2 ; -2LL(profiled sigma) = n*log(SSE/n)* + *n*(1+log 2pi) + 2*sum log y_i. Weighting: equal weights on the log scale (equivalent to a lognormal/exponential residual). Optimizer least_squares (bounded, transformed space, max 300 evaluations/attempt); planner **rules** (seed 7); budget 10 attempts / 1800 s; stopping: plateau over 4 successful attempts at rel. tol 0.001, or best objective reproduced by 3 starts.

item	value
status	succeeded
stop_reason	numerical_success: best objective reproduced by 4 independent starts (rel tol 0.001)
n_attempts	6
n_succeeded	4
n_failed	2
n_timeout	0
elapsed_s	112.5
n_distinct_optima	1

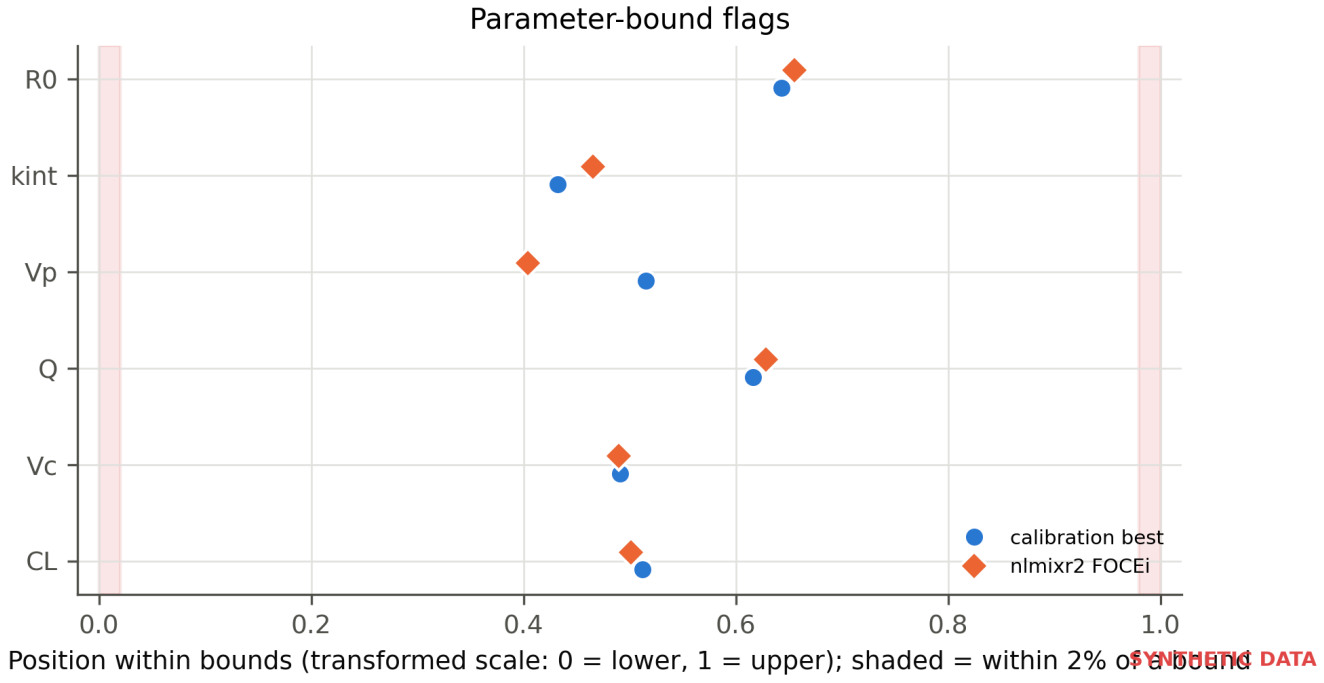
Best candidate (attempt 2, objective 187, near bounds: none):

CL	Vc	Q	Vp	kint	R0
0.01118	3.056	0.0291	4.127	0.03968	1.645

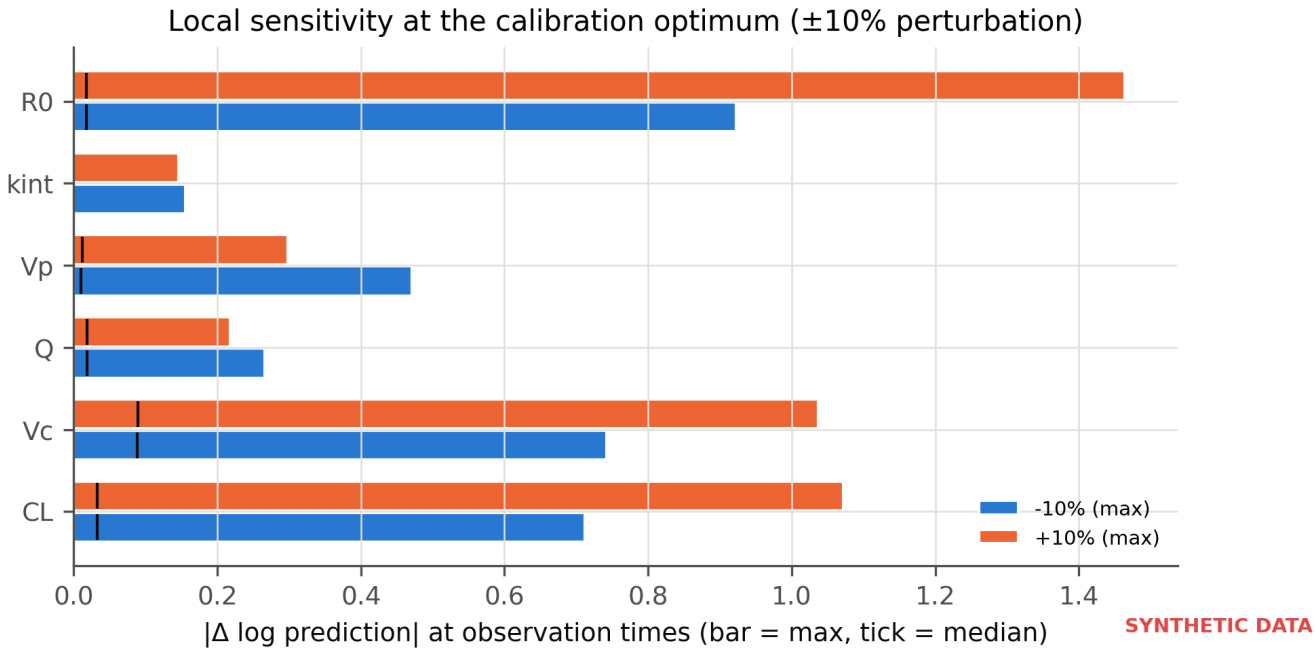
Practical identifiability at the pooled optimum (Gauss-Newton approximation J'J at pooled optimum (transformed scale)): condition number 346.6; concerns: none flagged.



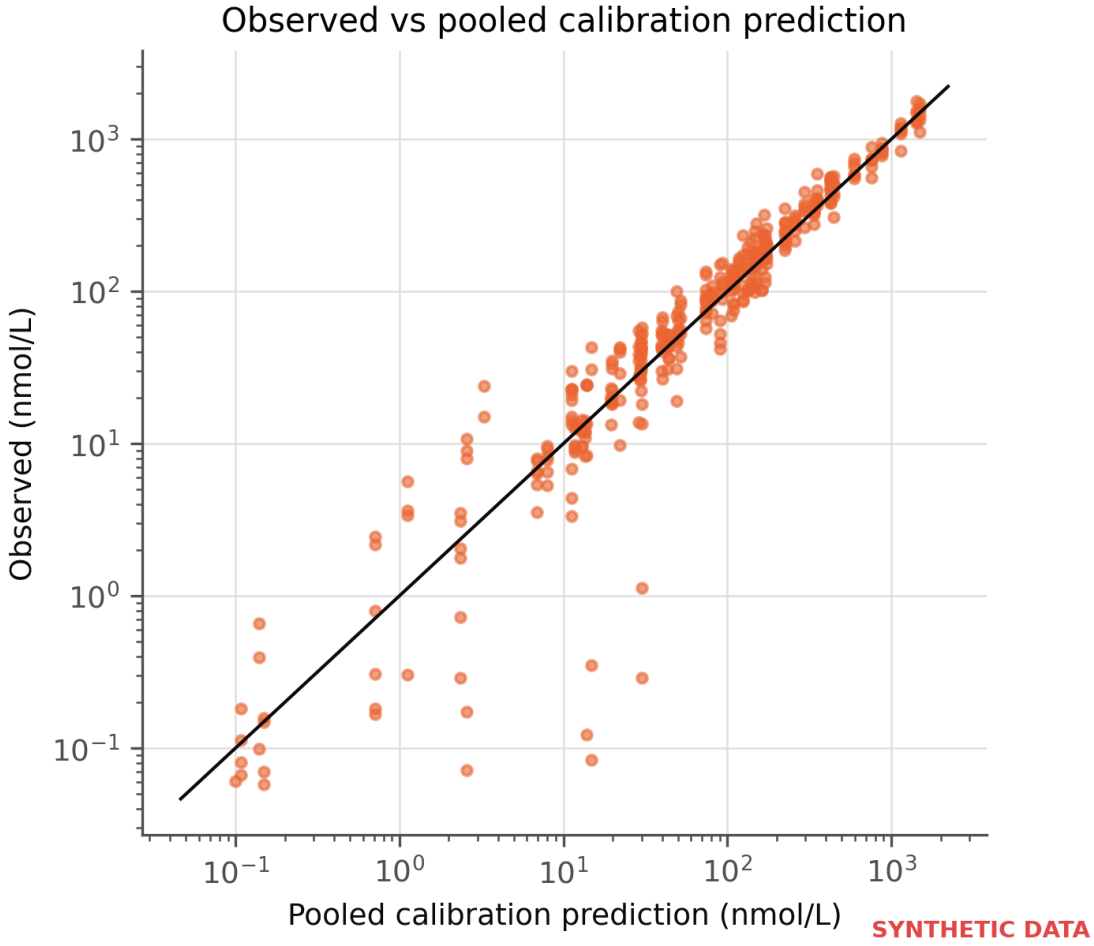
calibration history



parameter bounds



sensitivity



calibration gof

8.6 Population estimation (nlmixr2 FOCEi)

item	value
method	nlmixr2 FOCEi (est='focei')
OFV (nlmixr2)	2055
AIC	2786
BIC	2826
n observations	387
n profiles/subjects	36
optimizer message	relative convergence (4)
covariance method	r,s
covariance success	True
estimation attempts	1
selected attempt	1
runtime (s)	2494
CI method	Wald: back-transformed estimate(log/logit) +/- 1.96 SE on the estimation scale (nlmixr2 parFixedDf)

Parameter estimates (back-transformed by Kratos from the estimation scale; FIXED parameters were not estimated; RSE = delta-method relative standard error on the parameter scale, i.e. 100×SE of log θ, or 100×SE×(1−F) for logit F; BSV = between-profile variability as CV%):

parameter	unit	type	estimate	ci95_lower	ci95_upper	rse_pct	bsv_cv_pct	shrinkage_sd_pct
CL	L/h	estimated	0.01008	0.009186	0.01106	4.735	18.46	27.45
Vc	L	estimated	3.045	2.864	3.237	3.125	18.32	3.367
Q	L/h	estimated	0.03259	0.03071	0.03459	3.029	–	–
Vp	L	estimated	2.611	2.455	2.778	3.15	–	–
kon	L/(nmol*h)	fixed	1	–	–	–	–	–
koff	1/h	fixed	0.1	–	–	–	–	–
kint	1/h	estimated	0.05254	0.0491	0.05622	3.456	–	–
kdeg	1/h	fixed	0.1	–	–	–	–	–
R0	nmol/L	estimated	1.881	1.704	2.075	5.023	27.51	8.351
lnorm.sd	log-scale SD	residual	0.09864	–	–	–	–	–

The calibration objective and the nlmixr2 OFV are different quantities and are not compared.

8.7 Diagnostics and review flags

Review status (heuristic): no heuristic flags raised — scientist review still required. Convergence: {'converged': True, 'message': 'relative convergence (4)'}. Covariance: {'success': True, 'method': 'r,s'}.

Residual type: CWRES; trend statistics: {"CWRES_vs_TIME_spearman": 0.047, "CWRES_vs_PRED_spearman": -0.14, "CWRES_mean": -0.017, "CWRES_sd": 0.971, "frac_abs_CWRES_gt_2": 0.039}.

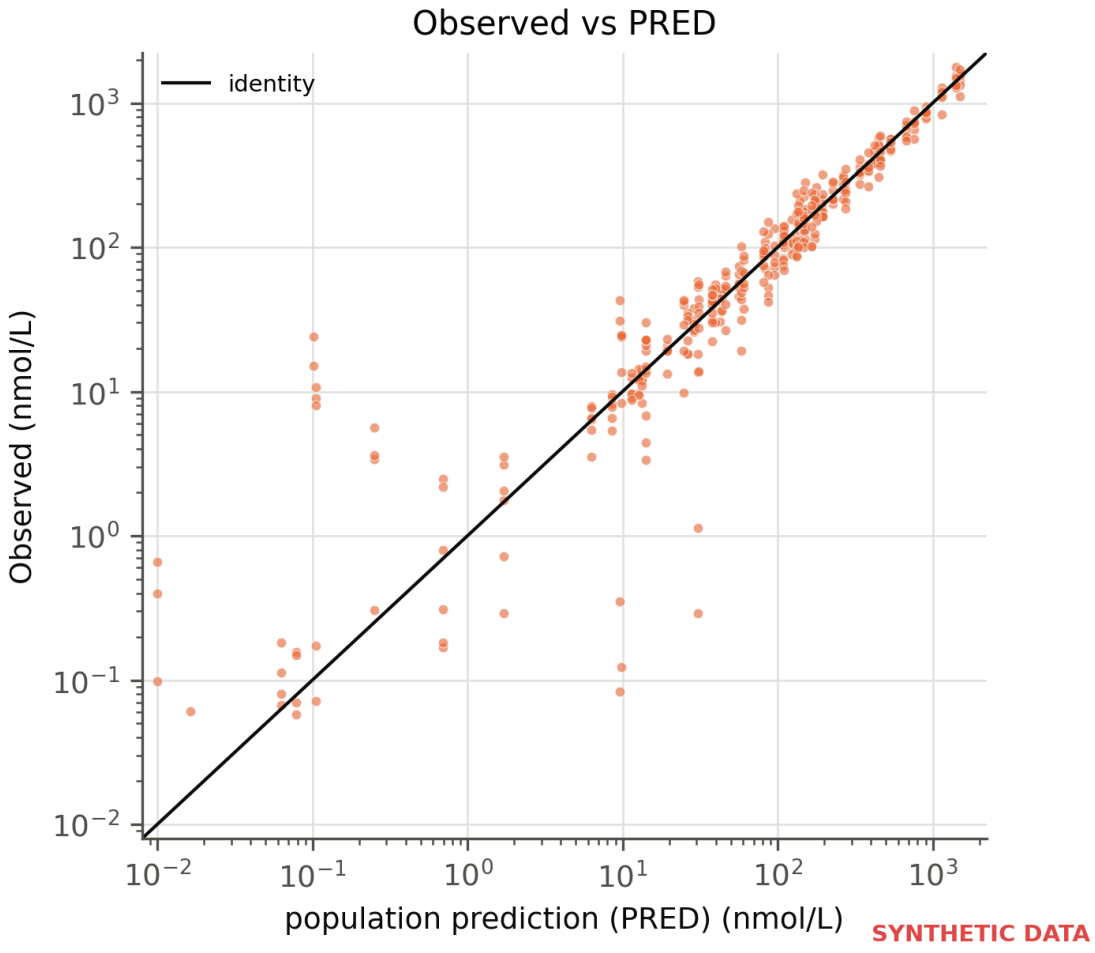
eta	omega	sd_eta	shrinkage_sd_pct
eta.CL	0.0335	0.1328	27.45
eta.Vc	0.03303	0.1756	3.367
eta.Rbase	0.07295	0.2475	8.351

VPC: succeeded — 200 replicates, binning: observed-time quantiles within stratum, strata: DOSEGRP, dose-normalized: False, prediction-corrected: False; fraction of observed percentile points outside simulated 95% intervals: 0.0381.

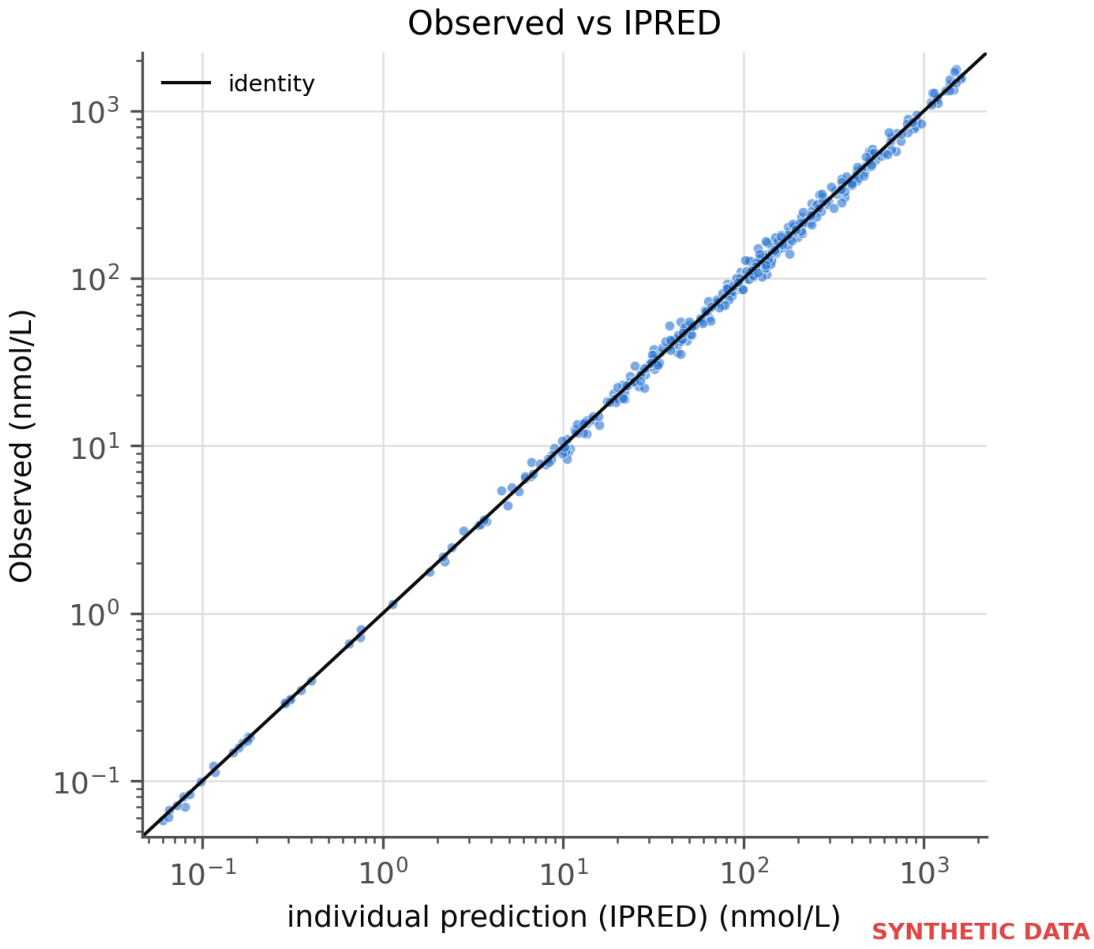
Review flags (thresholds are configurable heuristics: {"rse_max_pct": 50.0, "shrinkage_max_pct": 30.0, "corr_max_abs": 0.95, "bound_rel_tol": 0.02, "cwres_trend_max_abs_r": 0.3, "vpc_outside_max_frac": 0.2})

severity	category	message
info	data:coincident_dose_observation	12 observations share a time with a dose; convention 'pre_dose' applied identically in Python and R

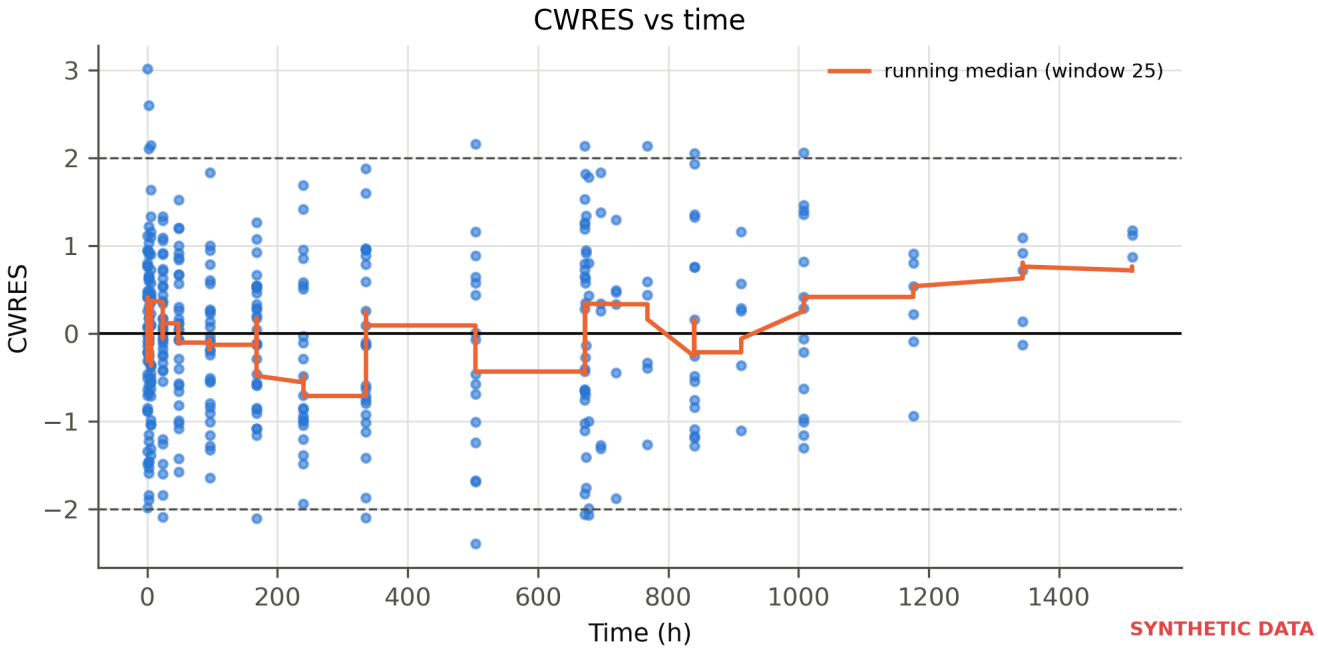
8.7.1 Model diagnostics graphics



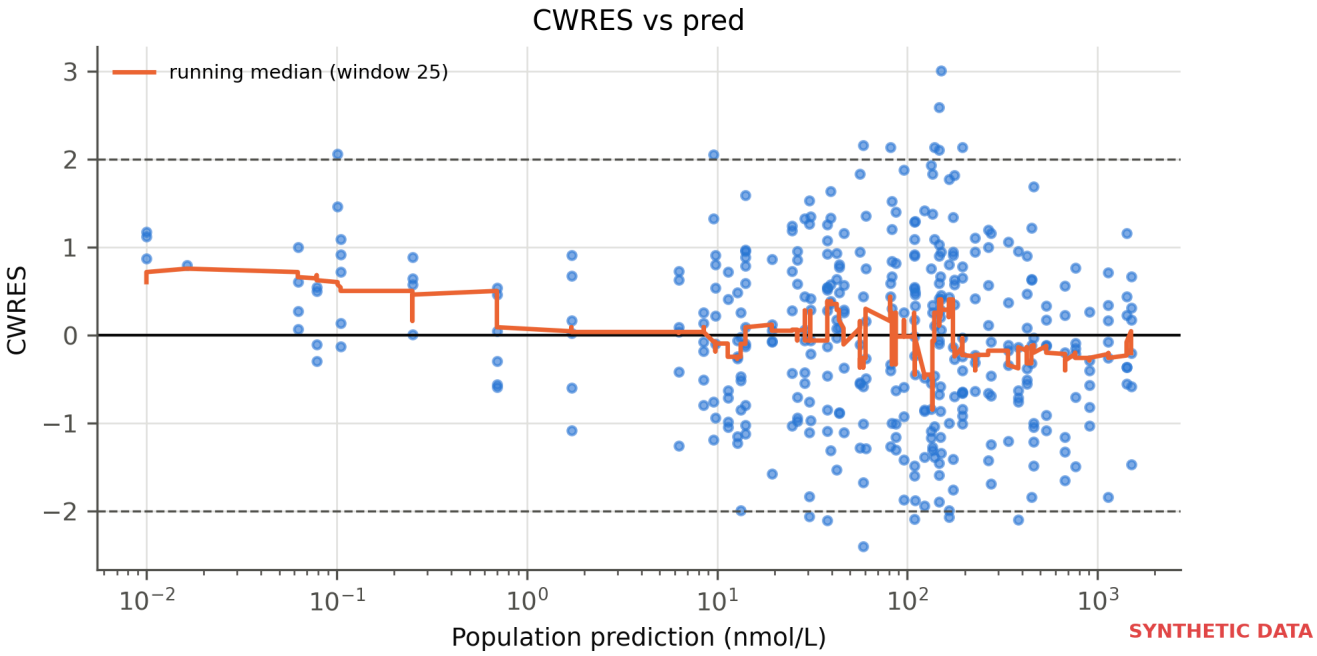
gof dv vs pred



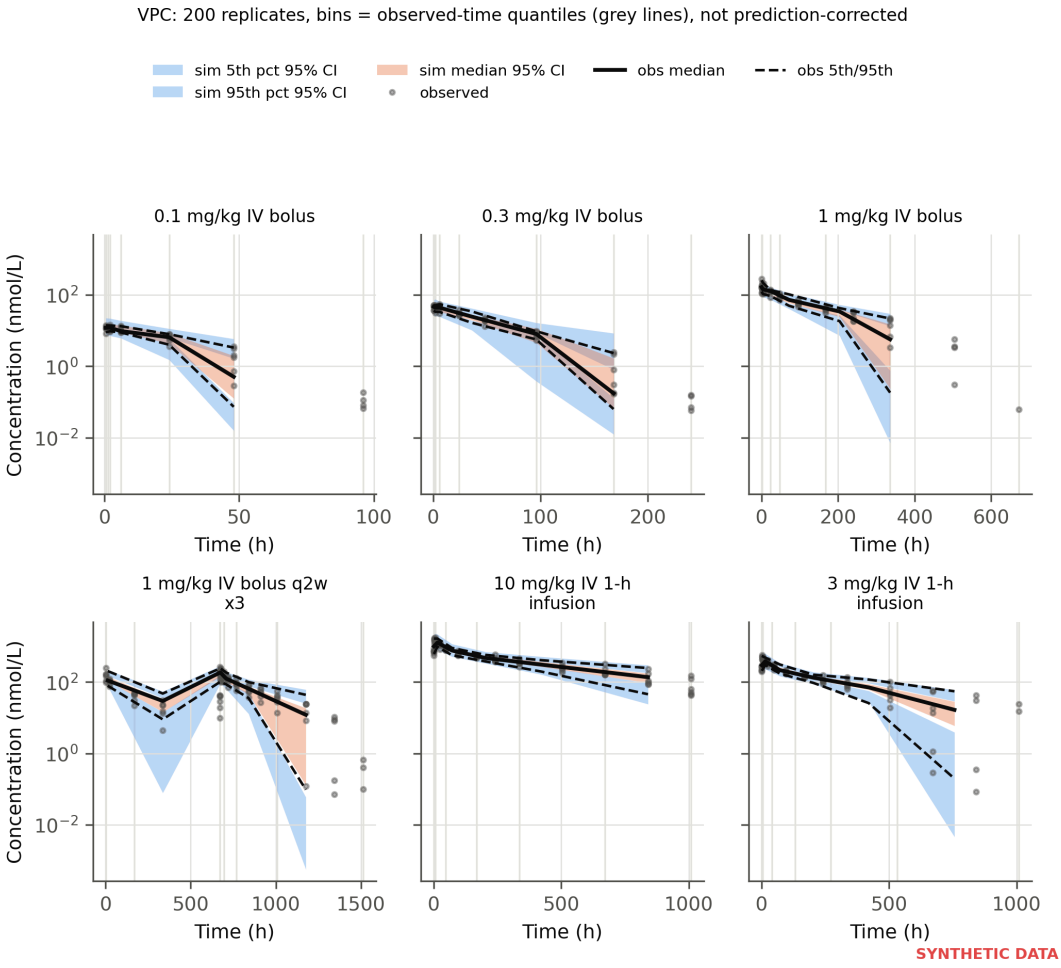
gof dv vs ipred



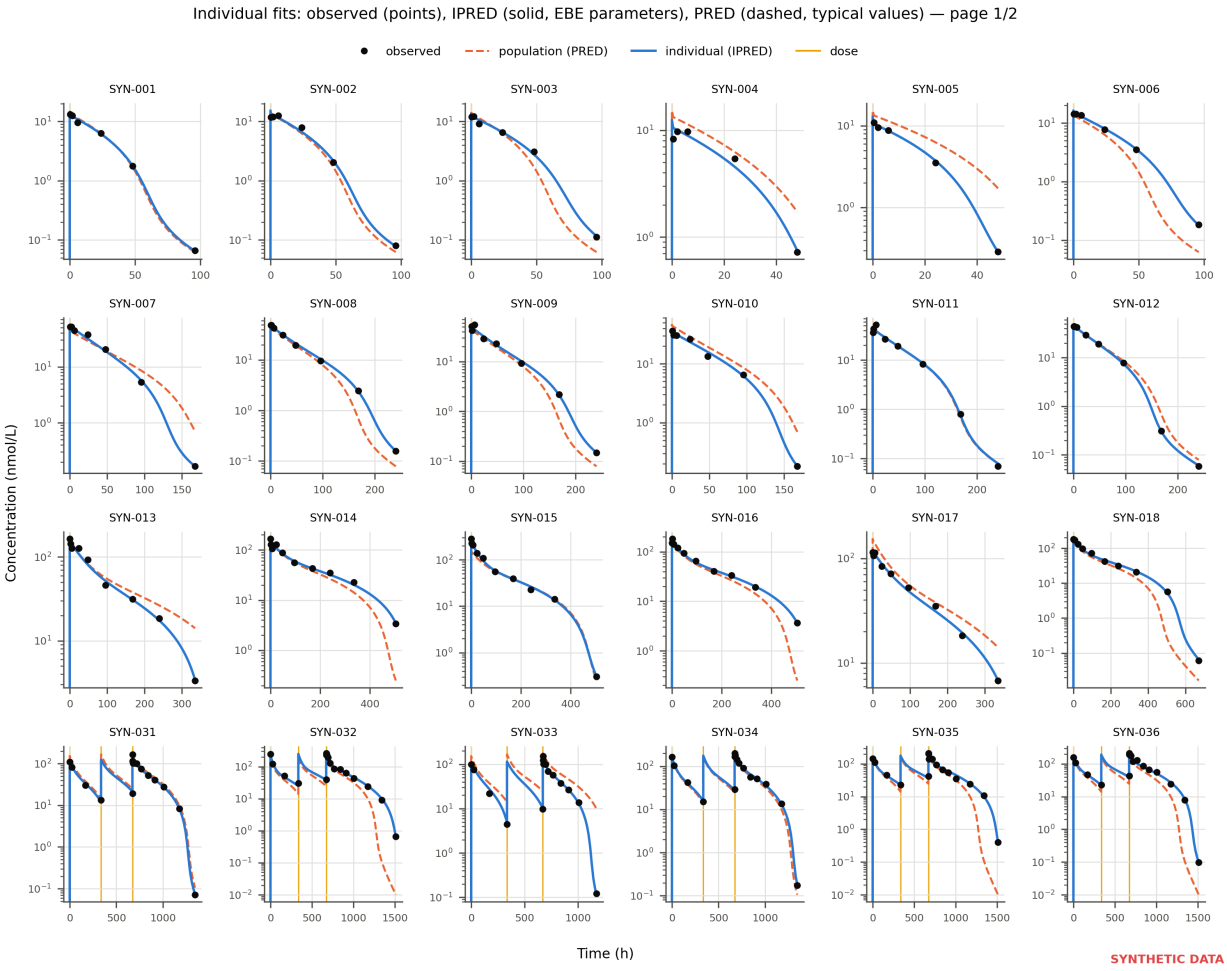
cwres vs time



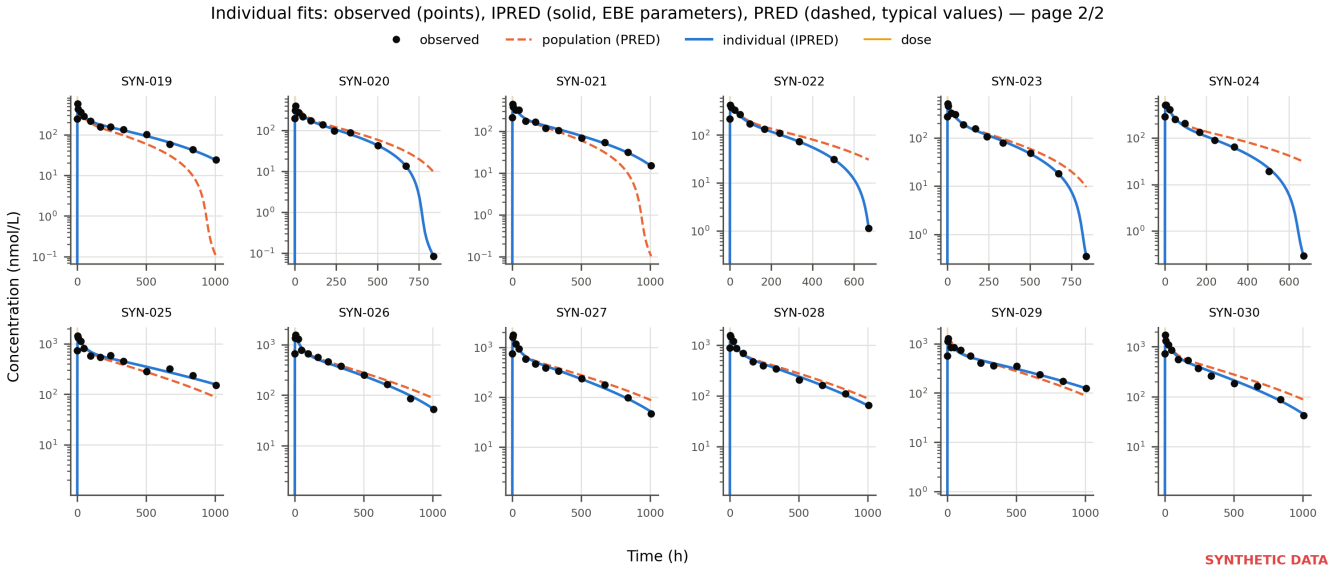
cwres vs pred



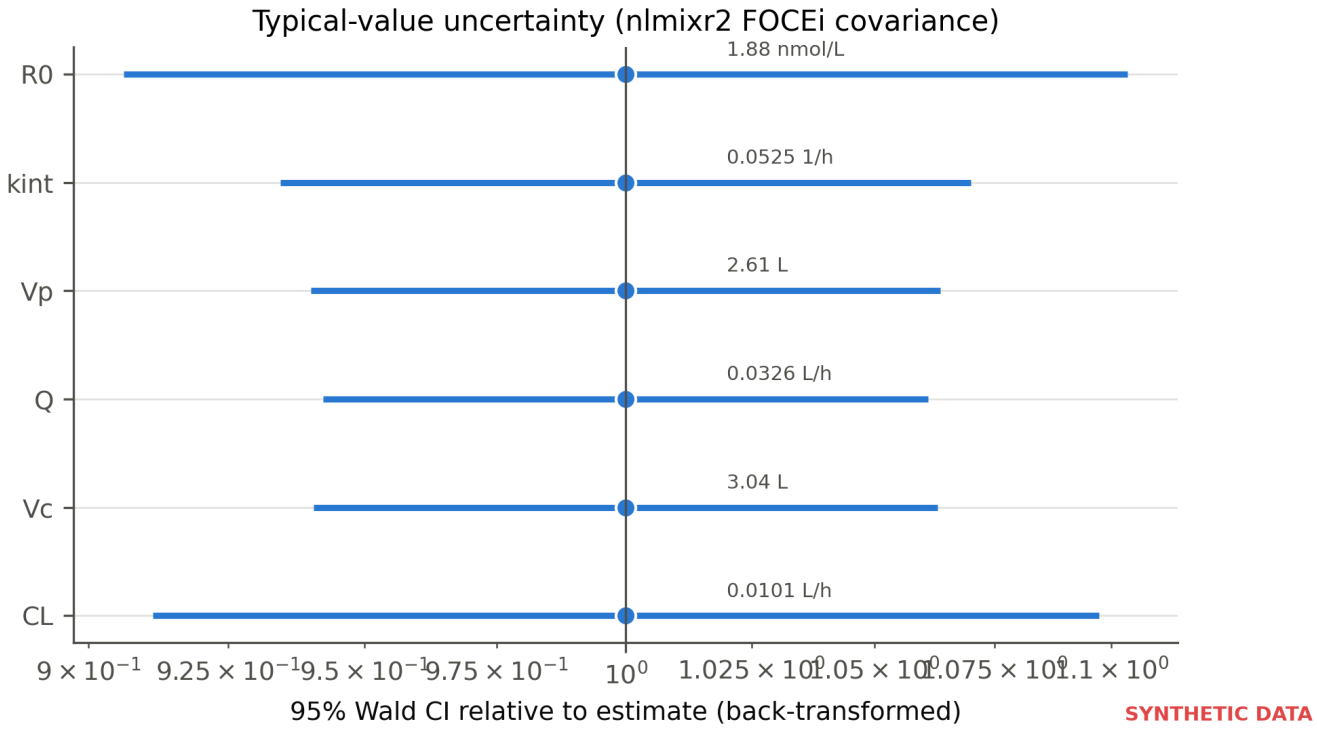
vpc



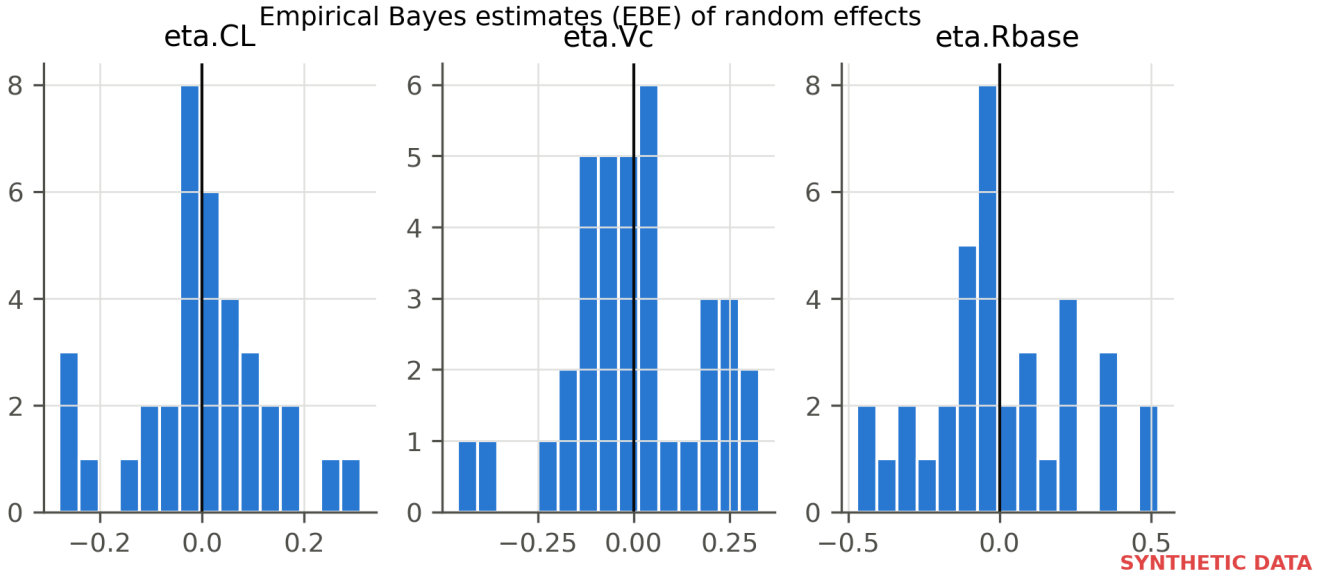
fit individual p01



fit individual p02



parameter uncertainty



eta distributions

8.8 Bootstrap uncertainty

subject-level resampling with replacement; dose histories preserved; duplicates relabelled; FOCEi refit from final estimates without covariance step; percentile 95% intervals. Requested 8, succeeded 5 (success rate 62%); failed replicates: 3; missing: [].

parameter	estimate	boot_median	boot_p2.5	boot_p97.5	wald_ci_lo	wald_ci_hi
CL	0.01008	0.01035	0.009506	0.0105	0.009186	0.01106

parameter	estimate	boot_median	boot_p2.5	boot_p97.5	wald_ci_lo	wald_ci_hi
Vc	3.045	3.066	2.899	3.235	2.864	3.237
Q	0.03259	0.03232	0.03122	0.03369	0.03071	0.03459
Vp	2.611	2.664	2.569	2.685	2.455	2.778
kint	0.05254	0.05352	0.05093	0.05592	0.0491	0.05622
R0	1.881	1.855	1.738	2.047	1.704	2.075

8.9 Parameter recovery (SYNTHETIC truth known)

Prespecified tolerance for identifiable typical values: |relative error| ≤ 25%.

parameter	true	estimate	ci95_lower	ci95_upper	rel_error_%	true_in_CI
CL	0.01	0.01008	0.009186	0.01106	0.7935	True
Vc	3	3.045	2.864	3.237	1.489	True
Q	0.03	0.03259	0.03071	0.03459	8.639	False
Vp	2.5	2.611	2.455	2.778	4.46	True
kint	0.05	0.05254	0.0491	0.05622	5.075	True
R0	2	1.881	1.704	2.075	-5.969	True

Random-effect variances (true vs estimated ω^2): eta.CL: 0.0335 (true 0.09); eta.Vc: 0.03303 (true 0.04); eta.Rbase: 0.07295 (true 0.09); residual SD true 0.1.

8.10 Run environment

item	value
Python	3.12.0
platform	Windows-10-10.0.19045-SP0
Python packages	{"numpy": "1.26.4", "scipy": "1.14.1", "pandas": "2.2.2", "matplotlib": "3.9.1", "PyYAML": "6.0.2", "jsonschema": "4.23.0", "streamlit": "1.44.1", "anthropic": null}
R	R version 4.5.1 (2025-06-13 ucrt)
R packages	{"rxode2": "5.1.6", "nlmixr2": "4.0.1", "nlmixr2est": "4.1.0", "jsonlite": "2.0.0", "ggplot2": "4.0.3", "data.table": "1.17.8"}

9 Appendix: run record `mab_partA_v1`

Human monoclonal antibody mean PK profiles (Part A of a molecule-grouped two-part ID split). Two-compartment full TMDD template with SC depot. Random effects describe between-profile (between-molecule/dose-group) variability of mean curves, not between-subject variability.

Analysis mode: population estimation (nlmixr2 FOCEi) + pooled calibration · **outcome:** completed

9.1 Stage status

stage	state	detail
validate	succeeded	
raw_plots	succeeded	
model	succeeded	
calibrate	succeeded	numerical_success: best objective reproduced by 8 independent starts (rel tol 0.001)
estimate	succeeded	
diagnose	succeeded	numerical checks passed with warnings — scientist review required
report	succeeded	

9.2 Results narrative

Numerical findings (sourced from artifacts)

- 72 mean profiles with 744 observations and 72 dose records were analysed.
- Data validation: passed with 5 review flags.
- Pooled calibration ran 8 attempts (0 failed, 0 timed out); stop reason: numerical_success: best objective reproduced by 8 independent starts (rel tol 0.001). Best objective 82.53 (attempt 3).
- nlmixr2 FOCEi converged to OFV 4005.4617 using calibration start rank 1 (1 estimation attempt(s)); covariance step succeeded.
- Diagnostics: 0 critical and 4 warning heuristic flags; numerical checks passed with warnings — scientist review required.
- VPC (200 replicates): 21% of observed percentile points outside simulated 95% intervals.
- Typical values: CL = 0.007832 L/h; Vc = 3.077 L; Q = 0.02144 L/h; Vp = 2.634 L; kint = 0.000396 1/h; R0 = 0.005798 nmol/L; ka = 0.008692 1/h; F = 0.7054 fraction

Interpretation

- Each ID is a published mean profile: random-effect variances describe between-profile (molecule/dose-group) variability, not between-patient variability, and standard errors do not reflect within-study patient numbers.
- The pooled calibration provides starting values only; its objective is not comparable with the nlmixr2 OFV.
- Heuristic checks are review aids; Kratos does not declare the model scientifically valid.

9.3 Data provenance and validation

item	value
source data	C:/Users/devamdesai/Downloads/Kratos/data/mab/mab_partA.csv
data sha256	3d4f7f72106a0952…
configuration	mab_partA.yaml
config sha256	5b1ae1c79efea32d…
Kratos code sha256	5e5f89f17e09a5c2…
seeds	{"global": 20260929, "calibration": 101, "estimation": 202, "vpc": 303}

item	value
n_subjects	72
n_records	816
n_observations	744
n_dose_records	72
n_doses_expanded_from_addl	0
routes	{"iv_bolus": 59, "sc": 13}
subjects_by_route	{"iv_bolus": 59, "sc": 13}
doses_per_subject	{"min": 1, "max": 1}
first_dose_nmol	{"min": 4.666666666666666, "max": 55999.99999999999, "n_levels": 38}
obs_per_subject	{"min": 4, "median": 10.5, "max": 25}
time_range_h	0, 4374
dv_range_nmol_per_L	0.06878, 1.391e+04
endpoints	{"free_drug": 744}
aggregate_data	True
synthetic	False

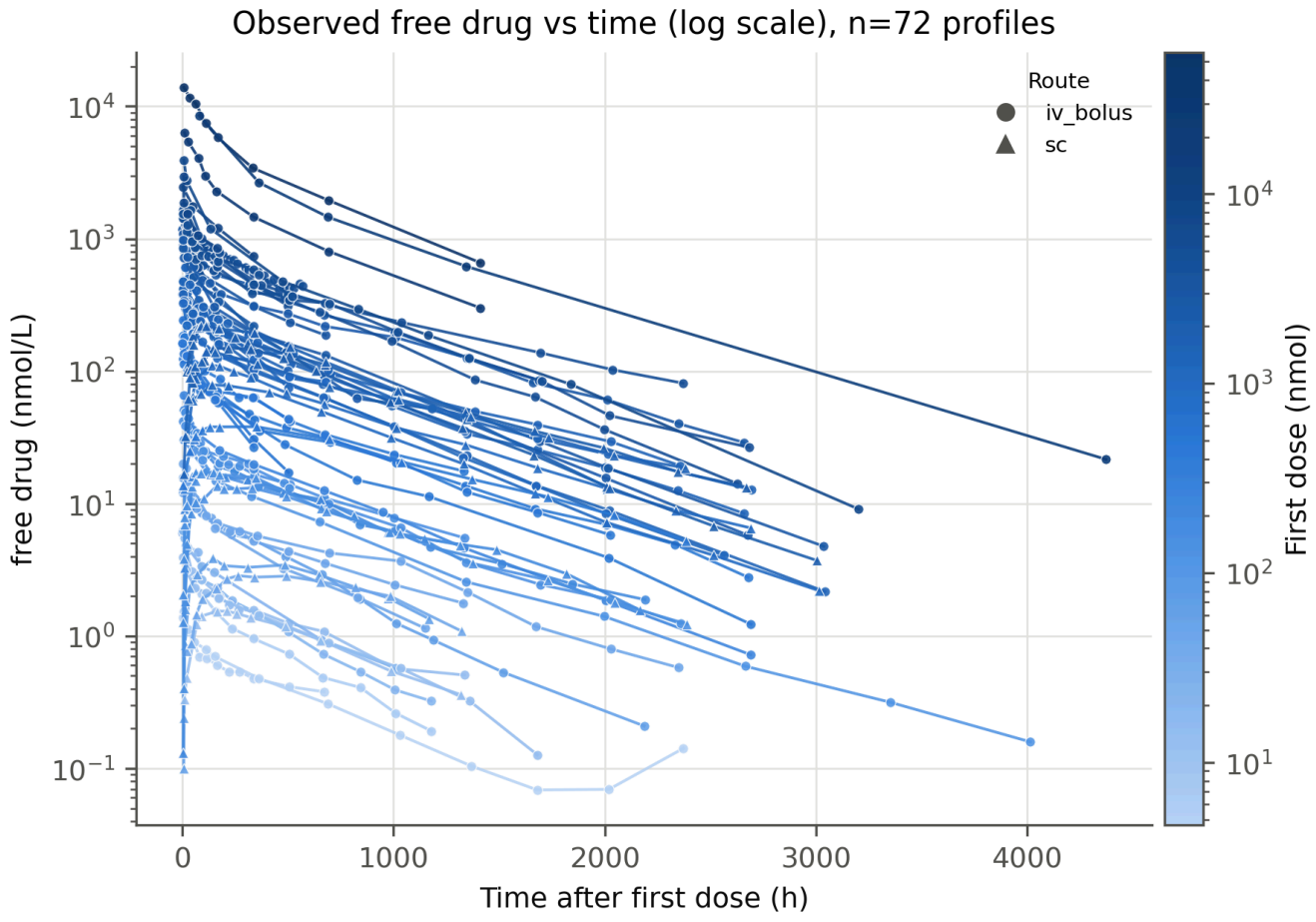
Unit conversions applied

quantity	from	to	factor	basis
TIME	h	h	1	definition
AMT	ug	nmol	0.006667	drug molecular weight 150000 g/mol
DV[default:free_drug]	ug/mL	nmol/L	6.667	drug molecular weight 150000 g/mol
RATE	ug/h	nmol/h	0.006667	amount factor divided by time factor

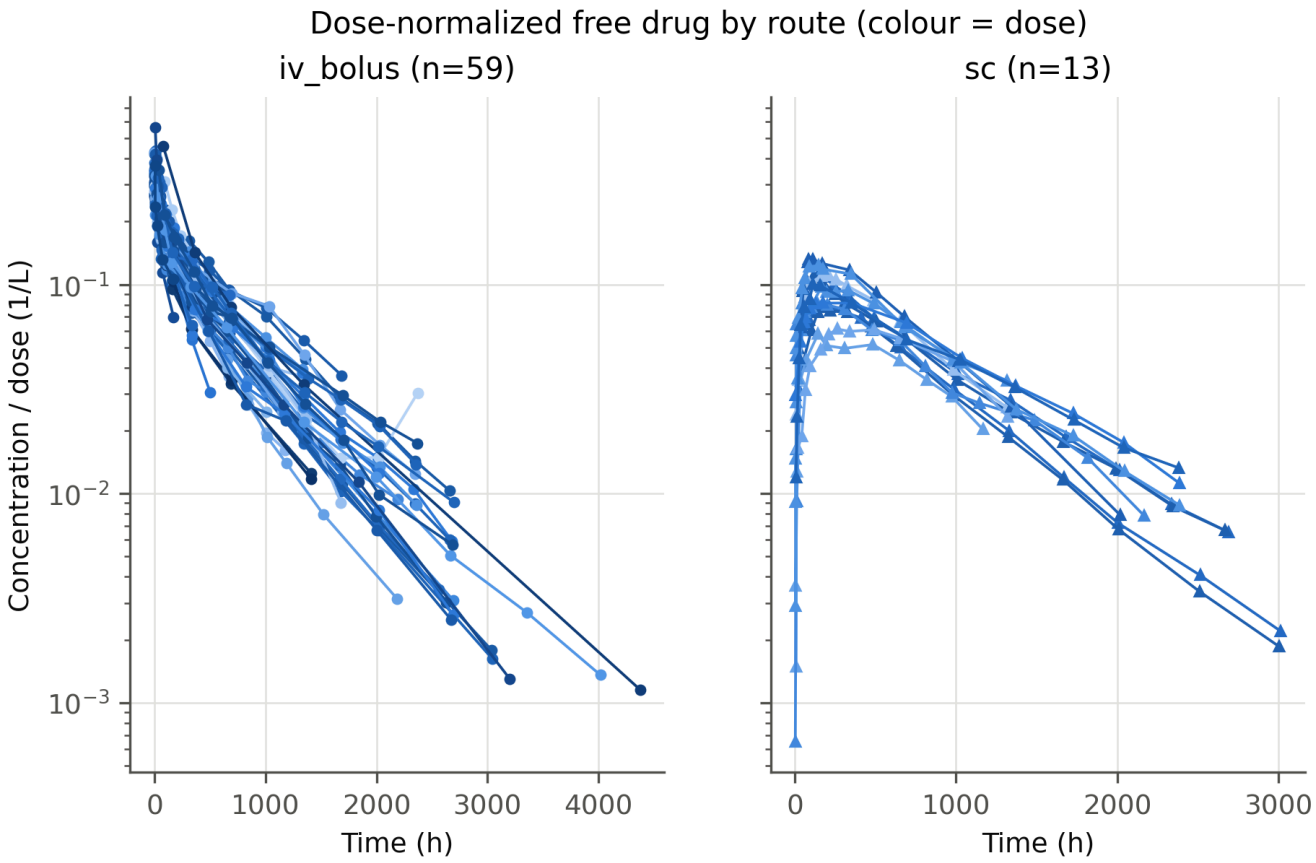
Validation: passed — flags raised for scientist review (nothing deleted):

level	code	message	where
flag	combined_dose_observation_split	rows carrying both a dose and a DV were split into a dose record and an observation record at the same time (declared policy; convention 'post_dose')	rows [15, 29, 132]
flag	unsorted_time	records not in time order within subject; normalized copy sorted by TIME (stable), original file unchanged — review source digitization	Gevokizumab_0.1mgkg, Fulranumab_10mg_SC, Fulranumab_30mg_SC, Gevokizumab_21mg_SC
flag	suspicious_rise	single-dose IV drug concentration rises by >50% between consecutive post-dose samples	Olokizumab_0.01mgkg
flag	coincident_dose_observation	3 observations share a time with a dose; convention 'post_dose' applied identically in Python and R	
flag	aggregate_data	each ID is an aggregate (mean) profile, not an individual; random effects describe between-profile variability and must not be interpreted as between-subject variability	

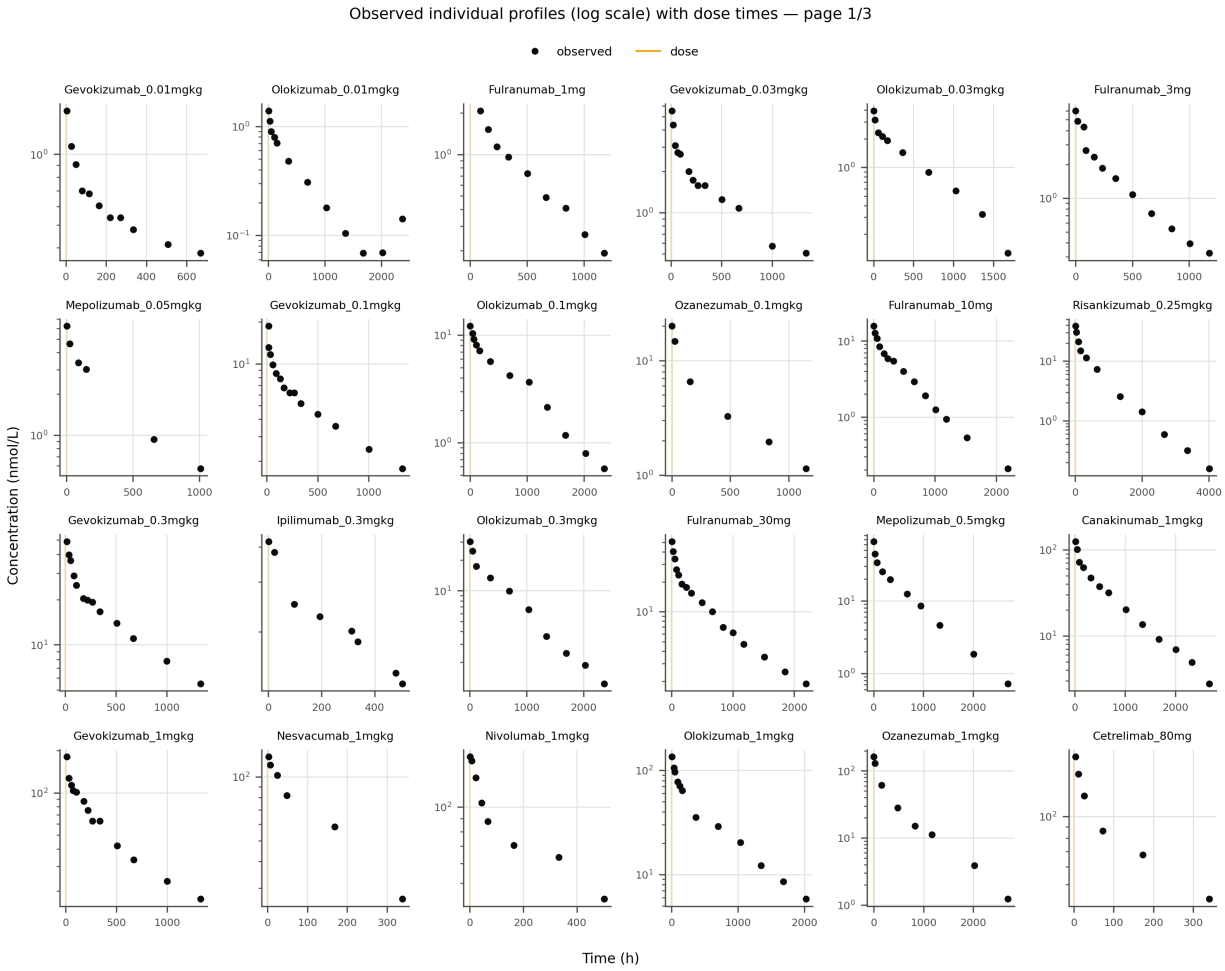
9.3.1 Raw data graphics



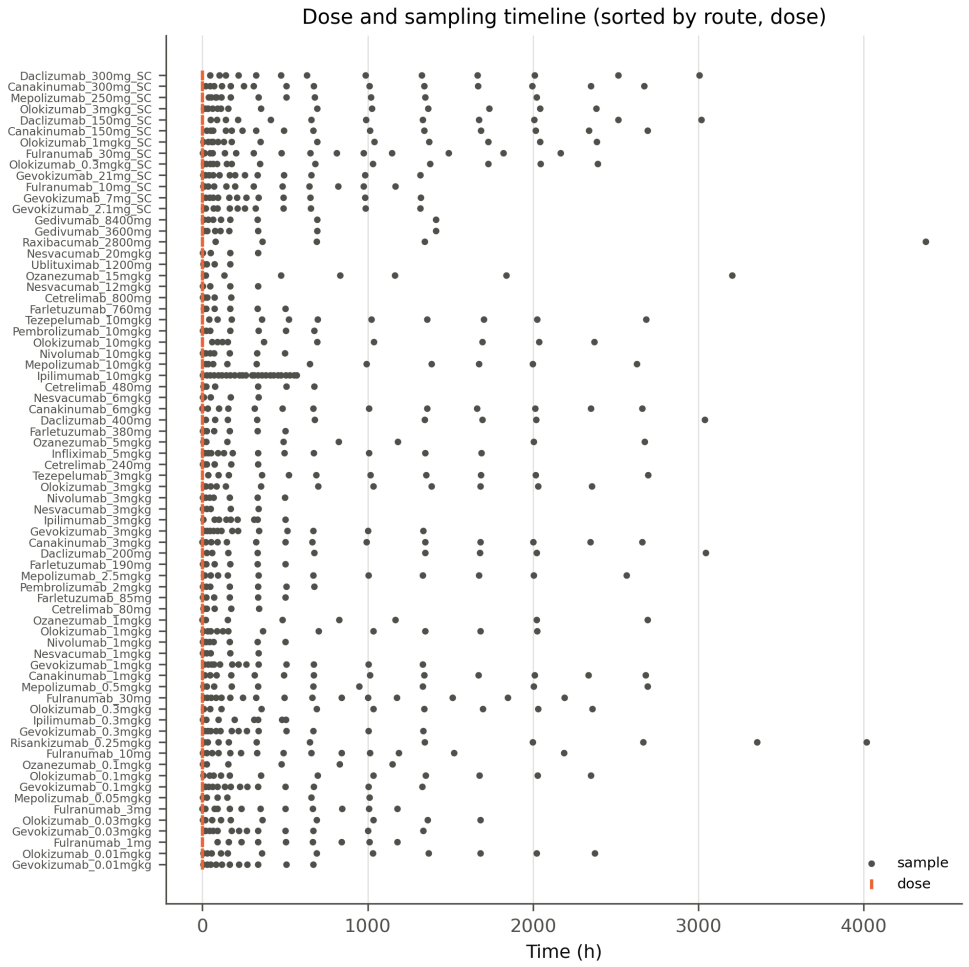
raw free drug log



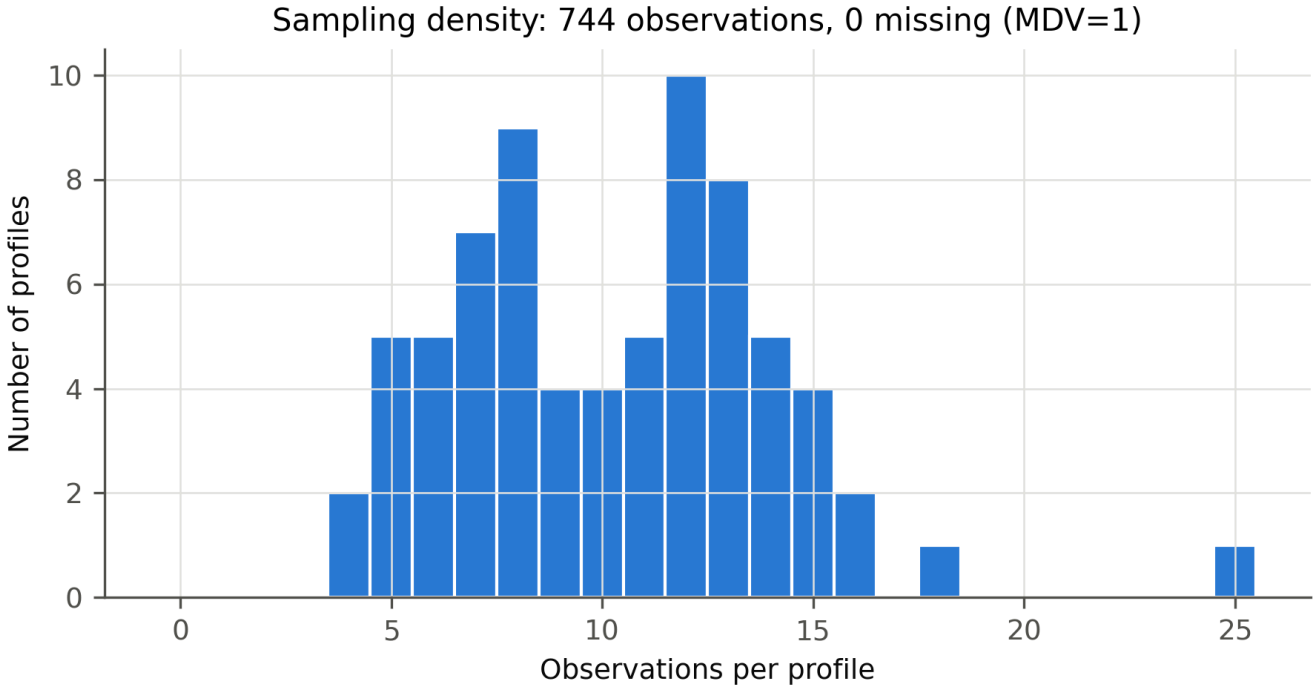
raw free drug dose normalized



raw individual p01



raw timeline



raw missingness

9.4 Model configuration

parameter	unit	status	scale	initial	lower	upper	IIV ω² init	justification
CL	L/h	estimated	log	0.008	1.000e-04	0.5	0.3	
Vc	L	estimated	log	3.5	0.5	20	0.1	
Q	L/h	estimated	log	0.02	1.000e-04	1	–	
Vp	L	estimated	log	3	0.3	30	–	
kon	L/(nmol*h)	FIXED	log	1	0.01	100	–	fixed: typical mAb association ~2.8e5 1/(Ms) = 1 L/(nmolh)
koff	1/h	FIXED	log	0.1	1.000e-04	10	–	fixed: KD = koff/kon = 0.1 nmol/L, typical high-affinity mAb
kint	1/h	estimated	log	0.05	1.000e-04	5	–	
kdeg	1/h	FIXED	log	0.1	1.000e-04	5	–	fixed: generic soluble/membrane target turnover half-life ~7 h
R0	nmol/L	estimated	log	1	1.000e-04	1000	1	
ka	1/h	estimated	log	0.01	1.000e-04	1	–	
F	fraction	estimated	logit	0.7	0.05	1	–	

Residual model: `{'model': 'lognormal', 'sd_init': 0.3, 'units': 'log(nmol/L)'}` · endpoints: `{'default': 'free_drug'}` · SC depot active: True · coincident dose/observation convention: **post_dose** · template tmdd2c_full v1.1.0 (sha256 18d11fe5facf…).

Recorded scientist/analyst decisions

decision	by	date	rationale
enable SC first-order absorption depot (template v1.1.0) because 13 profiles are SC	analyst (Kratos configuration for D. A. Desai)	2026-09-29	SC route present in data; TMDD equations unchanged
fix kon, koff, kdeg (binding/turnover not identifiable from drug-only mean data)	analyst (Kratos configuration for D. A. Desai)	2026-09-29	brief section 3: do not estimate every TMDD parameter from sparse drug-only data

9.5 Calibration (pooled; starting values only)

Objective **lognormal_sse**: SSE = sum_i (log y_i - log f_i)^2 ; -2LL(profiled sigma) = n*log(SSE/n)* + *n*(1+log 2pi) + 2*sum log y_i. Weighting: equal weights on the log scale (equivalent to a lognormal/exponential residual). Optimizer least_squares (bounded, transformed space, max 200 evaluations/attempt); planner **rules** (seed 101);

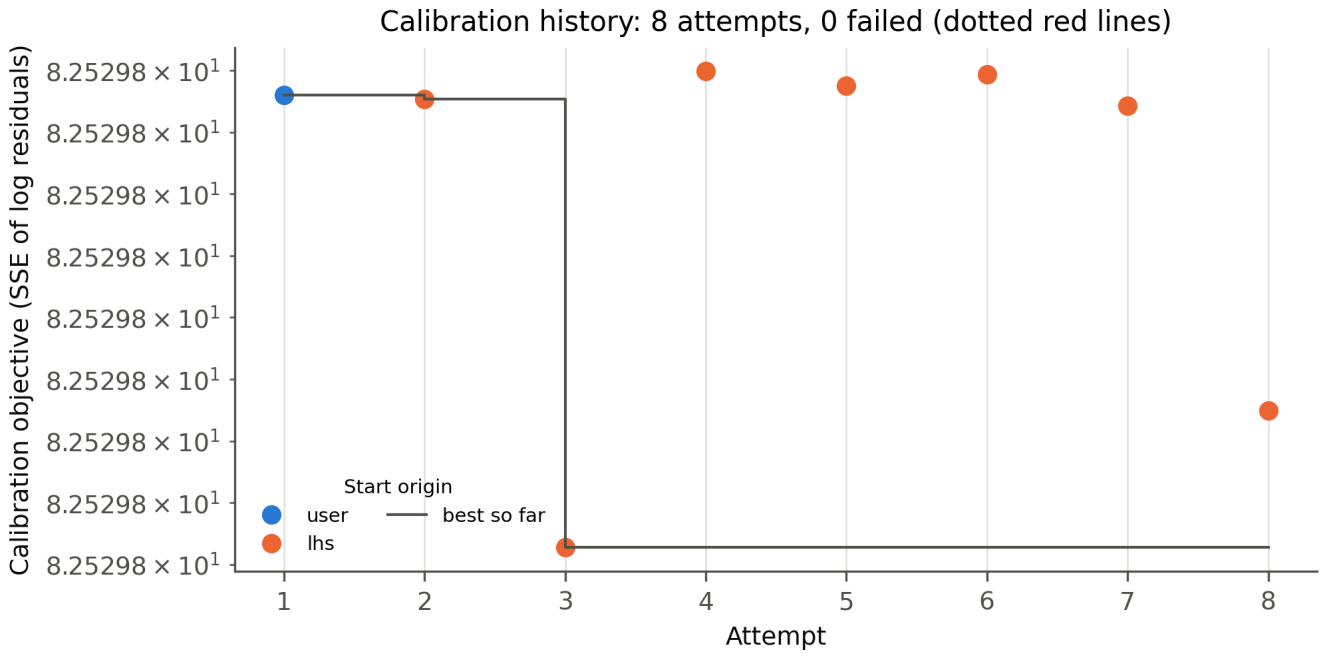
budget 16 attempts / 3600 s; stopping: plateau over 4 successful attempts at rel. tol 0.001, or best objective reproduced by 3 starts.

item	value
status	succeeded
stop_reason	numerical_success: best objective reproduced by 8 independent starts (rel tol 0.001)
n_attempts	8
n_succeeded	8
n_failed	0
n_timeout	0
elapsed_s	262.9
n_distinct_optima	1

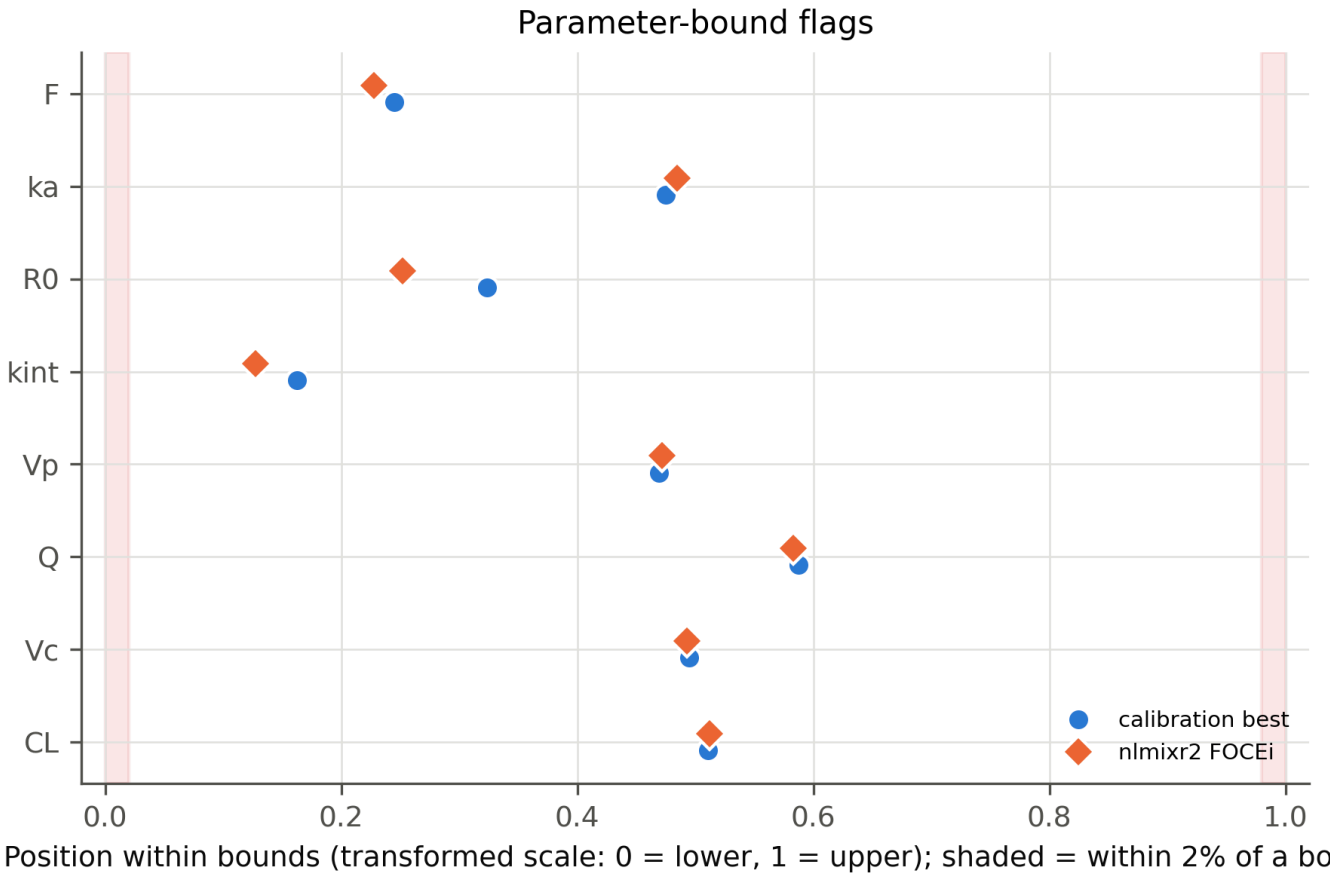
Best candidate (attempt 3, objective 82.53, near bounds: none):

CL	Vc	Q	Vp	kint	R0	ka	F
0.007733	3.104	0.0224	2.606	5.790e-04	0.0184	0.00792	0.762

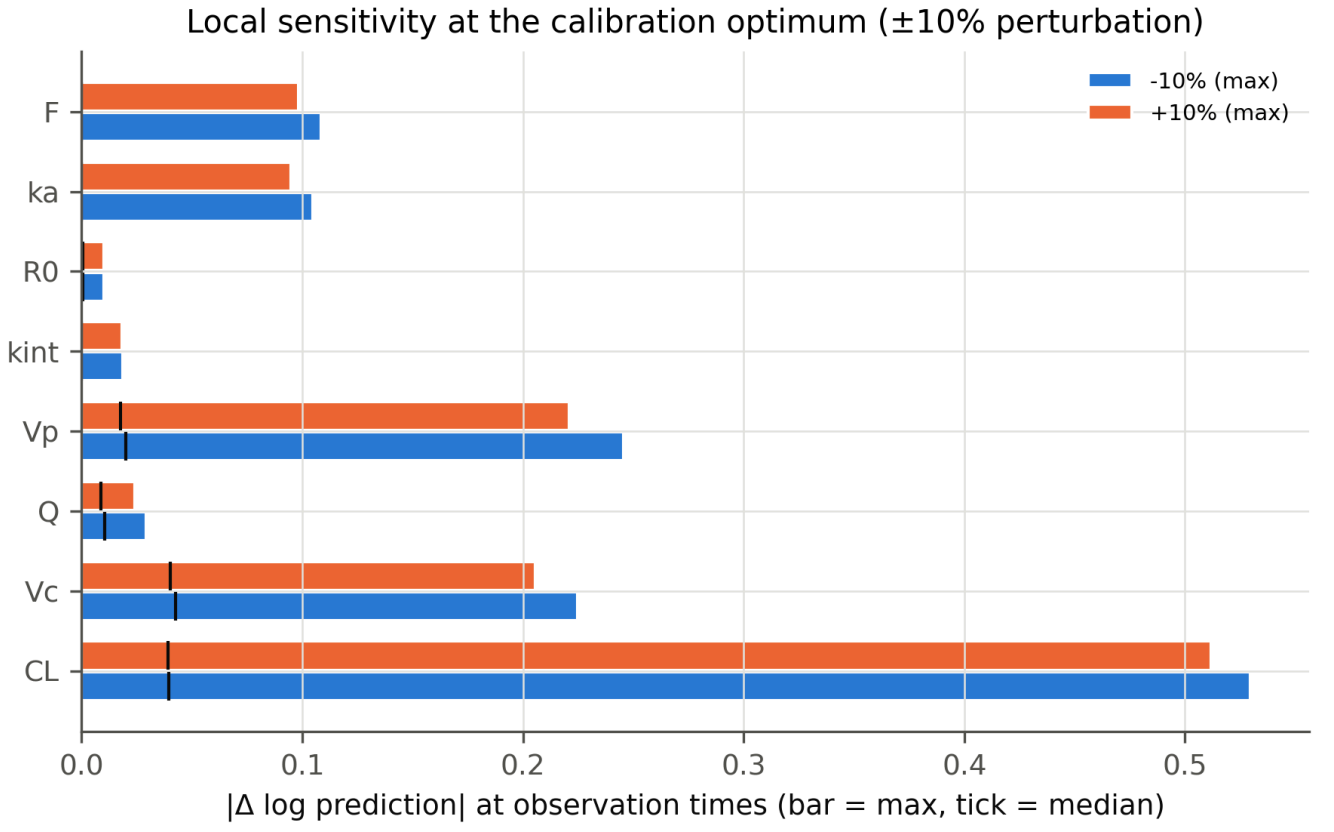
Practical identifiability at the pooled optimum (Gauss-Newton approximation J'J at pooled optimum (transformed scale)): condition number 6060; concerns: none flagged.



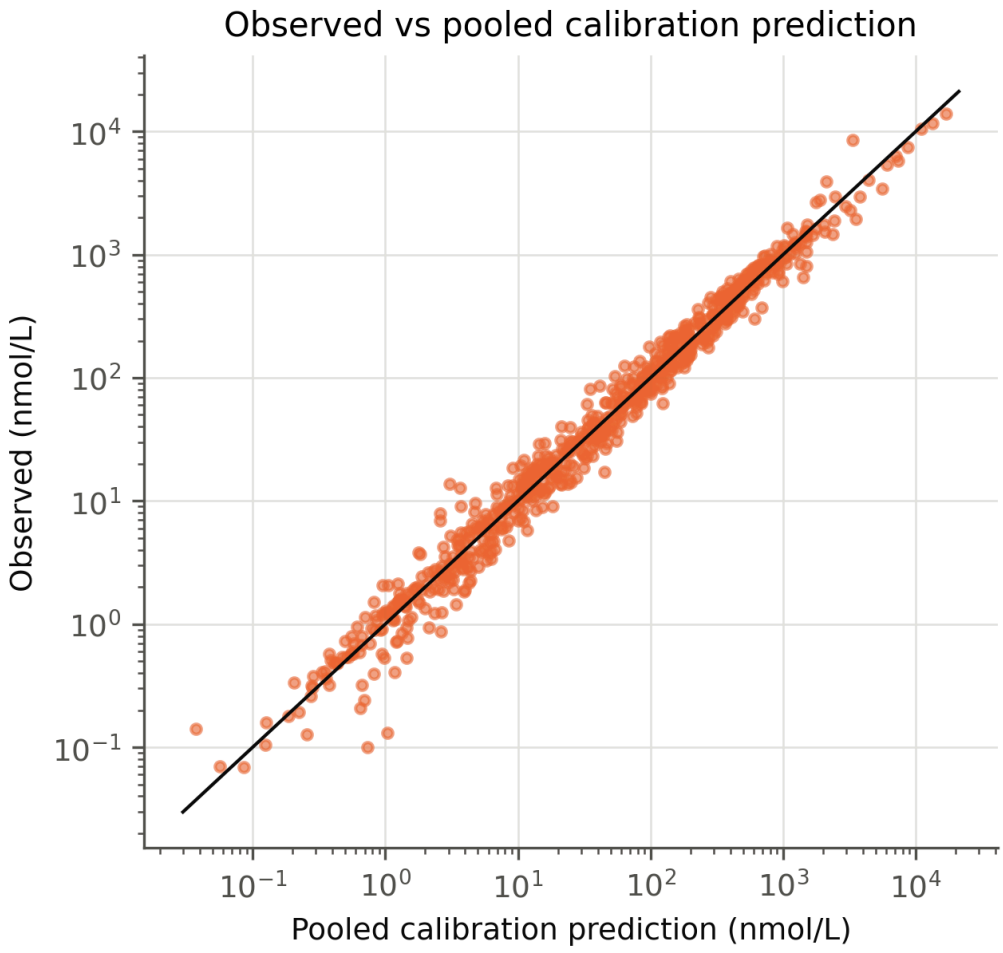
calibration history



parameter bounds



sensitivity



calibration gof

9.6 Population estimation (nlmixr2 FOCEi)

item	value
method	nlmixr2 FOCEi (est='focei')
OFV (nlmixr2)	4005
AIC	5397
BIC	5452
n observations	744
n profiles/subjects	72
optimizer message	relative convergence (4)
covariance method	r,s
covariance success	True
estimation attempts	1
selected attempt	1
runtime (s)	3786
CI method	Wald: back-transformed estimate(log/logit) +/- 1.96 SE on the estimation scale (nlmixr2 parFixedDf)

Parameter estimates (back-transformed by Kratos from the estimation scale; FIXED parameters were not estimated; RSE = delta-method relative standard error on the parameter scale, i.e. 100×SE of log θ, or 100×SE×(1−F) for logit F; BSV = between-profile variability as CV%):

parameter	unit	type	estimate	ci95_lower	ci95_upper	rse_pct	bsv_cv_pct	shrinkage_sd_pct
CL	L/h	estimated	0.007832	0.007272	0.008435	3.782	26.07	8.594
Vc	L	estimated	3.077	2.885	3.283	3.296	28.09	9.331
Q	L/h	estimated	0.02144	0.01615	0.02848	14.48	–	–
Vp	L	estimated	2.634	2.389	2.905	4.995	–	–
kon	L/(nmol*h)	fixed	1	–	–	–	–	–
koff	1/h	fixed	0.1	–	–	–	–	–
kint	1/h	estimated	3.960e-04	6.396e-05	0.002452	93.02	–	–
kdeg	1/h	fixed	0.1	–	–	–	–	–
R0	nmol/L	estimated	0.005798	0.001593	0.0211	65.92	162	86.38
ka	1/h	estimated	0.008692	0.006964	0.01085	11.31	–	–
F	fraction	estimated	0.7054	0.6181	0.78	5.892	–	–
lnorm.sd	log-scale SD	residual	0.1974	–	–	–	–	–

The calibration objective and the nlmixr2 OFV are different quantities and are not compared.

9.7 Diagnostics and review flags

Review status (heuristic): numerical checks passed with warnings — scientist review required. Convergence: {'converged': True, 'message': 'relative convergence (4)'}. Covariance: {'success': True, 'method': 'r,s'}.

Residual type: CWRES; trend statistics: {"CWRES_vs_TIME_spearman": 0.062, "CWRES_vs_PRED_spearman": -0.028, "CWRES_mean": 0.023, "CWRES_sd": 0.983, "frac_abs_CWRES_gt_2": 0.039}.

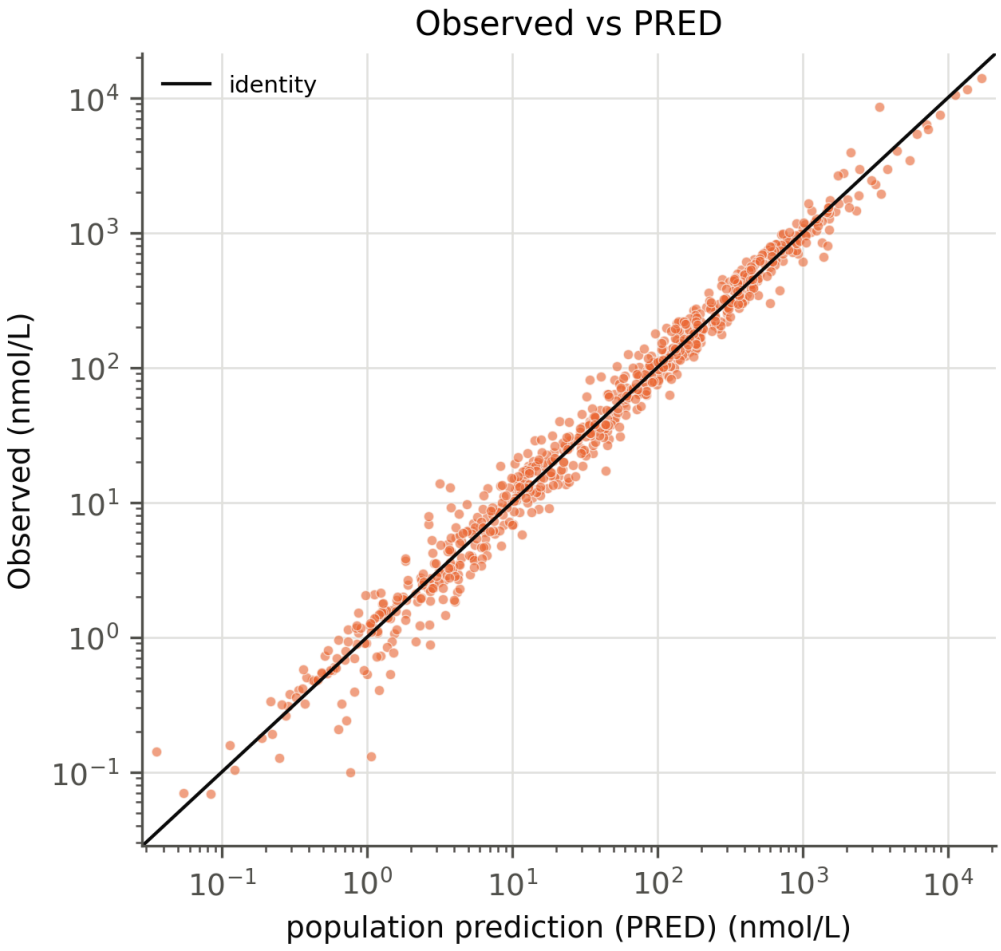
eta	omega	sd_eta	shrinkage_sd_pct
eta.CL	0.06578	0.2344	8.594
eta.Vc	0.07596	0.2499	9.331
eta.Rbase	1.288	0.1545	86.38

VPC: succeeded — 200 replicates, binning: observed-time quantiles within stratum, strata: ROUTE, dose-normalized: True, prediction-corrected: False; fraction of observed percentile points outside simulated 95% intervals: 0.2143.

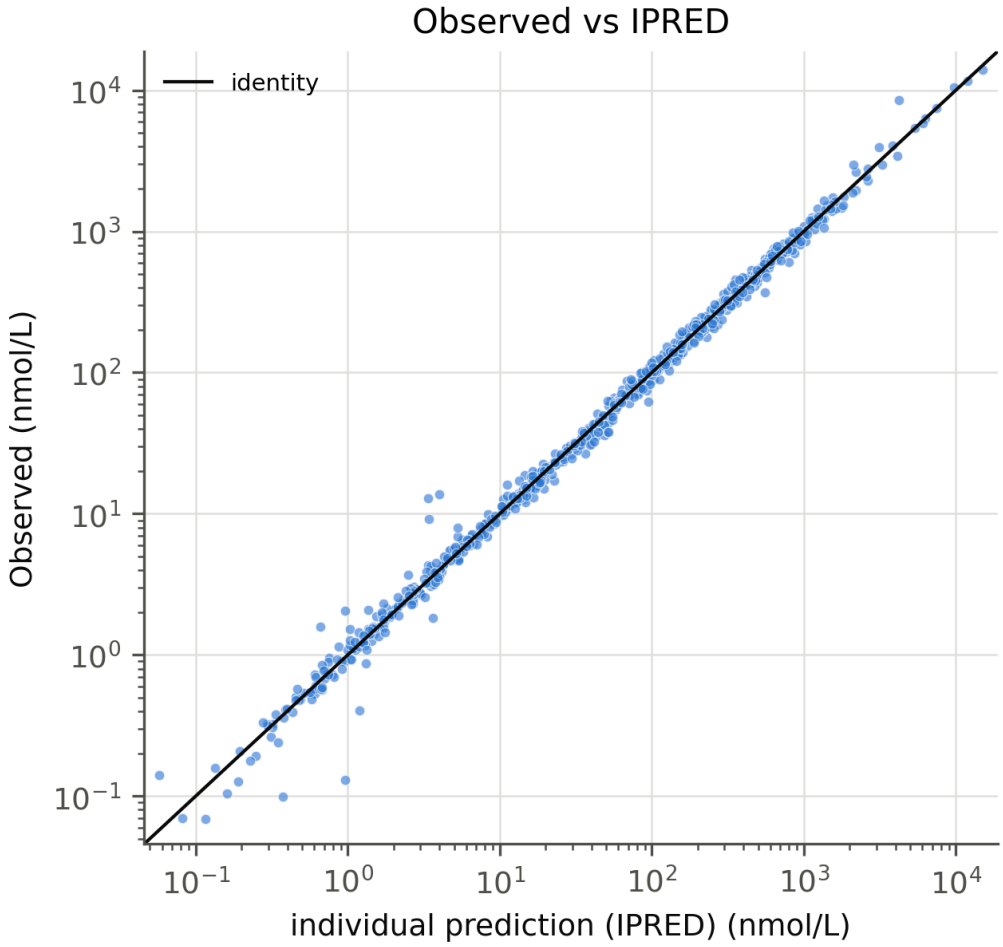
Review flags (thresholds are configurable heuristics: {"rse_max_pct": 50, "shrinkage_max_pct": 30, "corr_max_abs": 0.95, "bound_rel_tol": 0.02, "cwres_trend_max_abs_r": 0.3, "vpc_outside_max_frac": 0.2})

severity	category	message
warning	data:combined_dose_observation_split	rows carrying both a dose and a DV were split into a dose record and an observation record at the same time (declared policy; convention 'post_dose')
warning	data:unsorted_time	records not in time order within subject; normalized copy sorted by TIME (stable), original file unchanged — review source digitization
warning	data:suspicious_rise	single-dose IV drug concentration rises by >50% between consecutive post-dose samples
warning	precision	kint: RSE 93.0% (delta method, parameter scale) > 50%
warning	precision	R0: RSE 65.9% (delta method, parameter scale) > 50%
warning	shrinkage	eta.Rbase: SD shrinkage 86.4% > 30% (EBE-based diagnostics for this eta are less informative)
warning	vpc	21% of observed percentiles fall outside the simulated 95% intervals (> 20%)
info	data:coincident_dose_observation	3 observations share a time with a dose; convention 'post_dose' applied identically in Python and R
info	data:aggregate_data	each ID is an aggregate (mean) profile, not an individual; random effects describe between-profile variability and must not be interpreted as between-subject variability

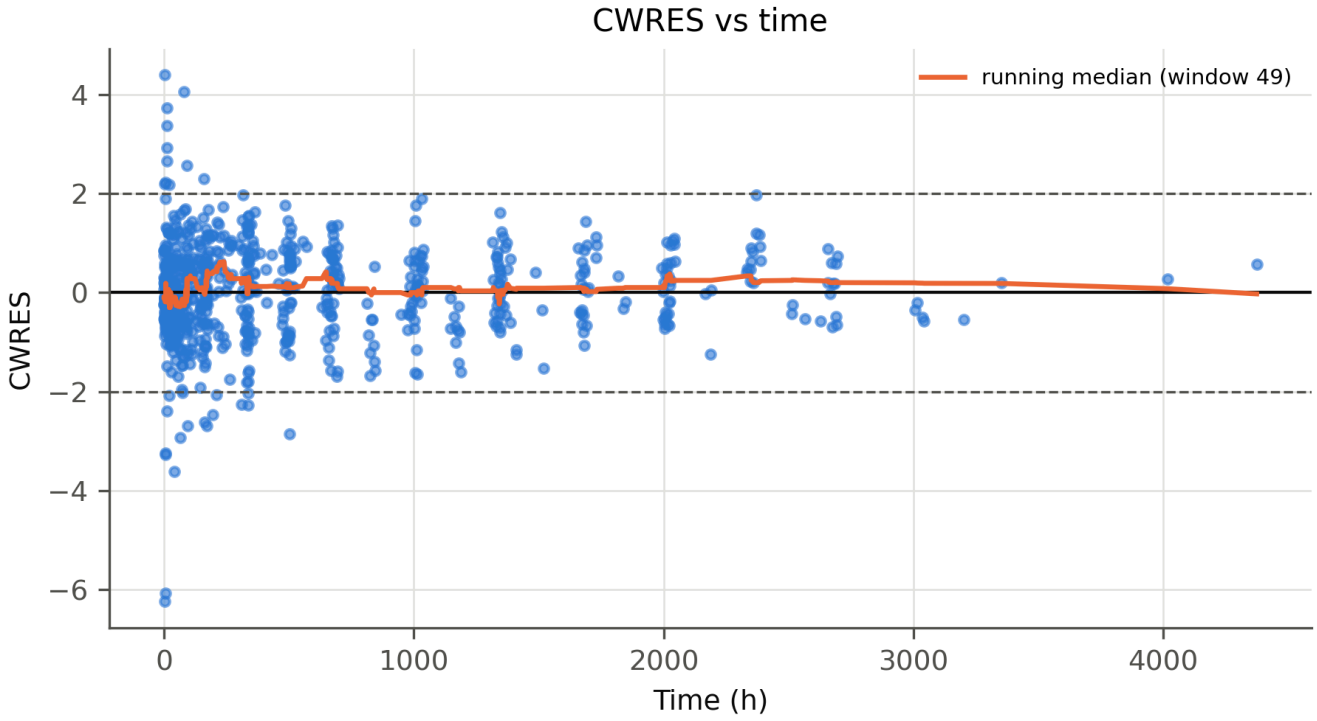
9.7.1 Model diagnostics graphics



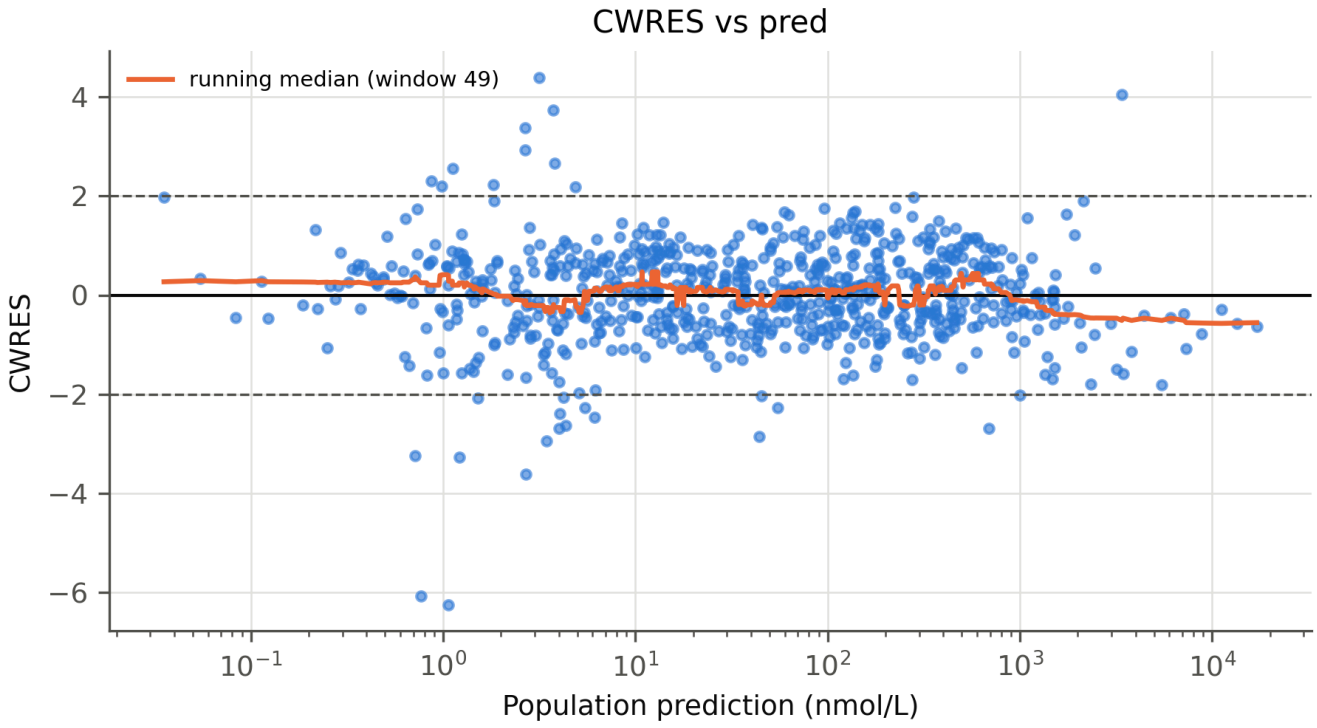
gof dv vs pred



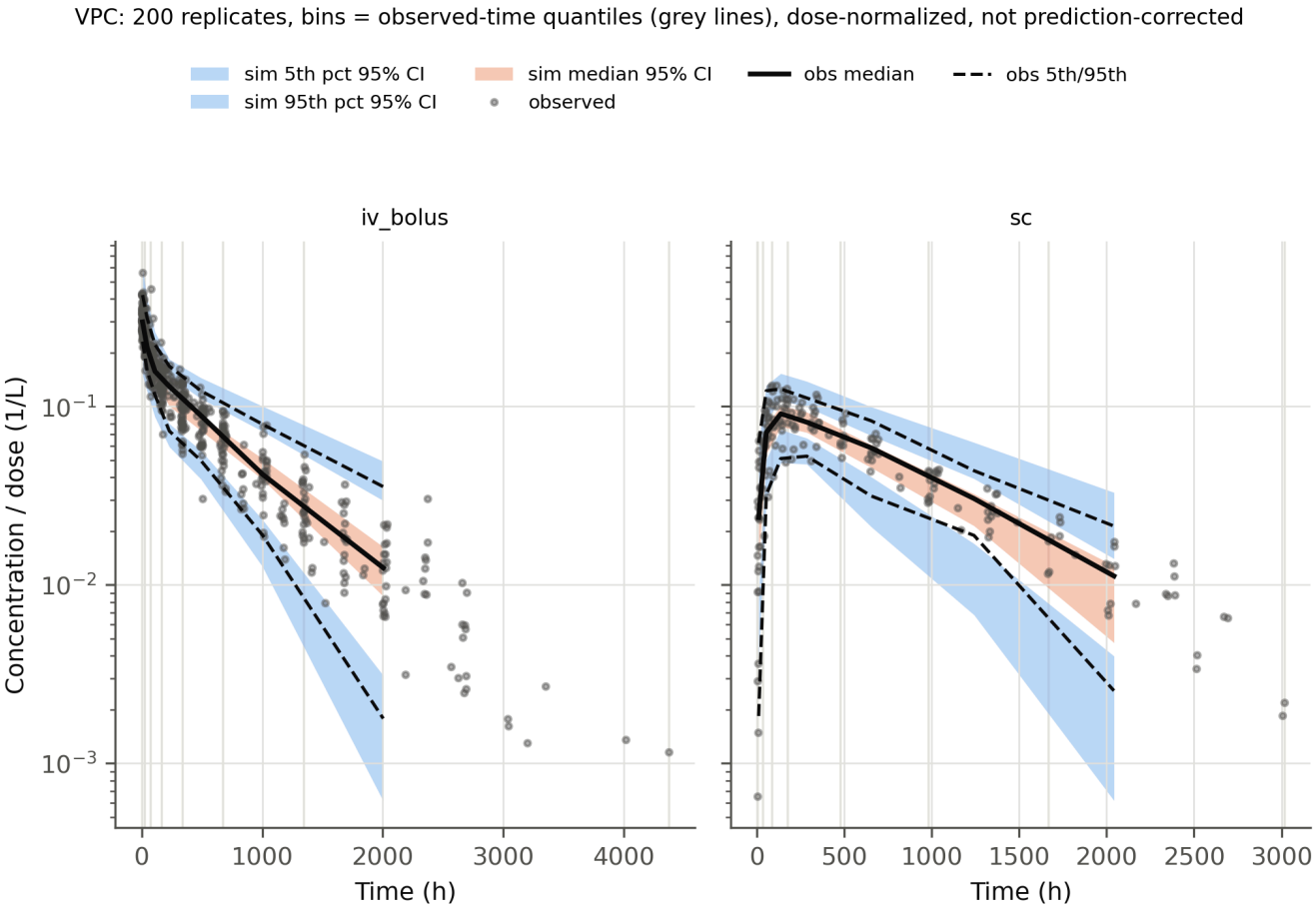
gof dv vs ipred



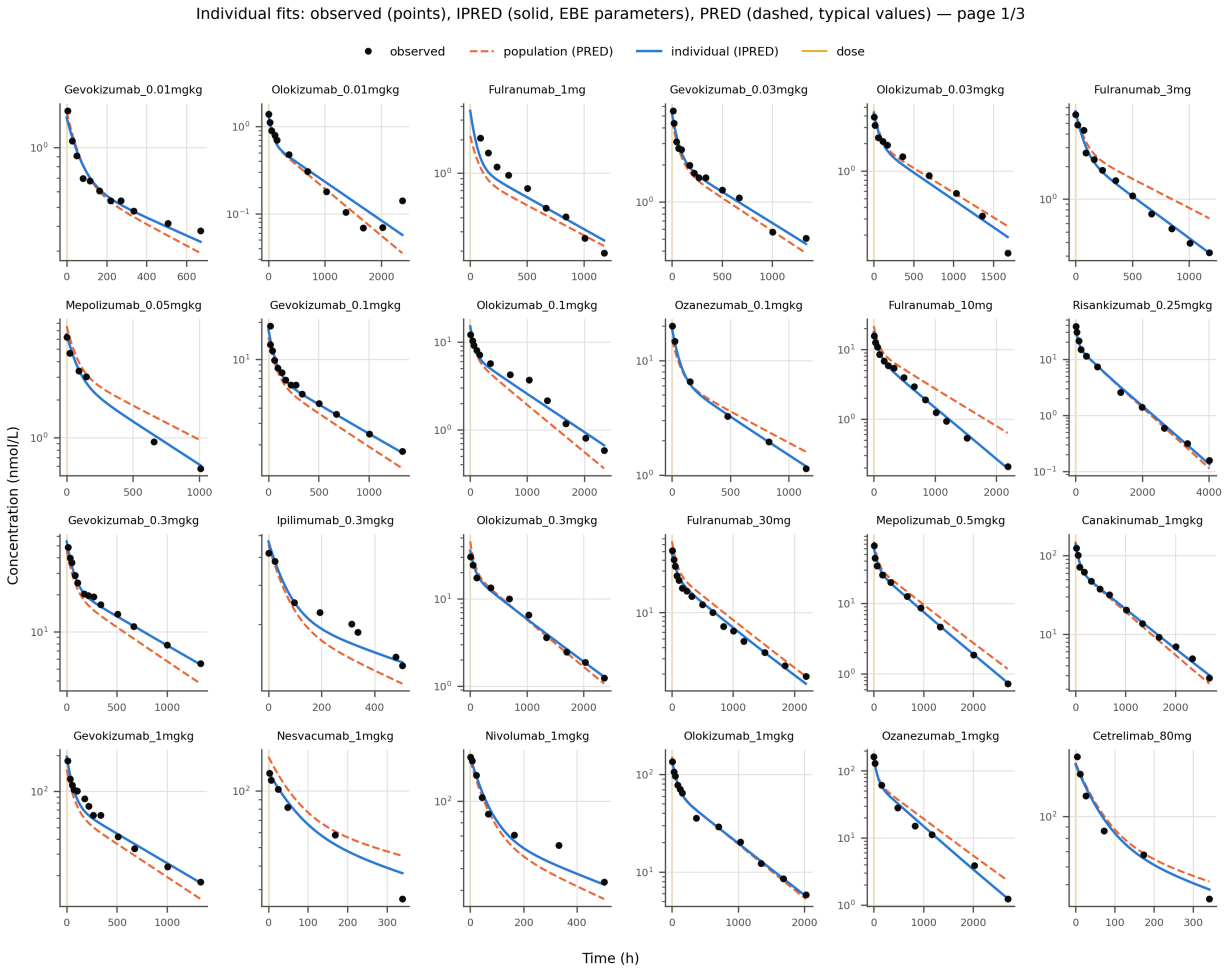
cwres vs time



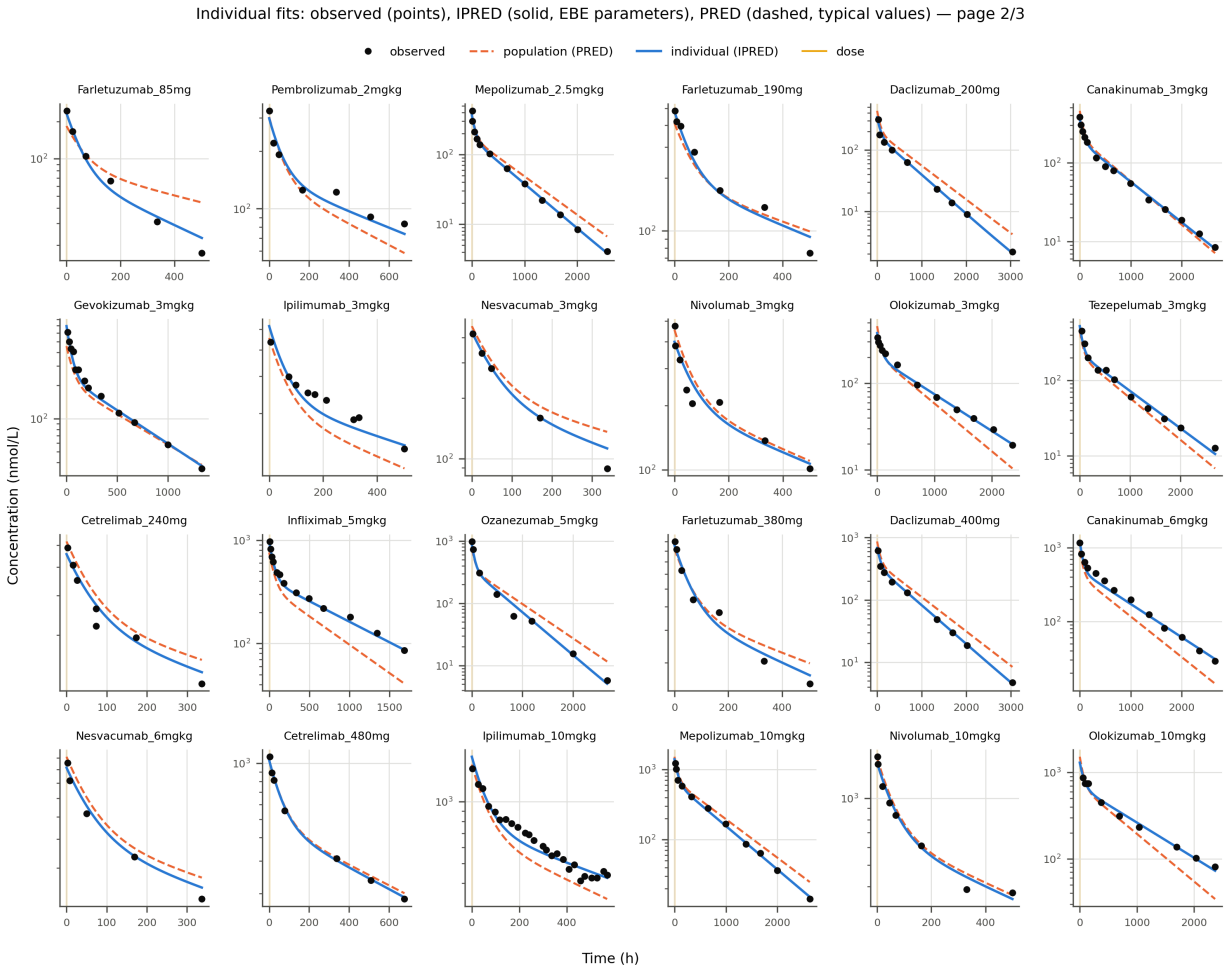
cwres vs pred



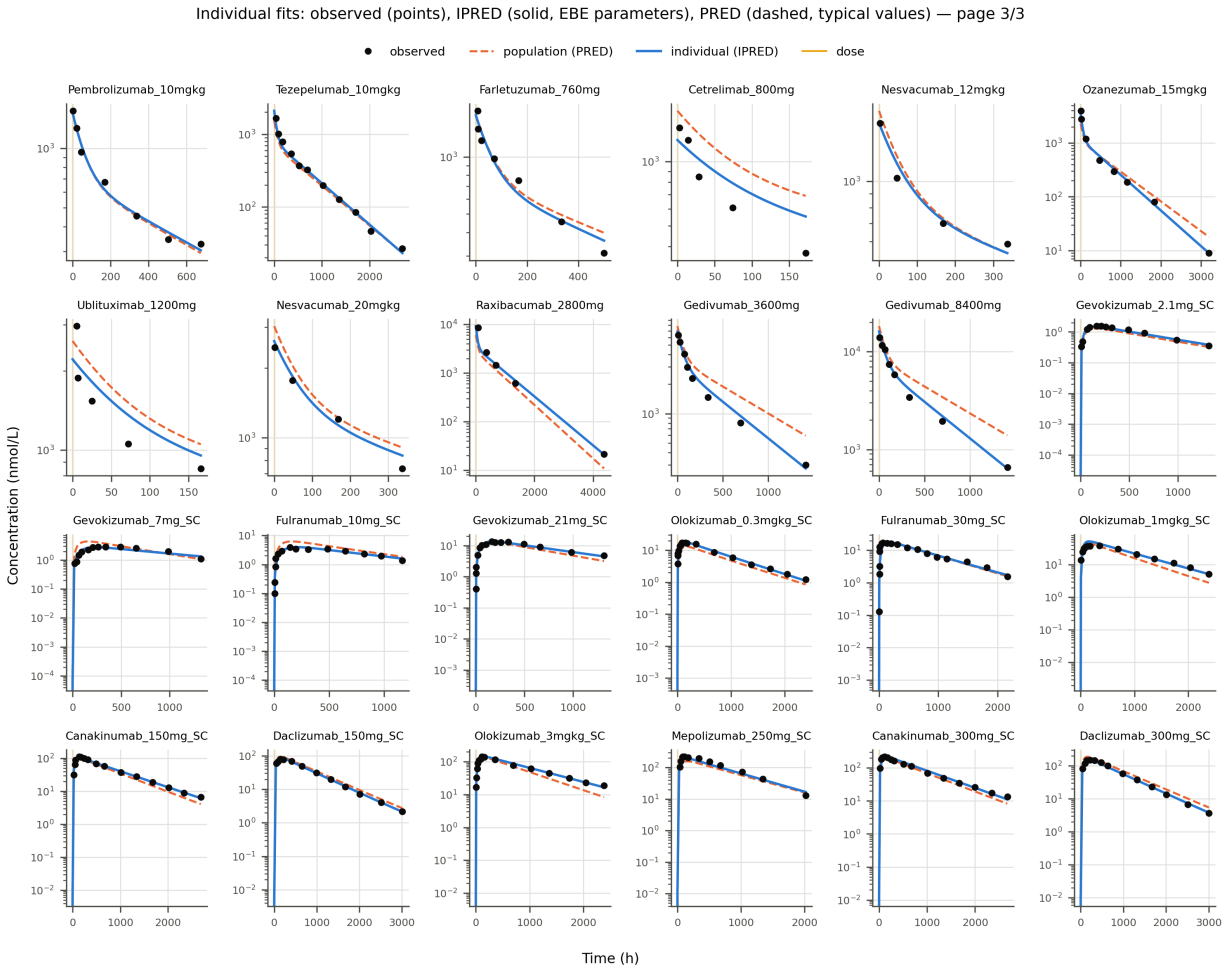
vpc



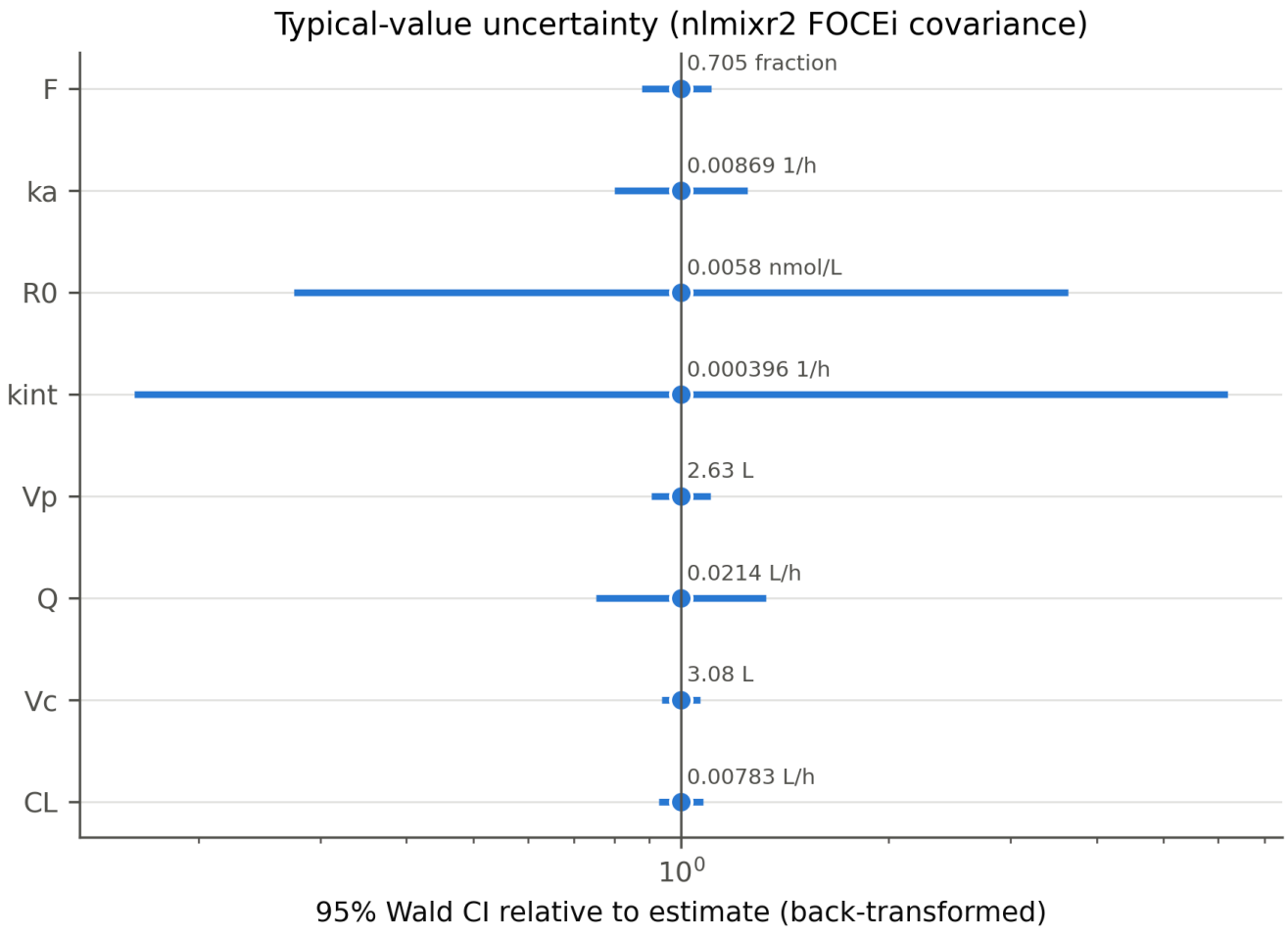
fit individual p01



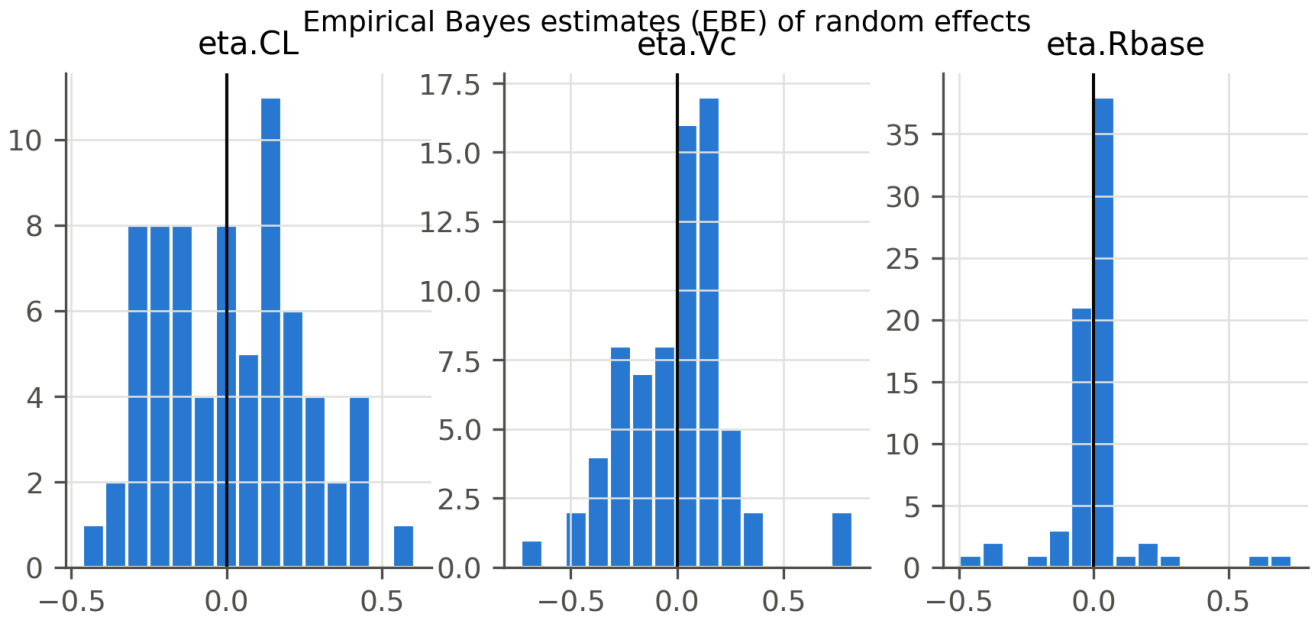
fit individual p02



fit individual p03



parameter uncertainty



eta distributions

9.8 Run environment

item	value
Python	3.12.0
platform	Windows-10-10.0.19045-SP0
Python packages	{"numpy": "1.26.4", "scipy": "1.14.1", "pandas": "2.2.2", "matplotlib": "3.9.1", "PyYAML": "6.0.2", "jsonschema": "4.23.0", "streamlit": "1.44.1", "anthropic": null}
R	R version 4.5.1 (2025-06-13 ucrt)
R packages	{"rxode2": "5.1.6", "nlmixr2": "4.0.1", "nlmixr2est": "4.1.0", "jsonlite": "2.0.0", "ggplot2": "4.0.3", "data.table": "1.17.8"}

10 Appendix: run record `mab_partB_v1`

Human monoclonal antibody mean PK profiles (Part B of a molecule-grouped two-part ID split). Two-compartment full TMDD template with SC depot. Random effects describe between-profile (between-molecule/dose-group) variability of mean curves, not between-subject variability.

Analysis mode: population estimation (nlmixr2 FOCEi) + pooled calibration · **outcome:** completed

10.1 Stage status

stage	state	detail
validate	succeeded	
raw_plots	succeeded	
model	succeeded	
calibrate	succeeded	plateau: no improvement > rel 0.001 over last 4 successful attempts
estimate	succeeded	
diagnose	succeeded	numerical checks raised critical issues — not suitable without resolution
report	succeeded	

10.2 Results narrative

Numerical findings (sourced from artifacts)

- 71 mean profiles with 679 observations and 71 dose records were analysed.
- Data validation: passed with 4 review flags.
- Pooled calibration ran 8 attempts (1 failed, 0 timed out); stop reason: plateau: no improvement > rel 0.001 over last 4 successful attempts. Best objective 273.3 (attempt 6).
- nlmixr2 FOCEi converged to OFV 4890.6868 using calibration start rank 1 (1 estimation attempt(s)); covariance step FAILED.
- Diagnostics: 1 critical and 4 warning heuristic flags; numerical checks raised critical issues — not suitable without resolution.
- VPC (200 replicates): 43% of observed percentile points outside simulated 95% intervals.
- Typical values: CL = 0.01115 L/h; Vc = 3.498 L; Q = 0.01844 L/h; Vp = 2.258 L; kint = 0.0001 1/h; R0 = 0.0382 nmol/L; ka = 0.01267 1/h; F = 0.8354 fraction

Interpretation

- Each ID is a published mean profile: random-effect variances describe between-profile (molecule/dose-group) variability, not between-patient variability, and standard errors do not reflect within-study patient numbers.
- The pooled calibration provides starting values only; its objective is not comparable with the nlmixr2 OFV.
- Critical numerical flags remain and must be resolved before any scientific use: covariance step failed or unavailable (method 'Boundary issue; Get SEs with `getVarCov()`: "tkint"')

- Heuristic checks are review aids; Kratos does not declare the model scientifically valid.

10.3 Data provenance and validation

item	value
source data	C:/Users/devamdesai/Downloads/Kratos/data/mab/mab_partB.csv
data sha256	1dfe778c913e6c8b...
configuration	mab_partB.yaml
config sha256	e36152db5cfc5595...
Kratos code sha256	5e5f89f17e09a5c2...
seeds	{"global": 20260929, "calibration": 111, "estimation": 212, "vpc": 313}

item	value
n_subjects	71
n_records	750
n_observations	679
n_dose_records	71
n_doses_expanded_from_addl	0
routes	{"iv_bolus": 58, "sc": 13}
subjects_by_route	{"iv_bolus": 58, "sc": 13}
doses_per_subject	{"min": 1, "max": 1}
first_dose_nmol	{"min": 13.999999999999998, "max": 72000.0, "n_levels": 29}
obs_per_subject	{"min": 5, "median": 9.0, "max": 21}
time_range_h	0, 9404
dv_range_nmol_per_L	0.1511, 4.457e+04
endpoints	{"free_drug": 679}
aggregate_data	True
synthetic	False

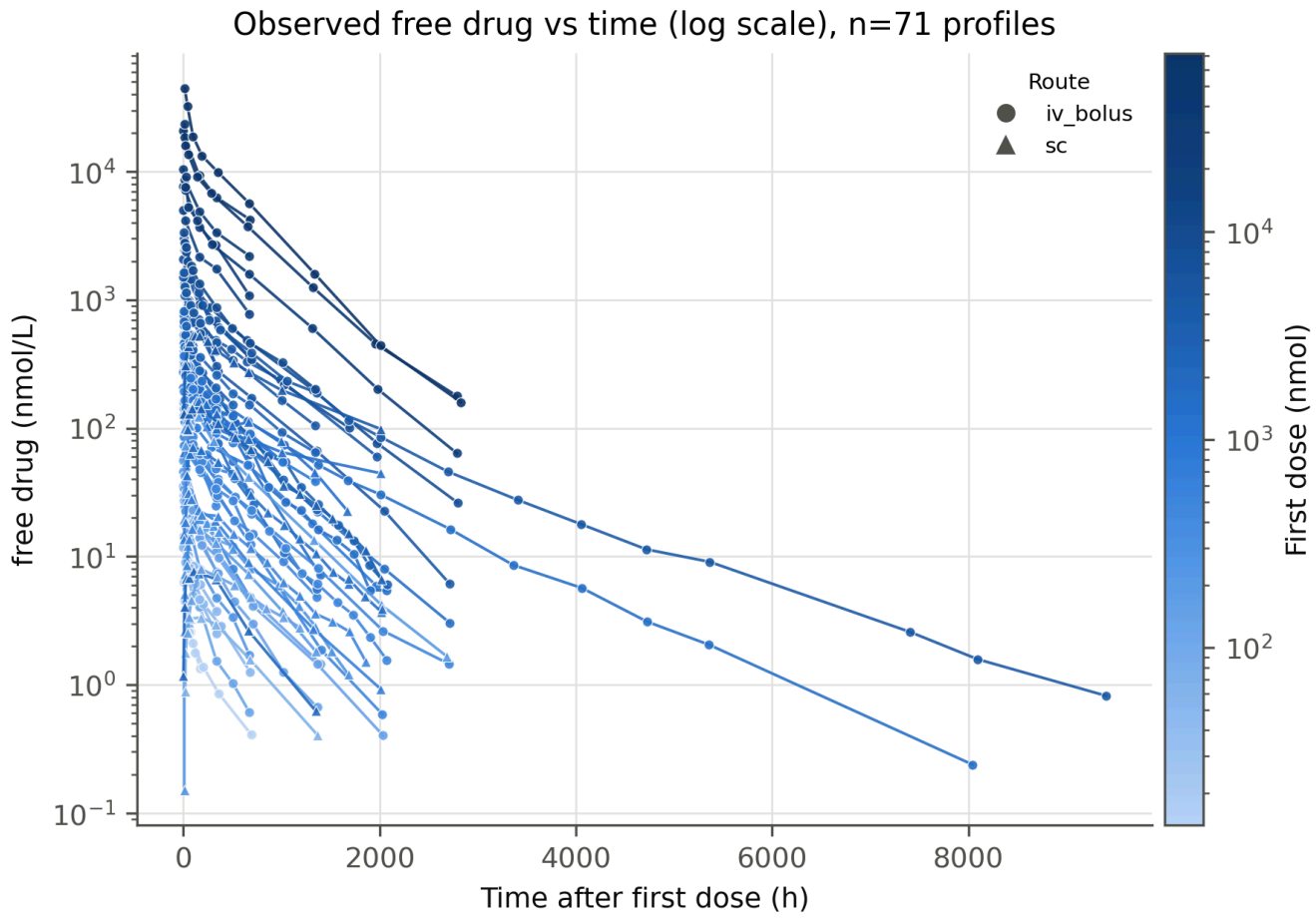
Unit conversions applied

quantity	from	to	factor	basis
TIME	h	h	1	definition
AMT	ug	nmol	0.006667	drug molecular weight 150000 g/mol
DV[default:free_drug]	ug/mL	nmol/L	6.667	drug molecular weight 150000 g/mol
RATE	ug/h	nmol/h	0.006667	amount factor divided by time factor

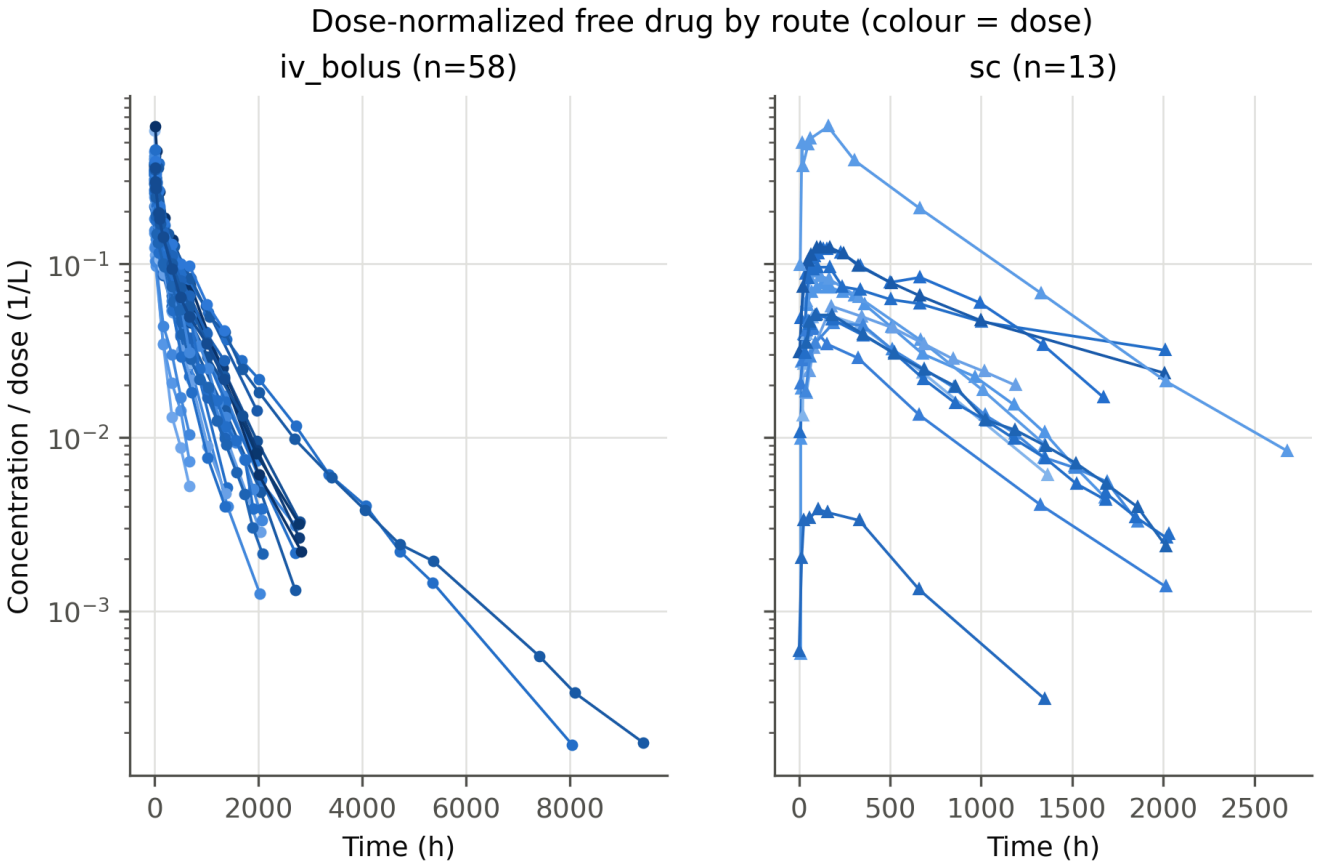
Validation: passed — flags raised for scientist review (nothing deleted):

level	code	message	where
flag	combined_dose_observation_split	rows carrying both a dose and a DV were split into a dose record and an observation record at the same time (declared policy; convention 'post_dose')	rows [119]
flag	unsorted_time	records not in time order within subject; normalized copy sorted by TIME (stable), original file unchanged — review source digitization	Lesofavumab_3600mg, Tefibazumab_5mgkg, Guselkumab_30mg_SC
flag	coincident_dose_observation	1 observations share a time with a dose; convention 'post_dose' applied identically in Python and R	
flag	aggregate_data	each ID is an aggregate (mean) profile, not an individual; random effects describe between-profile variability and must not be interpreted as between-subject variability	

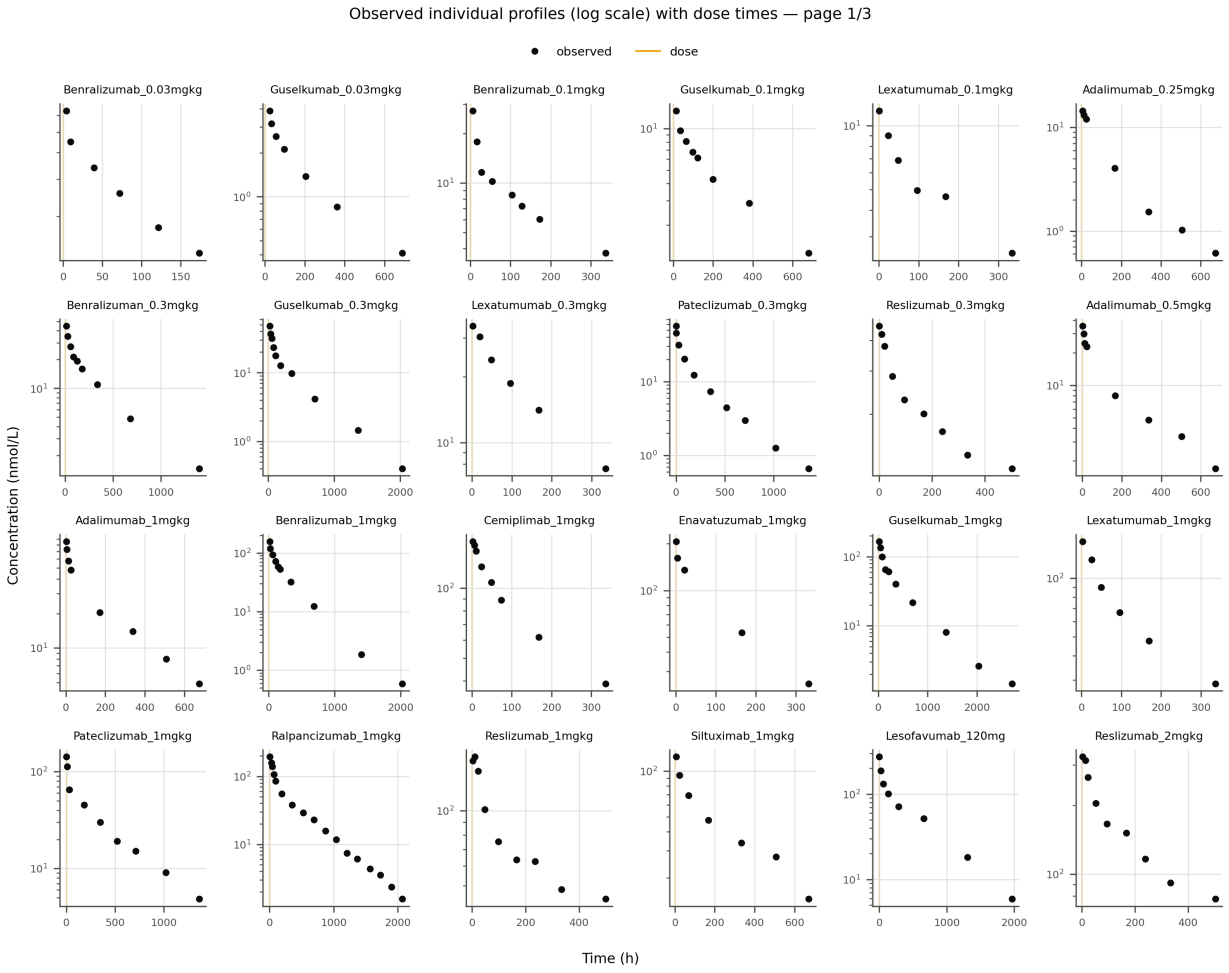
10.3.1 Raw data graphics



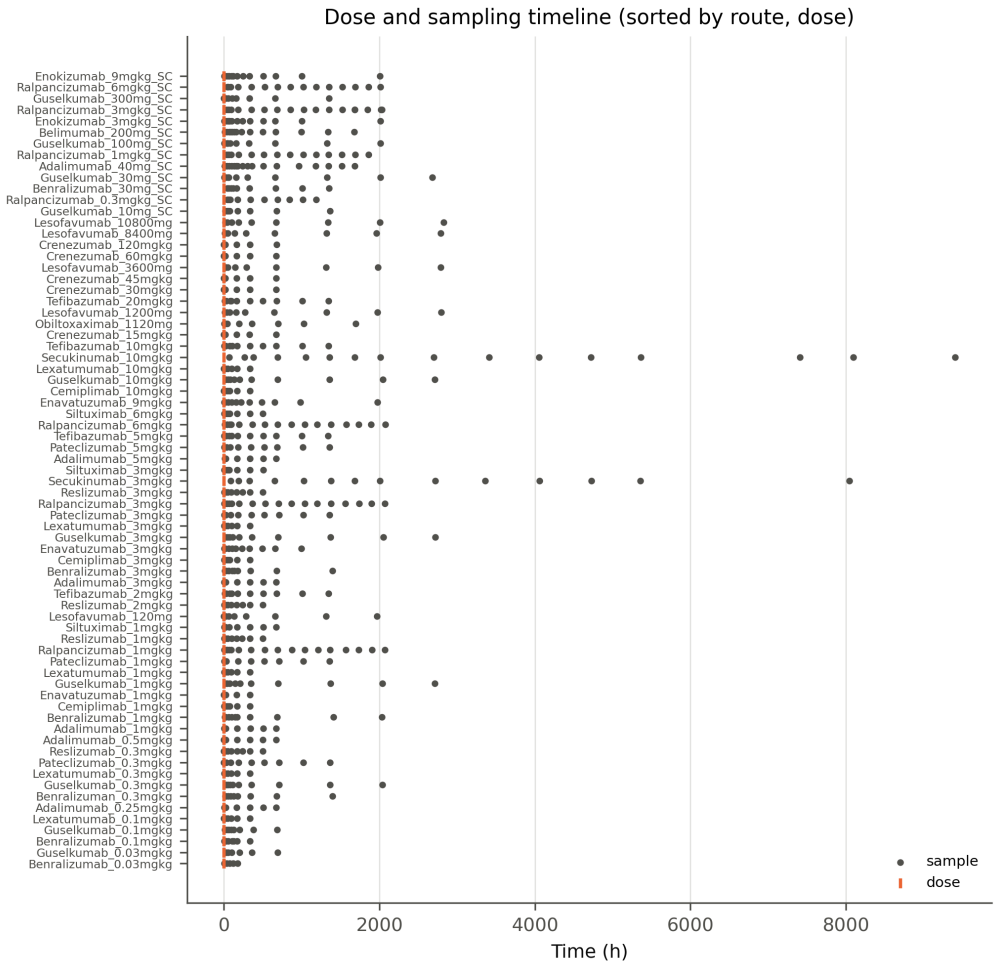
raw free drug log



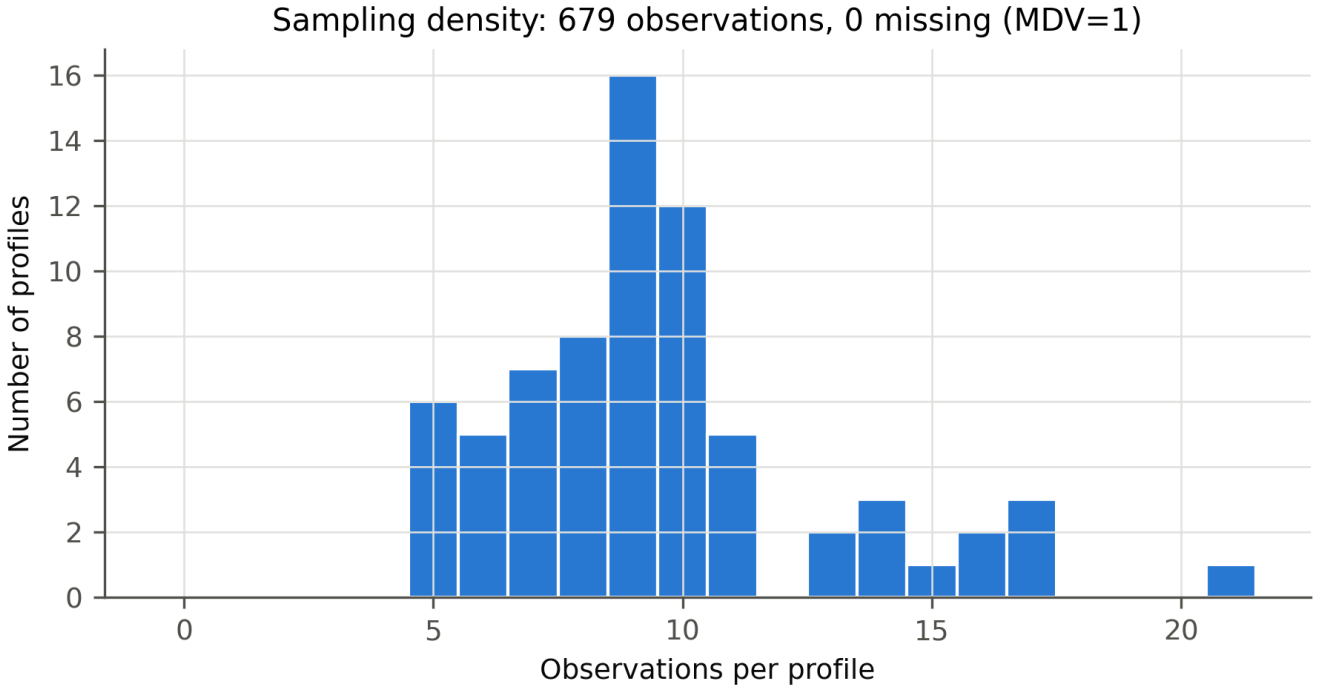
raw free drug dose normalized



raw individual p01



raw timeline



raw missingness

10.4 Model configuration

parameter	unit	status	scale	initial	lower	upper	IIV ω² init	justification
CL	L/h	estimated	log	0.008	1.000e-04	0.5	0.3	
Vc	L	estimated	log	3.5	0.5	20	0.1	
Q	L/h	estimated	log	0.02	1.000e-04	1	–	
Vp	L	estimated	log	3	0.3	30	–	
kon	L/(nmol*h)	FIXED	log	1	0.01	100	–	fixed: typical mAb association ~2.8e5 1/(M*s) = 1 L/(nmol*h)
koff	1/h	FIXED	log	0.1	1.000e-04	10	–	fixed: KD = koff/kon = 0.1 nmol/L, typical high-affinity mAb
kint	1/h	estimated	log	0.05	1.000e-04	5	–	
kdeg	1/h	FIXED	log	0.1	1.000e-04	5	–	fixed: generic soluble/membrane target turnover half-life ~7 h
R0	nmol/L	estimated	log	1	1.000e-04	1000	1	
ka	1/h	estimated	log	0.01	1.000e-04	1	–	
F	fraction	estimated	logit	0.7	0.05	1	–	

Residual model: `{'model': 'lognormal', 'sd_init': 0.3, 'units': 'log(nmol/L)'}` · endpoints: `{'default': 'free_drug'}` · SC depot active: True · coincident dose/observation convention: **post_dose** · template tmdd2c_full v1.1.0 (sha256 18d11fe5facf…).

Recorded scientist/analyst decisions

decision	by	date	rationale
enable SC first-order absorption depot (template v1.1.0) because 13 profiles are SC	analyst (Kratos configuration for D. A. Desai)	2026-09-29	SC route present in data; TMDD equations unchanged
fix kon, koff, kdeg (binding/turnover not identifiable from drug-only mean data)	analyst (Kratos configuration for D. A. Desai)	2026-09-29	brief section 3: do not estimate every TMDD parameter from sparse drug-only data

10.5 Calibration (pooled; starting values only)

Objective **lognormal_sse**: SSE = sum_i (log y_i - log f_i)^2 ; -2LL(profiled sigma) = n*log(SSE/n)* + *n*(1+log 2pi) + 2*sum log y_i. Weighting: equal weights on the log scale (equivalent to a lognormal/exponential residual). Optimizer least_squares (bounded, transformed space, max 200 evaluations/attempt); planner **rules** (seed 111);

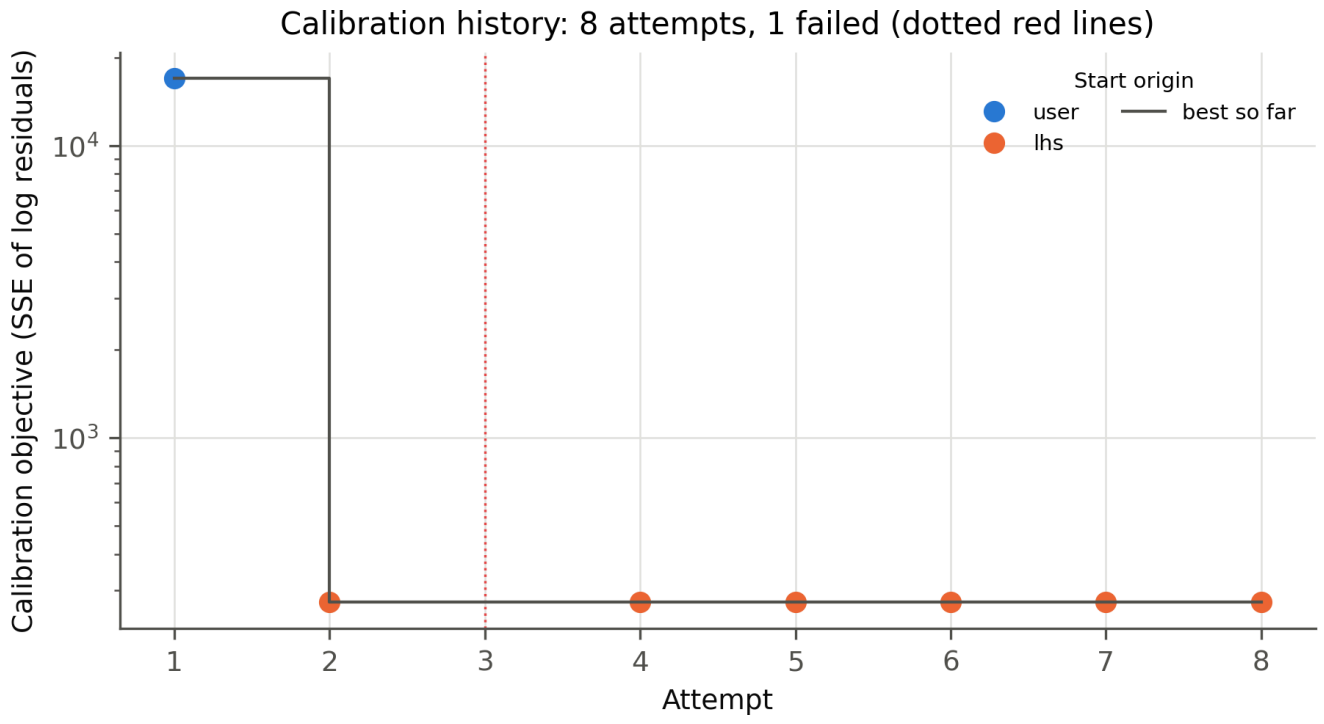
budget 16 attempts / 3600 s; stopping: plateau over 4 successful attempts at rel. tol 0.001, or best objective reproduced by 3 starts.

item	value
status	succeeded
stop_reason	plateau: no improvement > rel 0.001 over last 4 successful attempts
n_attempts	8
n_succeeded	7
n_failed	1
n_timeout	0
elapsed_s	438.1
n_distinct_optima	3

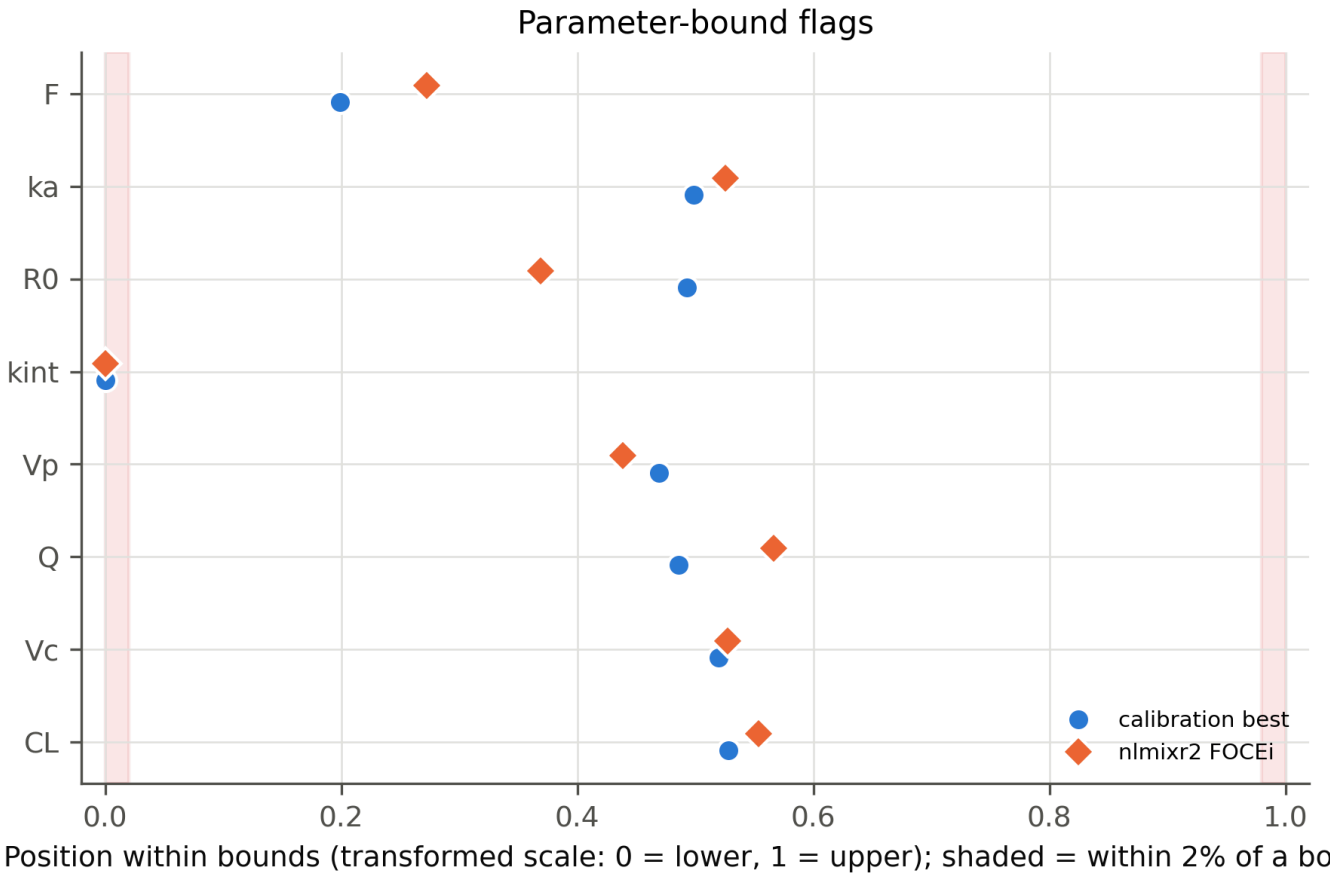
Best candidate (attempt 6, objective 273.3, near bounds: ['kint']):

CL	Vc	Q	Vp	kint	R0	ka	F
0.00896	3.398	0.00879	2.601	1.000e-04	0.2825	0.009877	0.5973

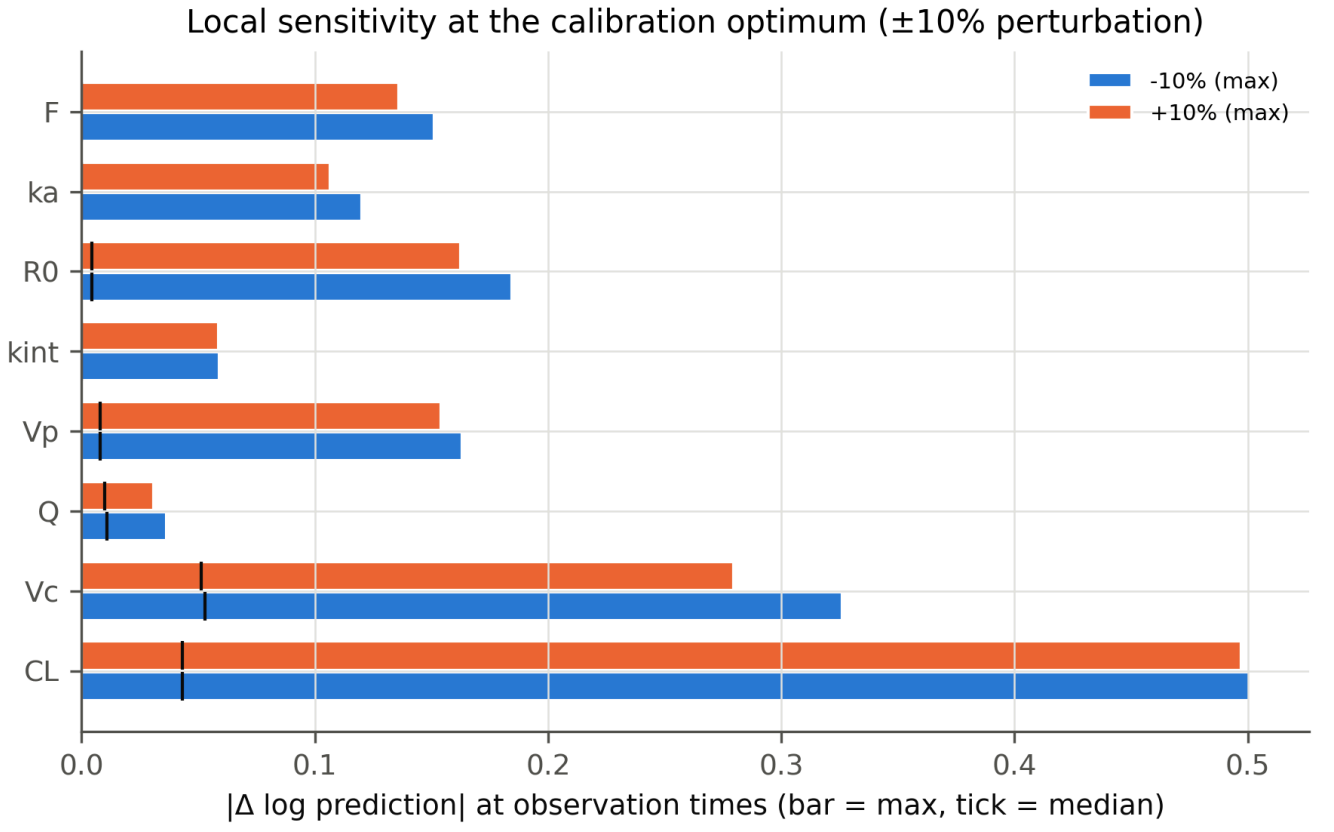
Practical identifiability at the pooled optimum (Gauss-Newton approximation J'J at pooled optimum (transformed scale)): condition number 1225; concerns: none flagged.



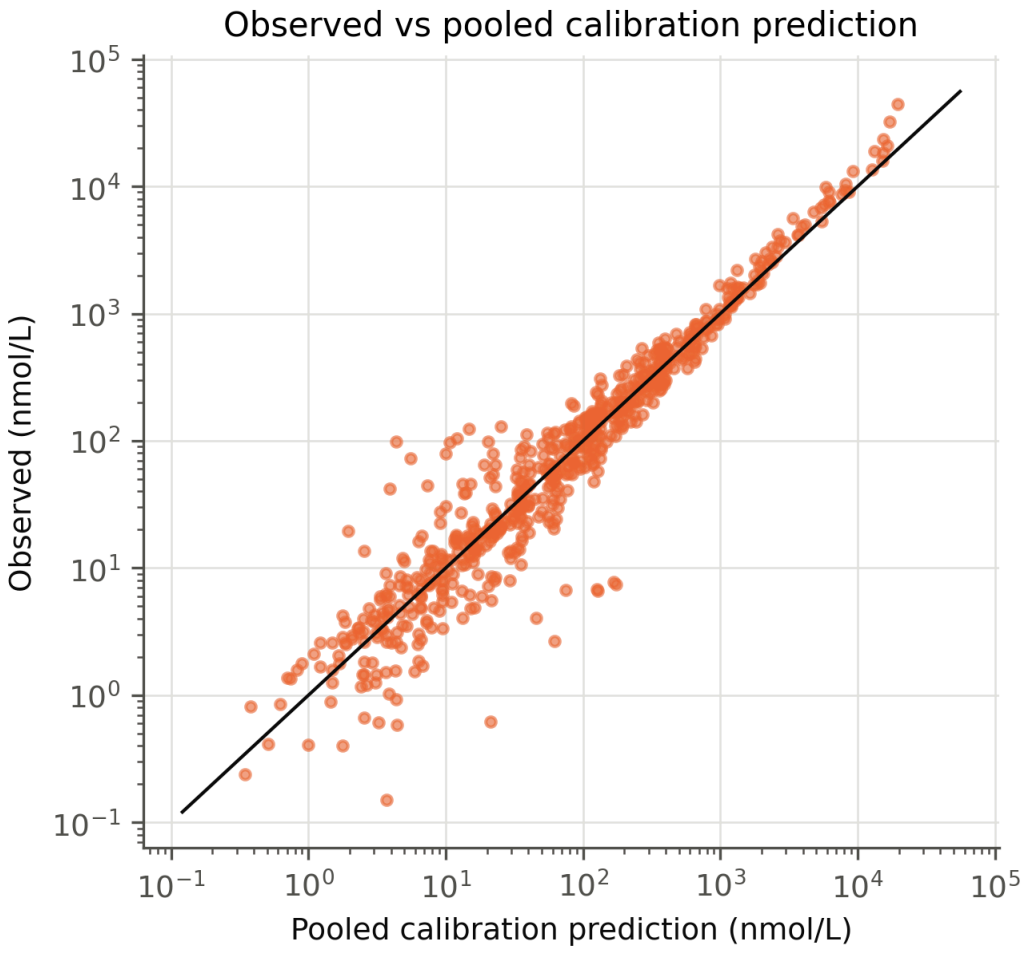
calibration history



parameter bounds



sensitivity



calibration gof

10.6 Population estimation (nlmixr2 FOCEi)

item	value
method	nlmixr2 FOCEi (est='focei')
OFV (nlmixr2)	4891
AIC	6163
BIC	6217
n observations	679
n profiles/subjects	71
optimizer message	relative convergence (4)
covariance method	Boundary issue; Get SEs with `getVarCov()` : "tkint"
covariance success	False
estimation attempts	1
selected attempt	1
runtime (s)	2018
CI method	Wald: back-transformed estimate(log/logit) +/- 1.96 SE on the estimation scale (nlmixr2 parFixedDf)

Parameter estimates (back-transformed by Kratos from the estimation scale; FIXED parameters were not estimated; RSE = delta-method relative standard error on the parameter scale, i.e. 100×SE of log θ, or 100×SE×(1−F) for logit F; BSV = between-profile variability as CV%):

parameter	unit	type	estimate	ci95_lower	ci95_upper	rse_pct	bsv_cv_pct	shrinkage_sd_pct
CL	L/h	estimated	0.01115	–	–	–	68.04	1.89
Vc	L	estimated	3.498	–	–	–	69.05	2.195
Q	L/h	estimated	0.01844	–	–	–	–	–
Vp	L	estimated	2.258	–	–	–	–	–
kon	L/(nmol*h)	fixed	1	–	–	–	–	–
koff	1/h	fixed	0.1	–	–	–	–	–
kint	1/h	estimated	1.000e-04	–	–	–	–	–
kdeg	1/h	fixed	0.1	–	–	–	–	–
R0	nmol/L	estimated	0.0382	–	–	–	118.7	66.51
ka	1/h	estimated	0.01267	–	–	–	–	–
F	fraction	estimated	0.8354	–	–	–	–	–
lnorm.sd	log-scale SD	residual	0.2215	–	–	–	–	–

The calibration objective and the nlmixr2 OFV are different quantities and are not compared.

10.7 Diagnostics and review flags

Review status (heuristic): numerical checks raised critical issues — not suitable without resolution. Convergence: {'converged': True, 'message': 'relative convergence (4)'}. Covariance: {'success': False, 'method': 'Boundary issue; Get SEs with `getVarCov()`: "tkint"'}.

Residual type: CWRES; trend statistics: {"CWRES_vs_TIME_spearman": 0.008, "CWRES_vs_PRED_spearman": 0.262, "CWRES_mean": -0.012, "CWRES_sd": 1.018, "frac_abs_CWRES_gt_2": 0.043}.

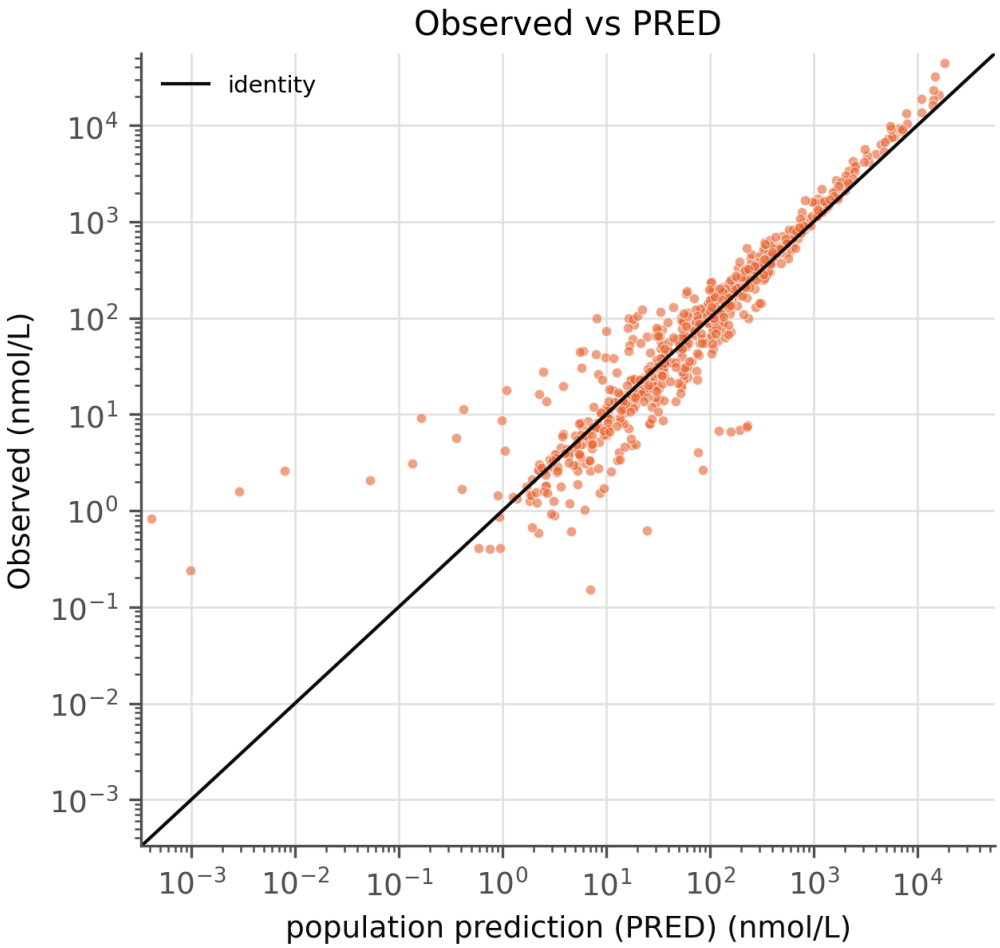
eta	omega	sd_eta	shrinkage_sd_pct
eta.CL	0.3805	0.6052	1.89
eta.Vc	0.3899	0.6107	2.195
eta.Rbase	0.8794	0.314	66.51

VPC: succeeded — 200 replicates, binning: observed-time quantiles within stratum, strata: ROUTE, dose-normalized: True, prediction-corrected: False; fraction of observed percentile points outside simulated 95% intervals: 0.4286.

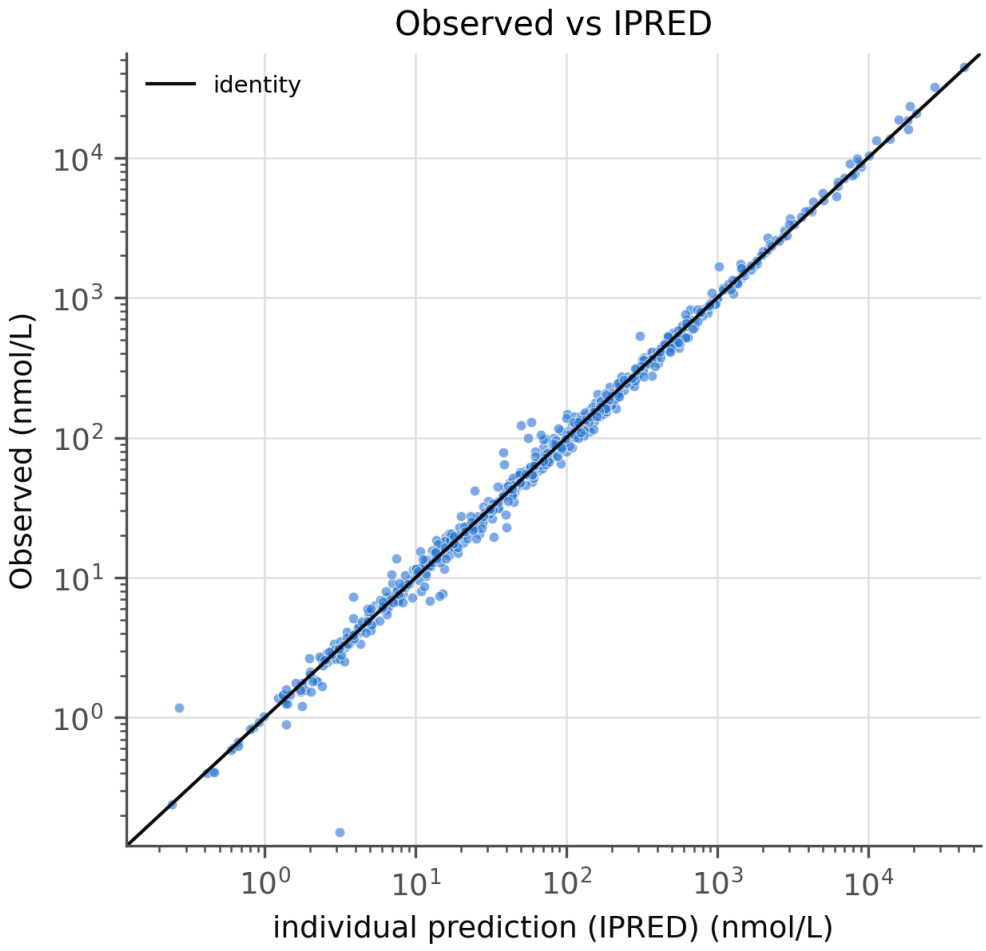
Review flags (thresholds are configurable heuristics: {"rse_max_pct": 50, "shrinkage_max_pct": 30, "corr_max_abs": 0.95, "bound_rel_tol": 0.02, "cwres_trend_max_abs_r": 0.3, "vpc_outside_max_frac": 0.2})

severity	category	message
critical	covariance	covariance step failed or unavailable (method 'Boundary issue; Get SEs with `getVarCov()`: "tkint"')
warning	data:combined_dose_observation_split	rows carrying both a dose and a DV were split into a dose record and an observation record at the same time (declared policy; convention 'post_dose')
warning	data:unsorted_time	records not in time order within subject; normalized copy sorted by TIME (stable), original file unchanged — review source digitization
warning	calibration_bound	calibration best fit: kint near bound
warning	boundary	kint = 0.0001 within 2% of a bound [0.0001, 5.0]
warning	shrinkage	eta.Rbase: SD shrinkage 66.5% > 30% (EBE-based diagnostics for this eta are less informative)
warning	vpc	43% of observed percentiles fall outside the simulated 95% intervals (> 20%)
info	data:coincident_dose_observation	1 observations share a time with a dose; convention 'post_dose' applied identically in Python and R
info	data:aggregate_data	each ID is an aggregate (mean) profile, not an individual; random effects describe between-profile variability and must not be interpreted as between-subject variability

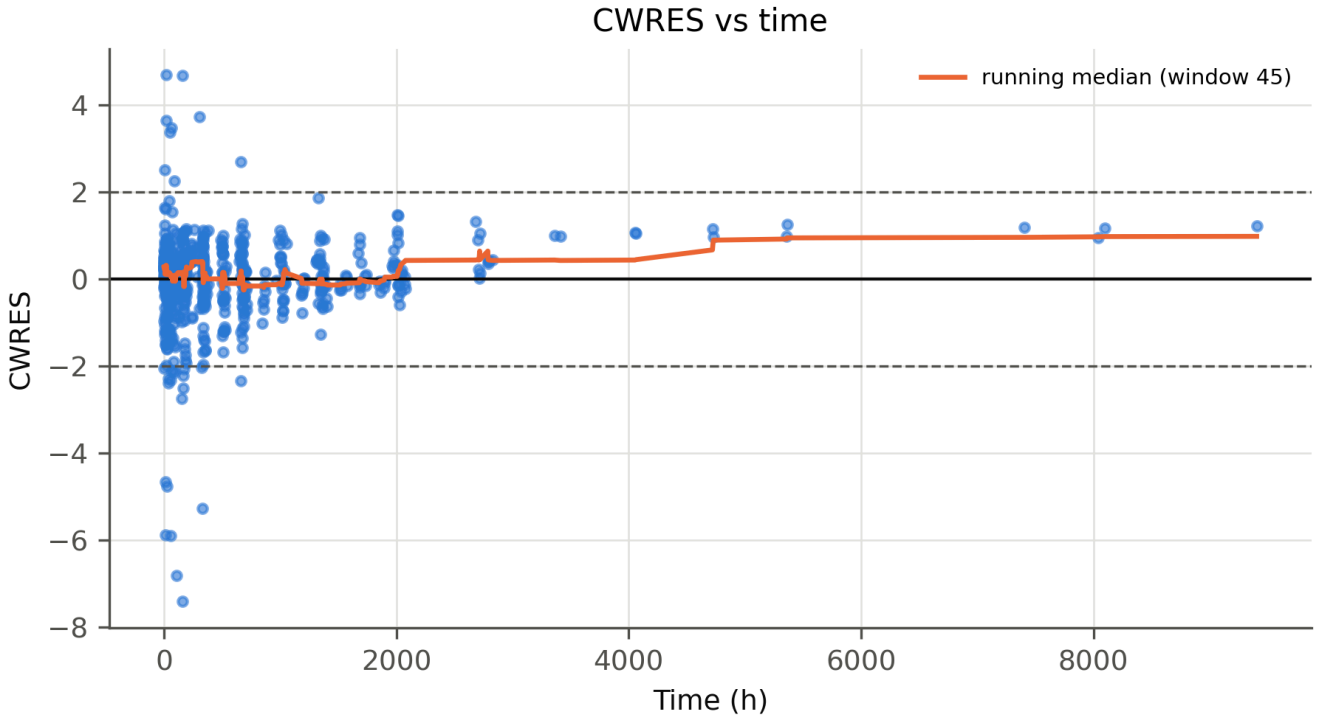
10.7.1 Model diagnostics graphics



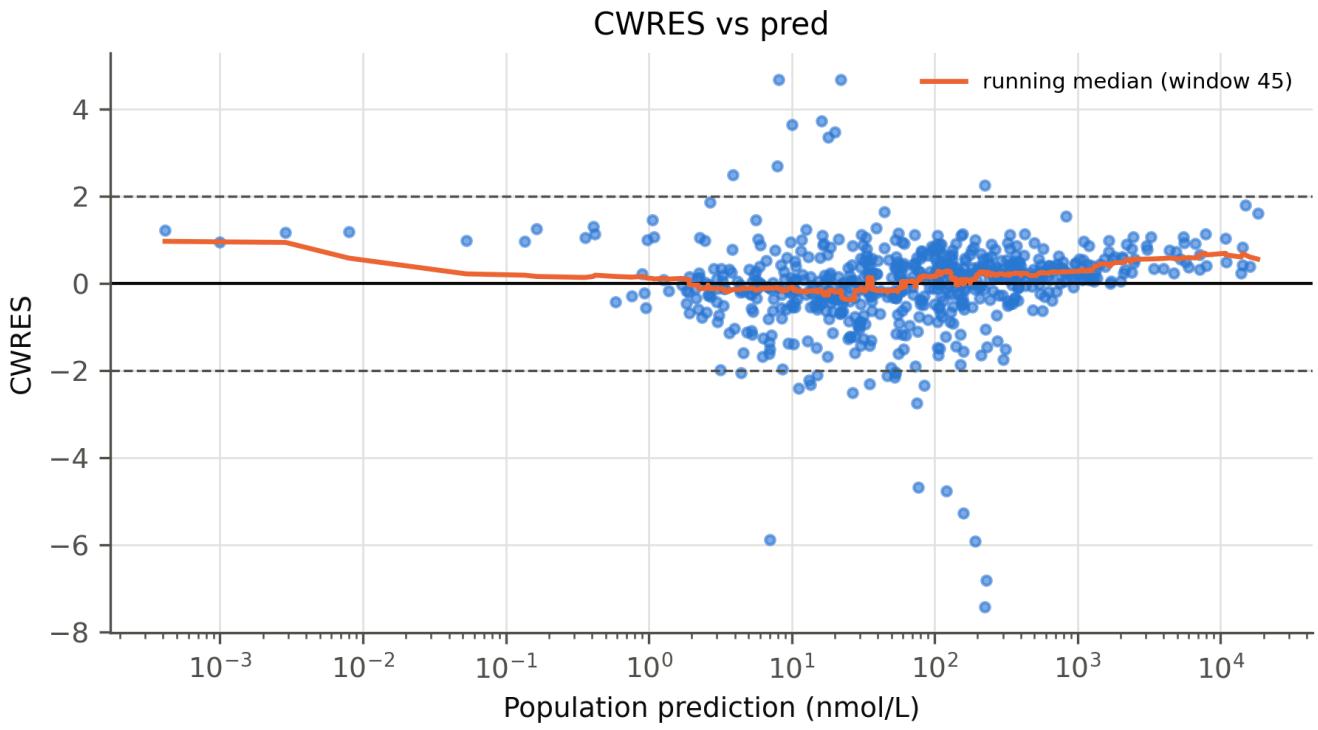
gof dv vs pred



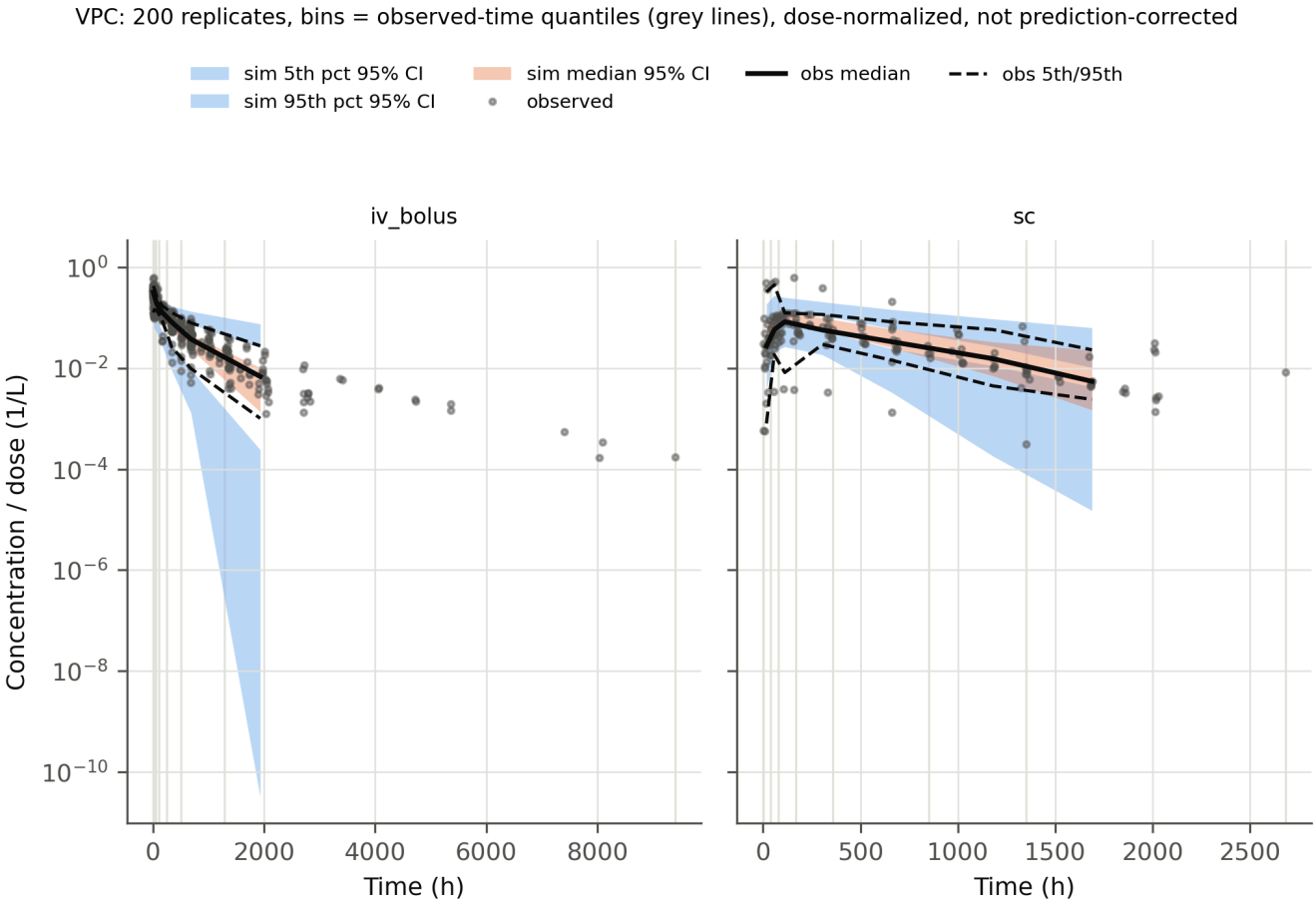
gof dv vs ipred



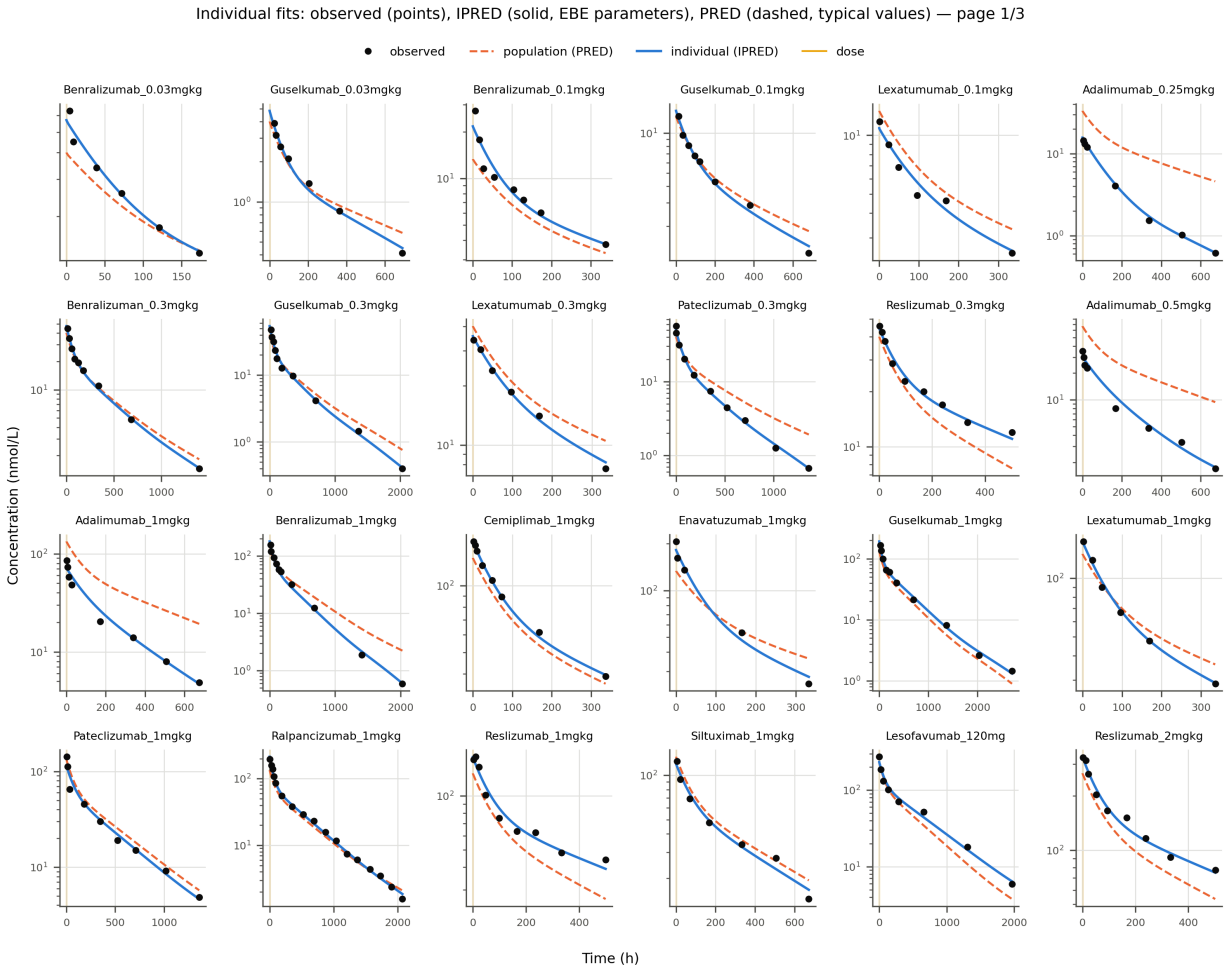
cwres vs time



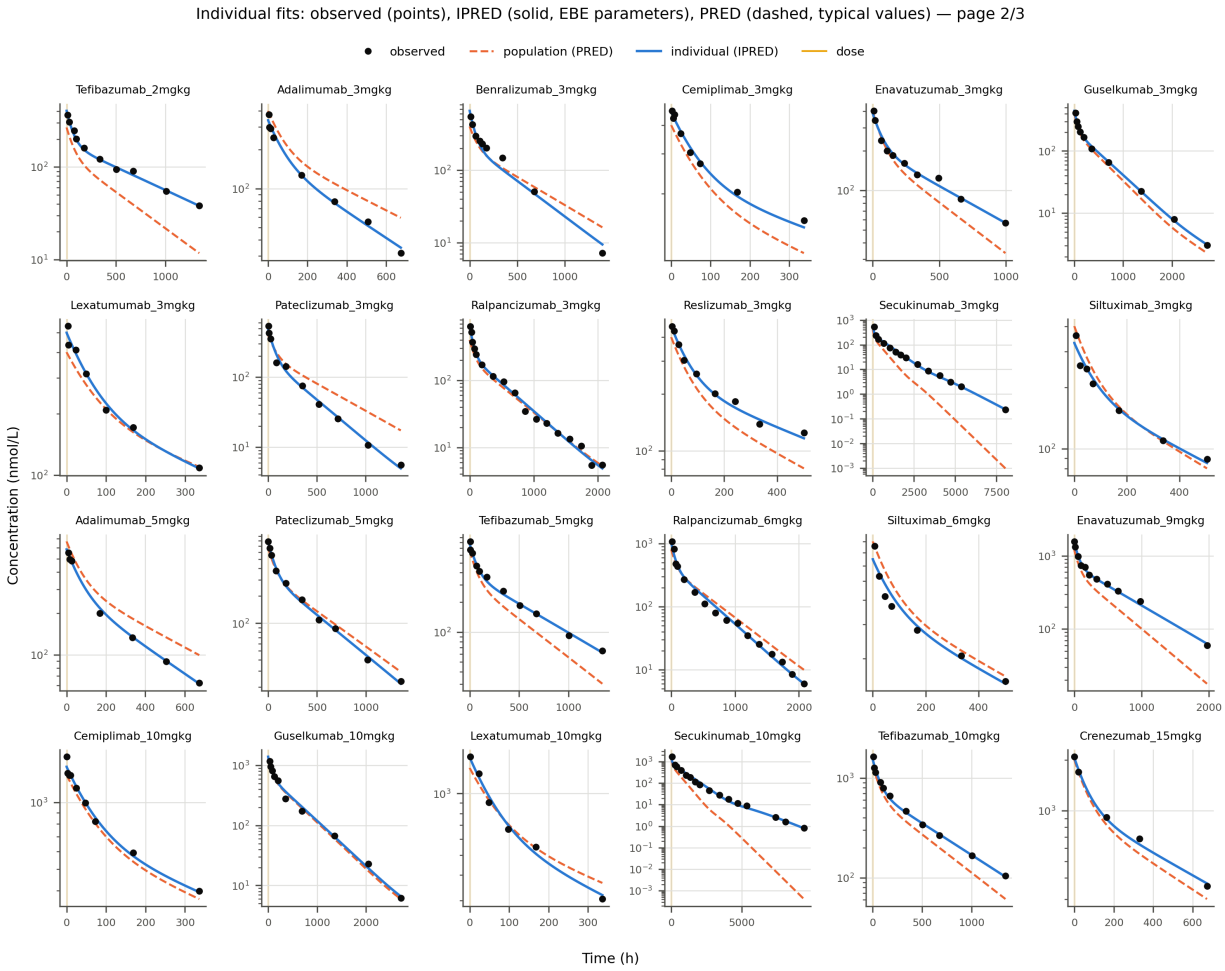
cwres vs pred



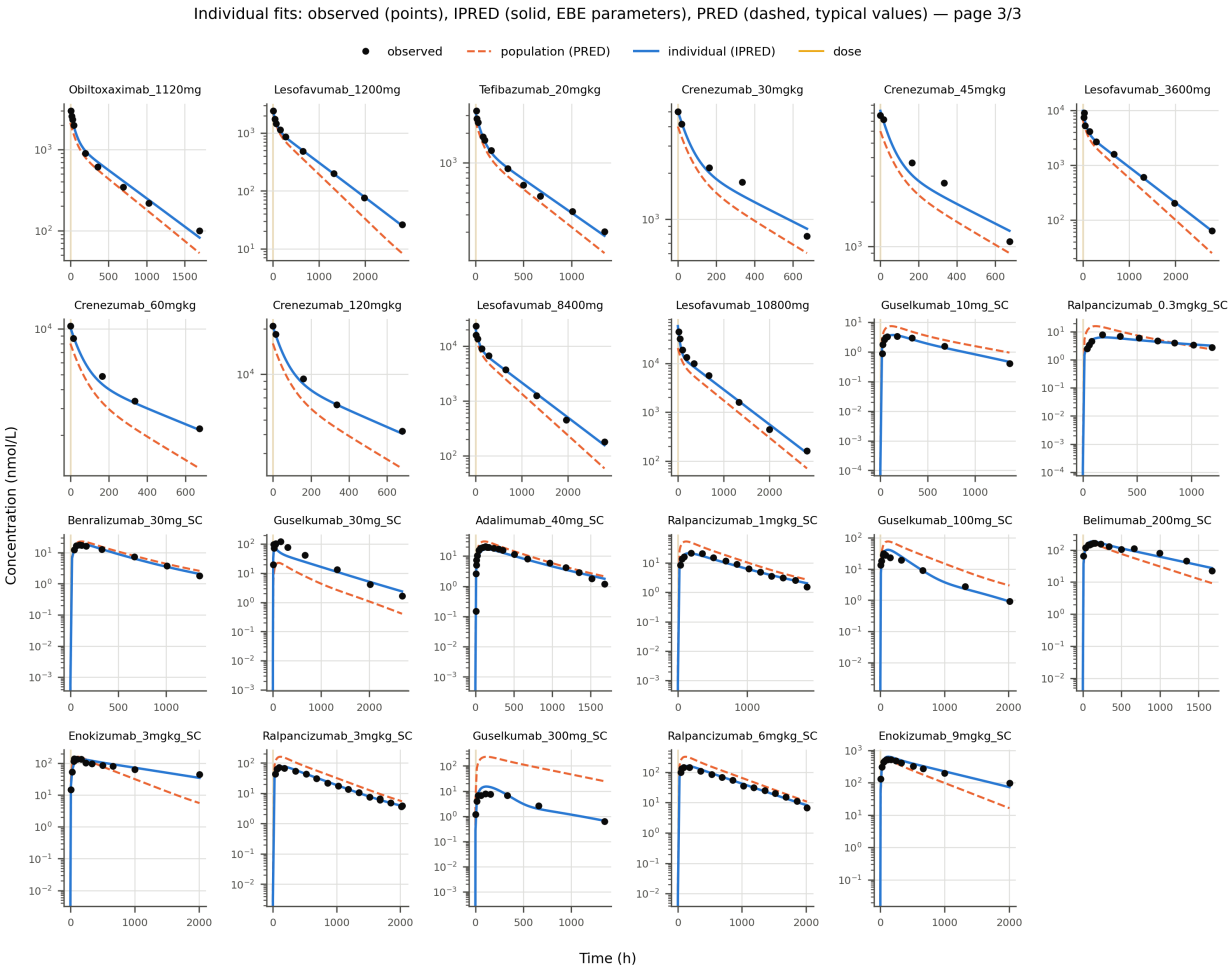
vpc



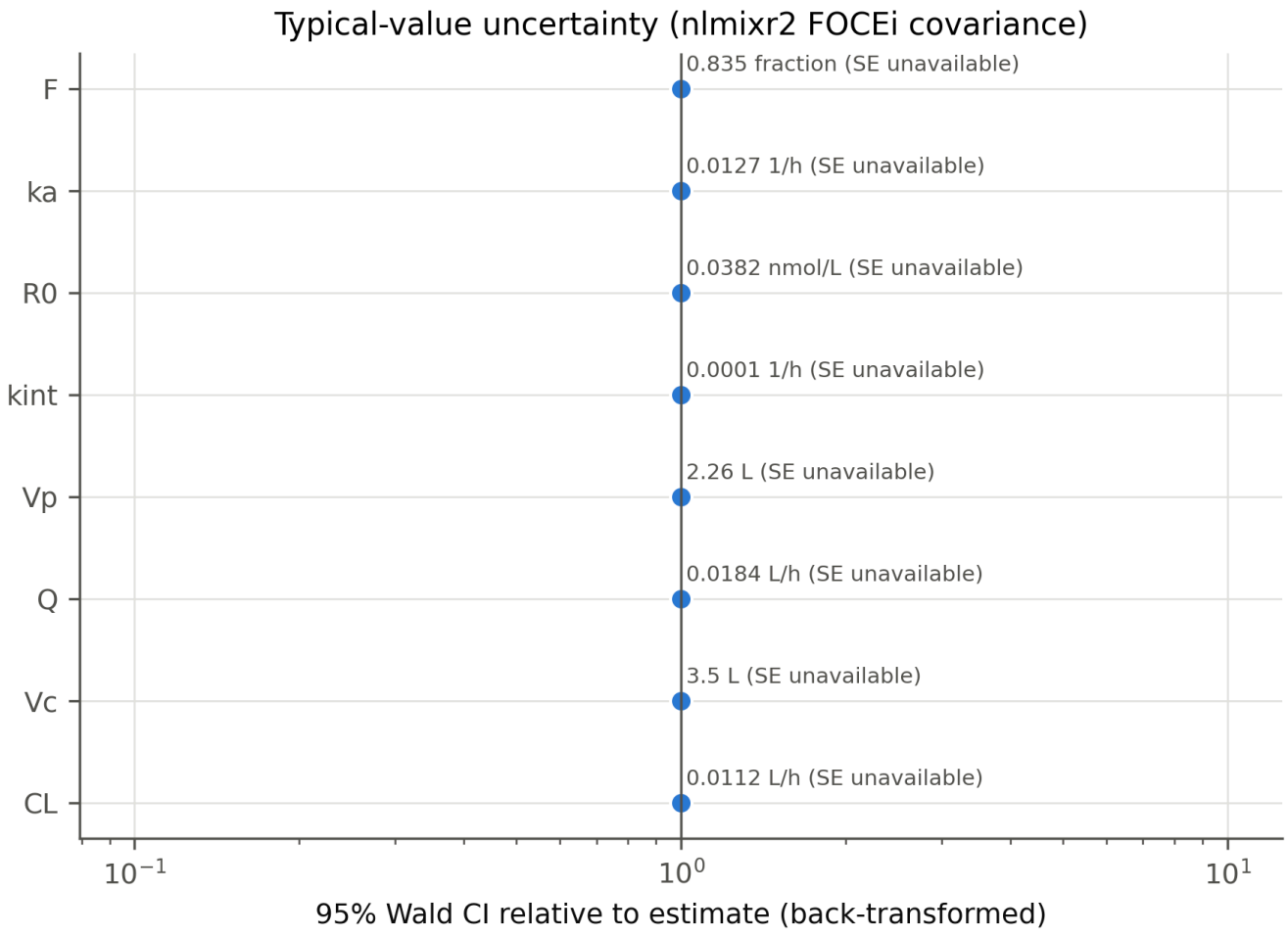
fit individual p01



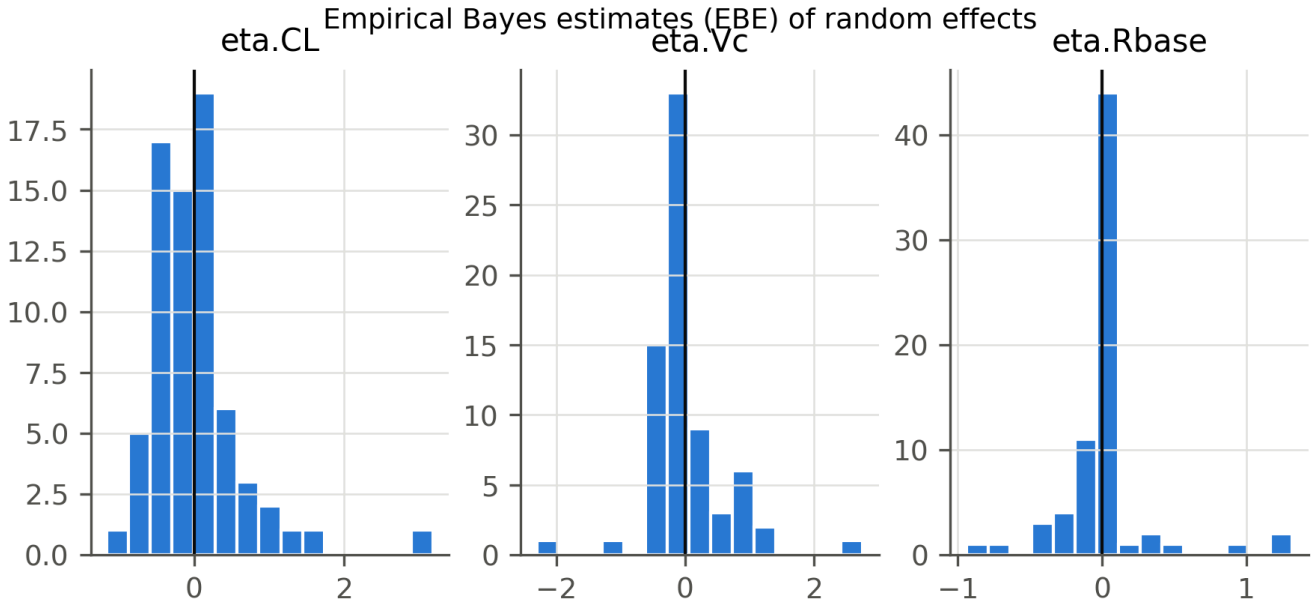
fit individual p02



fit individual p03



parameter uncertainty



eta distributions

10.8 Run environment

item	value
Python	3.12.0
platform	Windows-10-10.0.19045-SP0
Python packages	{"numpy": "1.26.4", "scipy": "1.14.1", "pandas": "2.2.2", "matplotlib": "3.9.1", "PyYAML": "6.0.2", "jsonschema": "4.23.0", "streamlit": "1.44.1", "anthropic": null}
R	R version 4.5.1 (2025-06-13 ucrt)
R packages	{"rxode2": "5.1.6", "nlmixr2": "4.0.1", "nlmixr2est": "4.1.0", "jsonlite": "2.0.0", "ggplot2": "4.0.3", "data.table": "1.17.8"}

11 Appendix: dataset split (two parts by ID)

Source `C:\Users\devamdesai\UF Dropbox\Devam Desai\Research\ML NODE-mPBPK Human PK\Data\mAb_dataset.csv` (sha256 8a554db895425004…), split seed 20260929. Every ID (one published mean profile) is assigned wholly to one part; IDs are grouped by molecule so all dose levels and routes of an antibody stay together (no leakage between parts). Grouping aliases (IDs themselves unchanged): {'Benralizuman': 'Benralizumab'}.

part	IDs	rows	SC IDs	molecules
A	72	813	13	Canakinumab, Cetrelimab, Daclizumab, Farletuzumab, Fulranumab, Gedivumab, Gevokizumab, Infliximab, Ipilimumab, Mepolizumab, Nesvacumab, Nivolumab, Olokizumab, Ozanezumab, Pembrolizumab, Raxibacumab, Risankizumab, Tezepelumab, Ublituximab
B	71	749	13	Adalimumab, Belimumab, Benralizumab, Cemiplimab, Crenezumab, Enavatuzumab, Enokizumab, Guselkumab, Lesofavumab, Lexatumumab, Obiltoxaximab, Pateclizumab, Ralpancizumab, Reslizumab, Secukinumab, Siltuximab, Tefibazumab

12 Appendix: cross-part comparison details

Python engine (cross-engine verified) with saved nlmixr2 typical values, Omega and residual SD; etas sampled from the model part's Omega; 90% PI from simulated replicates incl. residual error. *Caveat:* each ID is a mean profile of a different molecule/dose; out-of-part prediction tests transfer of a generic mAb TMDD model across molecules, not patient-level prediction.

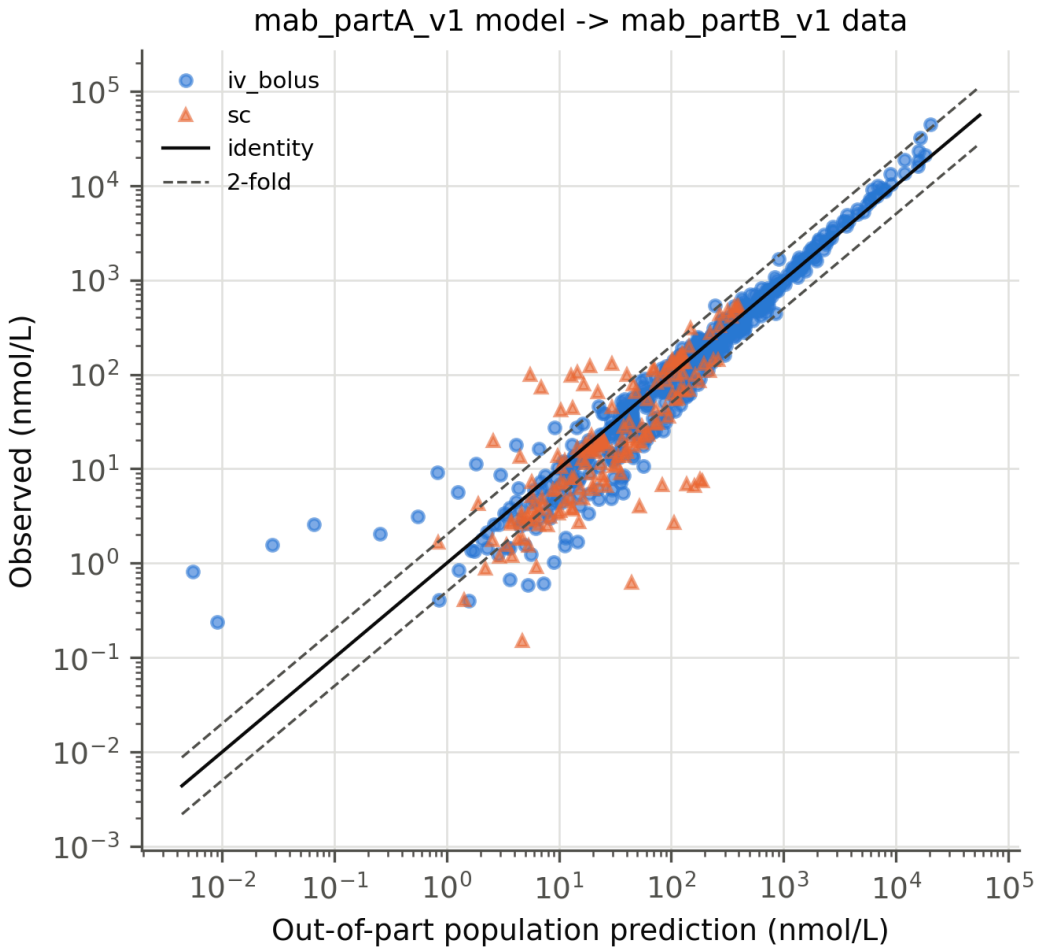
12.1 Parameter estimates: Part A vs Part B

parameter	type	unit	A_estimate	A_ci_lo	A_ci_hi	B_estimate	B_ci_lo	B_ci_hi	A_bsv_cv_pct	B_bsv_cv_pct	ratio_B_over_A	cis_overlap
CL	estimated	L/h	0.007832	0.007272	0.008435	0.01115	–	–	26.07	68.04	1.424	–
Vc	estimated	L	3.077	2.885	3.283	3.498	–	–	28.09	69.05	1.137	–
Q	estimated	L/h	0.02144	0.01615	0.02848	0.01844	–	–	–	–	0.8598	–
Vp	estimated	L	2.634	2.389	2.905	2.258	–	–	–	–	0.8571	–
kon	fixed	L/(nmol*h)	1	–	–	1	–	–	–	–	1	–
koff	fixed	1/h	0.1	–	–	0.1	–	–	–	–	1	–
kint	estimated	1/h	3.960e-04	6.396e-05	0.002452	1.000e-04	–	–	–	–	0.2525	–
kdeg	fixed	1/h	0.1	–	–	0.1	–	–	–	–	1	–
R0	estimated	nmol/L	0.005798	0.001593	0.0211	0.0382	–	–	162	118.7	6.588	–
ka	estimated	1/h	0.008692	0.006964	0.01085	0.01267	–	–	–	–	1.457	–
F	estimated	fraction	0.7054	0.6181	0.78	0.8354	–	–	–	–	1.184	–
lnorm.sd	residual	log-scale SD	0.1974	–	–	0.2215	–	–	–	–	1.122	–

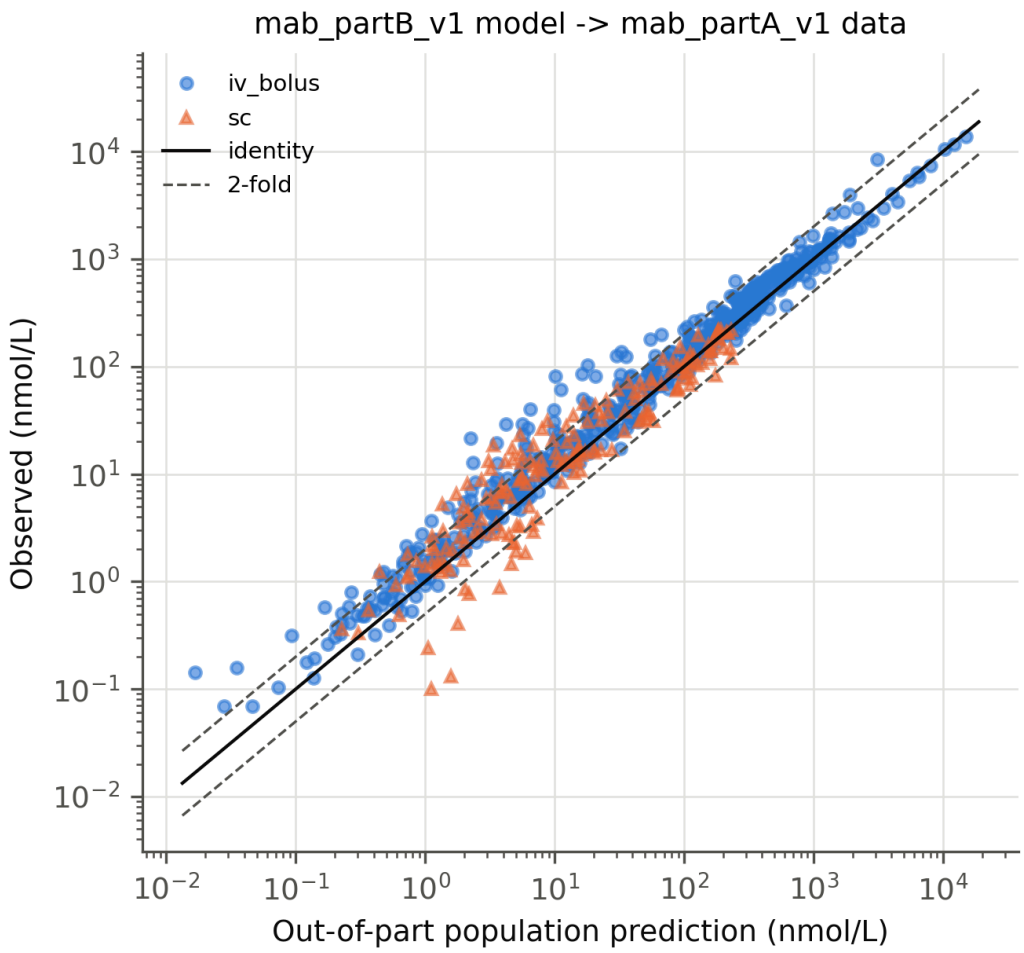
12.2 Prediction performance

model → data	n obs	median log error (bias)	median fold error	within 2-fold	within 3-fold	90% PI coverage	median per-profile PI coverage	replicates
mab_partA_v1 → mab_partB_v1	679	-0.09838	1.294	0.7629	0.894	0.7467	0.9	200

model → data	n obs	median log error (bias)	median fold error	within 2-fold	within 3-fold	90% PI coverage	median per-profile PI coverage	replicates
mab_partB_v1 → mab_partA_v1	744	0.244	1.352	0.836	0.9341	0.9718	1	200
mab_partA_v1 (own fit, PRED)	744	0.02255	1.205	0.9583	0.9906	–	–	–
mab_partA_v1 (own fit, IPRED)	744	0.002784	1.069	0.9866	0.9946	–	–	–
mab_partB_v1 (own fit, PRED)	679	0.07812	1.304	0.8144	0.8969	–	–	–
mab_partB_v1 (own fit, IPRED)	679	0.004048	1.072	0.9926	0.9971	–	–	–



xpred mab partA v1 on mab partB v1



xpred mab partB v1 on mab partA v1

13 Appendix: hypothesis-evaluation details

Kratos' adaptive rules planner and a non-adaptive Latin-hypercube multistart baseline were run on the same replicated SYNTHETIC datasets with the same model, bounds, per-attempt evaluation budget, attempt/time budgets and stopping rules (both arms share the first round). *Successful fit* = best objective within 0.1% of the best found by any arm for that replicate. Differences are shown with their spread; runtimes were measured on a shared, loaded machine and are not a reliable efficiency measure.

13.1 Setting `hypothesis`

Overrides relative to the synthetic configuration: `{"calibration": {"n_lhs": 3, "max_attempts": 12, "max_nfev": 60, "workers": 4, "batch_size": 4}}`.

Calibration-stage comparison on synthetic replicates (n_per_group=4). Differences are reported with their spread; efficiency/accuracy claims are made only where consistent across replicates. Budget: {'n_lhs': 3, 'max_attempts': 12, 'max_nfev': 60, 'time_budget_s': 1800, 'batch_size': 4}; stopping rules: {'plateau_window': 4, 'plateau_rel_tol': 0.001, 'success_repeats': 3}. LLM-assisted arm: **not_run: no Anthropic credentials in environment**.

planner	n_replicates	successful_fit_rate	attempts_mean	attempts_sd	runtime_s_mean	runtime_s_sd	failed_attempts_mean	median_abs_param_relerr_mean	median_abs_log_pred_error_mean	attempts_to_best_mean	analyst_interventions
fixed	5	1	4	0	129.8	31.67	0	0.2491	0.2409	3.2	0
rules	5	1	4	0	132.5	32.27	0	0.2491	0.2409	3.2	0

Paired differences (rules − fixed) across replicates

metric	mean_diff_rules_minus_fixed	sd	n	tie_rel_tol	rules_better_count	fixed_better_count	ties
best_objective	0	0	5	0.001	0	0	5
elapsed_s	2.739	28.11	5	0.001	2	3	0
n_attempts	0	0	5	0.001	0	0	5
median_abs_relerr	0	0	5	0.001	0	0	5

13.2 Setting `hypothesis_stress`

Overrides relative to the synthetic configuration: `{"calibration": {"n_lhs": 1, "max_attempts": 10, "max_nfev": 60, "workers": 2, "batch_size": 2}, "model": {"parameters": {"R0": {"init": 90.0, "lower": 0.001, "upper": 100.0, "estimate": true, "eta": 0.1}, "kint": {"init": 4.0, "lower": 0.001, "upper": 5.0, "estimate": true, "eta": null}}}}`.

Calibration-stage comparison on synthetic replicates (n_per_group=4). Differences are reported with their spread; efficiency/accuracy claims are made only where consistent across replicates. Budget: {'n_lhs': 1, 'max_attempts': 10, 'max_nfev': 60, 'time_budget_s': 1800, 'batch_size': 2}; stopping rules: {'plateau_window': 4, 'plateau_rel_tol': 0.001, 'success_repeats': 3}. LLM-assisted arm: **not requested**.

planner	n_replicates	successful_fit_rate	attempts_mean	attempts_sd	runtime_s_mean	runtime_s_sd	failed_attempts_mean	median_abs_param_relerr_mean	median_abs_log_pred_error_mean	attempts_to_best_mean	analyst_interventions
fixed	5	1	4	0	208.3	54.34	0.4	0.2458	0.2077	2.2	0
rules	5	1	4	0	190.6	37.15	0.4	0.2457	0.2076	2.6	0

Paired differences (rules − fixed) across replicates

metric	mean_diff_rules_minus_fixed	sd	n	tie_rel_tol	rules_better_count	fixed_better_count	ties
best_objective	-5.753e-07	3.052e-06	5	0.001	0	0	5
elapsed_s	-17.63	24.31	5	0.001	5	0	0
n_attempts	0	0	5	0.001	0	0	5
median_abs_relerr	-1.077e-05	2.900e-05	5	0.001	0	0	5

14 Appendix: feature status matrix

area	feature	status	evidence
Data	CSV + YAML/JSON configuration with JSON-Schema validation and actionable row/field errors	implemented and tested	test_data_validation (12 invalid fixtures; row/field message test)
Data	Original files preserved; normalized copy in h/nmol/nmol/L; every unit conversion recorded	implemented and tested	test_synthetic_normalization_units_and_provenance; data/unit_conversions.json in each run
Data	Molecular weight required for mass units (drug and target separately)	implemented and tested	test_mass_units_require_molecular_weight
Data	IV bolus; finite infusion (RATE precedence; DUR conversion; consistency check); repeat doses (ADDL/II expansion)	implemented and tested	test_model_math; test_invalid_inputs (inconsistent_infusion); synthetic ADDL expansion test
Data	SC route via versioned depot extension (template v1.1.0)	implemented and tested	test_sc_depot_bioavailability; cross-engine SC case; mAb runs (26 SC profiles)
Data	Rejection of unsupported EVID / SS / modeled rates / unknown routes / SC infusion	implemented and tested (EVID/SS/route); implemented but untested (modeled rate; SC infusion)	test_invalid_inputs_rejected_with_actionable_codes
Data	Censored (BLQ) records: explicit unsupported-data stop (no LLOQ/2)	implemented and tested	test_censoring_is_unsupported_not_imputed
Data	Censored likelihood (M3)	not implemented	-
Data	Pre-/post-dose convention at coincident times applied identically in both engines	implemented and tested	test_repeat_doses_superposition_and_trough_conventions; cross-engine pre/post cases
Data	Suspicious-record flags (unsorted times; rises; duplicates; missing DV) without deletion	implemented and tested (unsorted); implemented but lightly tested (others)	test_unsorted_can_be_flagged_and_sorted; mAb validation reports
Data	Explicit split of combined dose+DV rows (declared policy)	implemented and tested	test_combined_dose_observation_split_is_explicit; mAb runs
Data	Subject ID map (original -> R integer) saved	implemented and tested	data/subject_id_map.csv; synthetic provenance test
Data	Two-part split by ID grouped by molecule (no leakage)	implemented and executed	data/mab/mab_split_manifest.json
Data	Reproducible synthetic generator (known parameters; bolus/infusion/repeat; noise/variability; invalid fixtures)	implemented and tested	data/synthetic; tests/fixtures
Model	Versioned approved TMDD template; parameter metadata; fixed vs estimated; random effects	implemented and tested	test_model_spec_and_planner
Model	Dimension/unit consistency of all ODE terms	implemented and tested	model_agent.dimension_check; test_spec_dimensions_and_derived_ksyn

area	feature	status	evidence
Model	ksyn derived from kdeg*R0 (never estimated independently); guard against estimating all binding constants	implemented and tested	test_refuse_estimating_all_binding_parameters
Model	Observation mappings free/total drug; free/total target; complex	implemented and tested	test_endpoint_mappings; cross-engine all-endpoint cases
Model	Baseline target equilibrium and total-drug mass balance	implemented and tested	test_baseline_target_equilibrium; test_total_drug_mass_balance (IV and SC)
Model	Hand-checkable linear limit; bolus jump; infusion stop; repeat-dose superposition	implemented and tested	test_model_math
Engines	Python SciPy stiff solver (BDF; sparse Jacobian) with failure/negative-state/non-finite rejection (no clipping)	implemented and tested	test_nonfinite_parameters_rejected; calibration failure records
Engines	Python vs R (rxode2) agreement within 1e-4 relative (prespecified mixed tolerance)	implemented and tested	test_cross_engine (4 cases; 1362 comparisons; max rel diff 1.2e-6)
Engines	R bridge: allow-listed scripts; argument arrays; isolated dirs; captured logs; exit codes; timeouts; cancellation	implemented and tested (allowlist; failures; timeouts via code path)	test_r_bridge_allowlist; estimation/VPC/bootstrap logs
Calibration	Bounded transformed-space multistart (user start + Latin hypercube)	implemented and tested	test_rules_planner_is_bounded_and_deterministic
Calibration	Adaptive rules planner using failures and successes (recentering; bound escape; local refinement; exploration)	implemented and tested	test_rules_planner...; stress evaluation
Calibration	Transparent stopping rules (attempt/time budget; plateau; reproduced optimum)	implemented and tested	test_stopping_rules
Calibration	All attempts recorded incl. failures; ranking; near-bound flags; calibration_failed status	implemented and tested	calibration_history.csv in all runs; test_audit_and_manifest
Calibration	Local sensitivity and Gauss-Newton identifiability summary	implemented and executed	calibration/sensitivity.csv; calibration_summary.json
Calibration	R calibration adapter (rxode2 + optim)	not implemented	Python calibration used; R used for simulation cross-checks and estimation
Estimation	nlmixr2 FOCEi from ranked calibration starts with bounded retries	implemented and tested	synthetic_demo_v1; mab_partA_v1; mab_partB_v1 (all converged)
Estimation	Saved model code; fit RDS; estimates; SE; Wald CIs; Omega; residual SD; individual parameters; predictions; residuals; covariance status	implemented and tested	estimation/final in each run

area	feature	status	evidence
Estimation	Parameter recovery on identifiable synthetic data (25% prespecified tolerance)	implemented and tested (single replicate fit + bootstrap)	synthetic run: all typical values within 9%
Estimation	Replicate population-estimation bias/variability study (multiple simulated datasets refit)	not implemented	only one synthetic dataset fitted with FOCEi (bootstrap provides resampling variability)
Estimation	Alternative estimation methods (SAEM etc.)	not implemented	only FOCEi is enabled (brief: enable others only after testing)
Estimation	Multi-endpoint nlmixr2 estimation (e.g. drug + target)	not implemented	simulation/calibration support multiple endpoints; estimation stops with not_run
Estimation	Missing R -> explicit calibration-only; nothing synthesized	implemented and tested	test_missing_r_gives_calibration_only; test_failed_step_visible_in_report
Diagnostics	Convergence; covariance; delta-method RSE; bounds; correlations; shrinkage; plausibility; residual trends (heuristic flags)	implemented and tested	diagnostics_summary.json and review_flags.json in each run
Diagnostics	VPC (nlmixr2 vpcSim; design/dose histories; BSV + residual; bins; strata; replicates; dose-normalization)	implemented and executed	figures/vpc.png in each run
Diagnostics	Prediction-corrected VPC	not implemented	reported as prediction_corrected=false
Diagnostics	Subject-level bootstrap (dose histories preserved; relabelling; success rate; failed replicates)	implemented and executed on synthetic run	diagnostics/bootstrap (see report)
Diagnostics	Cross-part out-of-part prediction (fold error; 90% PI coverage)	implemented and executed	reports/crosscheck
Graphics	Raw data plots (linear/log; individual; dose-normalized overlays; timelines; missingness)	implemented and executed	figures/raw_* in each run
Graphics	Final model plots (DV vs PRED/IPRED; individual fits; CWRES; VPC; uncertainty; EBEs; calibration history; bound flags; sensitivity)	implemented and executed	figures/* in each run
Graphics	PNG 300 dpi + SVG (Python) / PDF (R) with data behind every figure	implemented and tested	test_regenerate_without_estimation (figure data files)
Reporting	moDoc.qmd from saved artifacts (HTML)	implemented and tested	run/report/moDoc.html; reports/project/moDoc.html
Reporting	PDF report	implemented and executed	HTML printed to PDF with headless Microsoft Edge (reporting_agent.html_to_pdf); moDoc.pdf
Reporting	scripts/results.py and R/results.R regenerate tables/figures without re-estimation	implemented and tested	test_regenerate_without_estimation; figures_R in runs
Reporting	Results narrative with values sourced from artifacts; numerical findings vs interpretation	implemented and tested	report/narrative.json
Reporting	Export bundle with secret scan	implemented and tested	test_secrets_not_exported

area	feature	status	evidence
Run store	One directory per run; atomic status; audit.jsonl; hashes; seeds; environment; manifest	implemented and tested	test_audit_and_manifest
Run store	Cancellation preserving records	implemented and tested	test_cancellation_preserves_records
Run store	Resume only when input/config/code compatible	implemented and tested	test_resume_requires_compatibility
Run store	Seed reproducibility	implemented and tested	test_seed_reproducibility
Interface	Streamlit UI (project; upload; mapping; units/assay; parameters; fixed/estimated; run launch; progress; cancel; history; estimates; diagnostics; figures; export; previous runs)	implemented; smoke-tested (renders both pages)	test_streamlit_app_renders; interactive use not tested
Interface	CLI commands validate/simulate/calibrate/estimate/diagnose/report/run/ui (+ split/crosscheck/bootstrap/evaluate/synth/status/cancel)	implemented and executed	commands used to produce all runs
LLM	Optional LLM planner (Anthropic SDK; schema-validated JSON; bounds/fixed-parameter checks; no subject data; fallback)	implemented; guardrails tested with mocks	test_llm_malformed_and_out_of_bounds_rejected_with_fallback
LLM	Live LLM calls and LLM-assisted evaluation arm	not run	no Anthropic credentials and anthropic package not installed
Evaluation	Hypothesis evaluation: adaptive vs fixed multistart (same data/model/bounds/budget/stopping)	implemented and executed (5 replicates; default and stress settings)	evaluation/hypothesis; evaluation/hypothesis_stress
Environment	Lockfiles	partially: requirements-lock.txt and R/r-packages-lock.json written; renv.lock not written (renv not installed)	-

15 Appendix: reproducibility

Each run directory contains the original inputs (unchanged), normalized data, subject-ID map, configuration snapshot, model specification and generated nlmixr2 model code, seeds, input/code hashes, environment versions, `audit.jsonl`, calibration history (including failures), accepted starts, the nlmixr2 fit object (`fit.rds`), parameter/prediction/diagnostic tables, figures (PNG 300 dpi + SVG/PDF with the data behind each figure), this report and a checksummed `manifest.json`. `python scripts/results.py --run <run>` and `Rscript R/results.R --run <run>` regenerate tables and figures without re-estimation.